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Schulze Gronover et al.

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(54) **METHOD TO OBTAIN LOW
TRITERPENE/TRITERPENOID-CONTAINING
NATURAL RUBBER LATEX**

(71) Applicant: **Fraunhofer-Gesellschaft zur
Foerderung der Angewandten
Forschung E.V., Munich (DE)**

(72) Inventors: **Christian Schulze Gronover, Muenster
(DE); Kristina Unland, Muenster (DE)**

(73) Assignee: **Fraunhofer-Gesellschaft zur
Foerderung der Angewandten
Forschung E.V., Munich (DE)**

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Primary Examiner — Amjad Abraham

Assistant Examiner — Santosh Sharma

(74) *Attorney, Agent, or Firm* — Kagan Binder, PLLC

(57) **ABSTRACT**

The invention provides a method of reducing or eliminating the content of triterpenes and/or triterpenoids of latex or rubber from a laticiferous plant or part thereof, comprising genetically modifying the laticiferous plant or part thereof, such that the activity of at least one oxidosqualene cyclase is reduced or eliminated, compared to the respective wild-type or untreated laticiferous plant or part thereof. The invention further provides a laticiferous plant or part thereof being genetically modified such that the activity of at least one oxidosqualene cyclase is reduced or eliminated, compared to the respective wild-type or untreated laticiferous plant or part thereof, and an RNAi being specifically directed against a mRNA transcript of at least one oxidosqualene cyclase, for use in a method of reducing or eliminating the content of triterpenes and/or triterpenoids of latex or rubber from a laticiferous plant or part thereof.

15 Claims, 10 Drawing Sheets

Specification includes a Sequence Listing.

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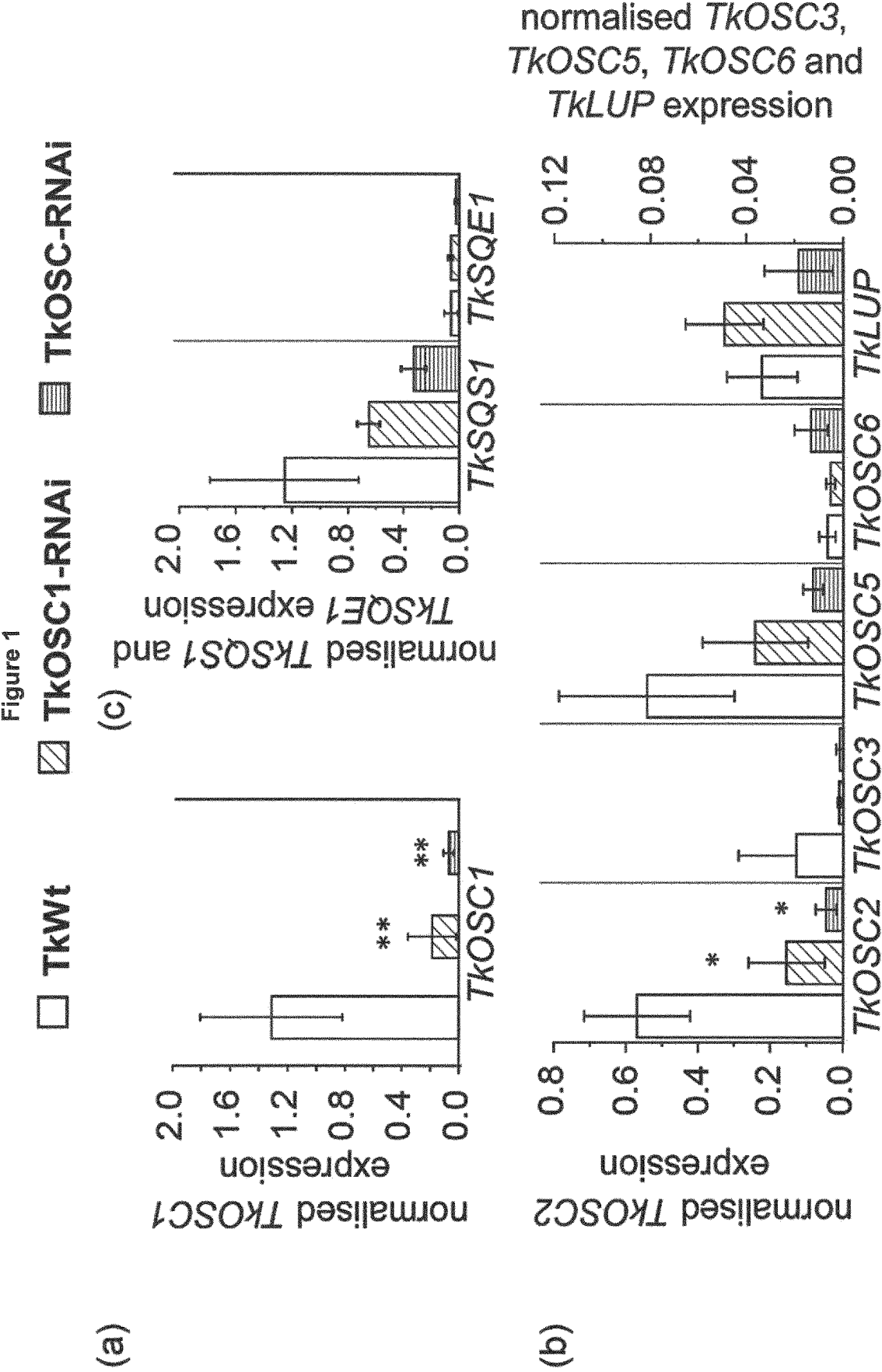


Figure 1 (cont.)

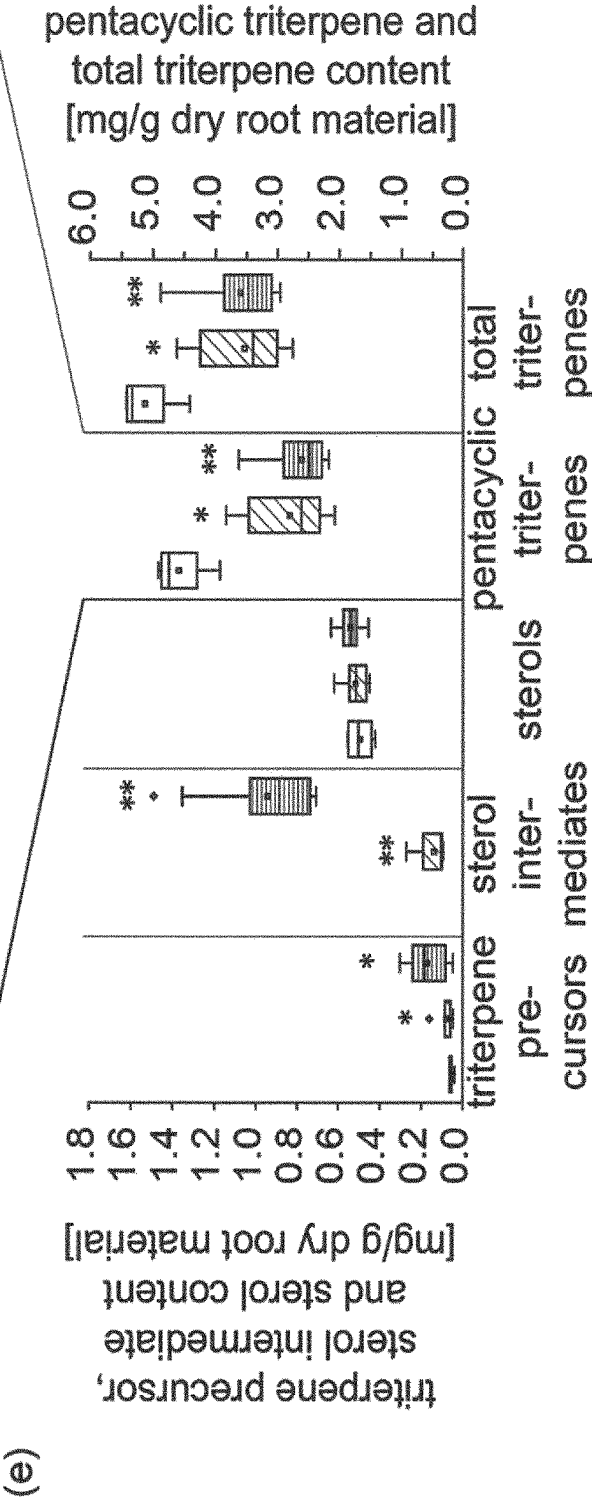
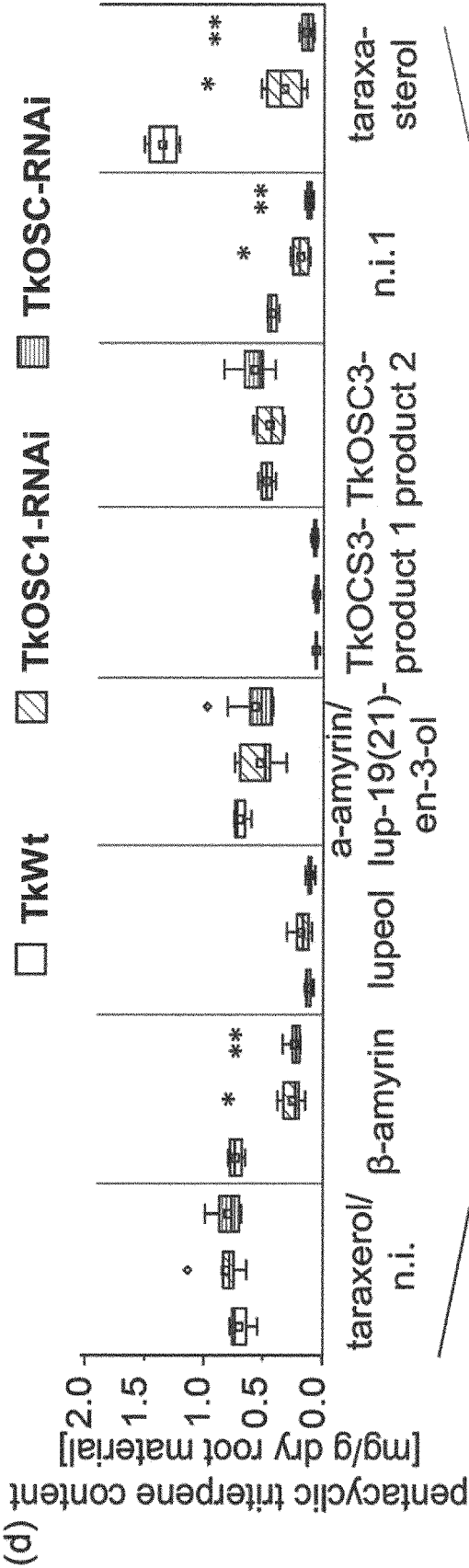


Figure 2

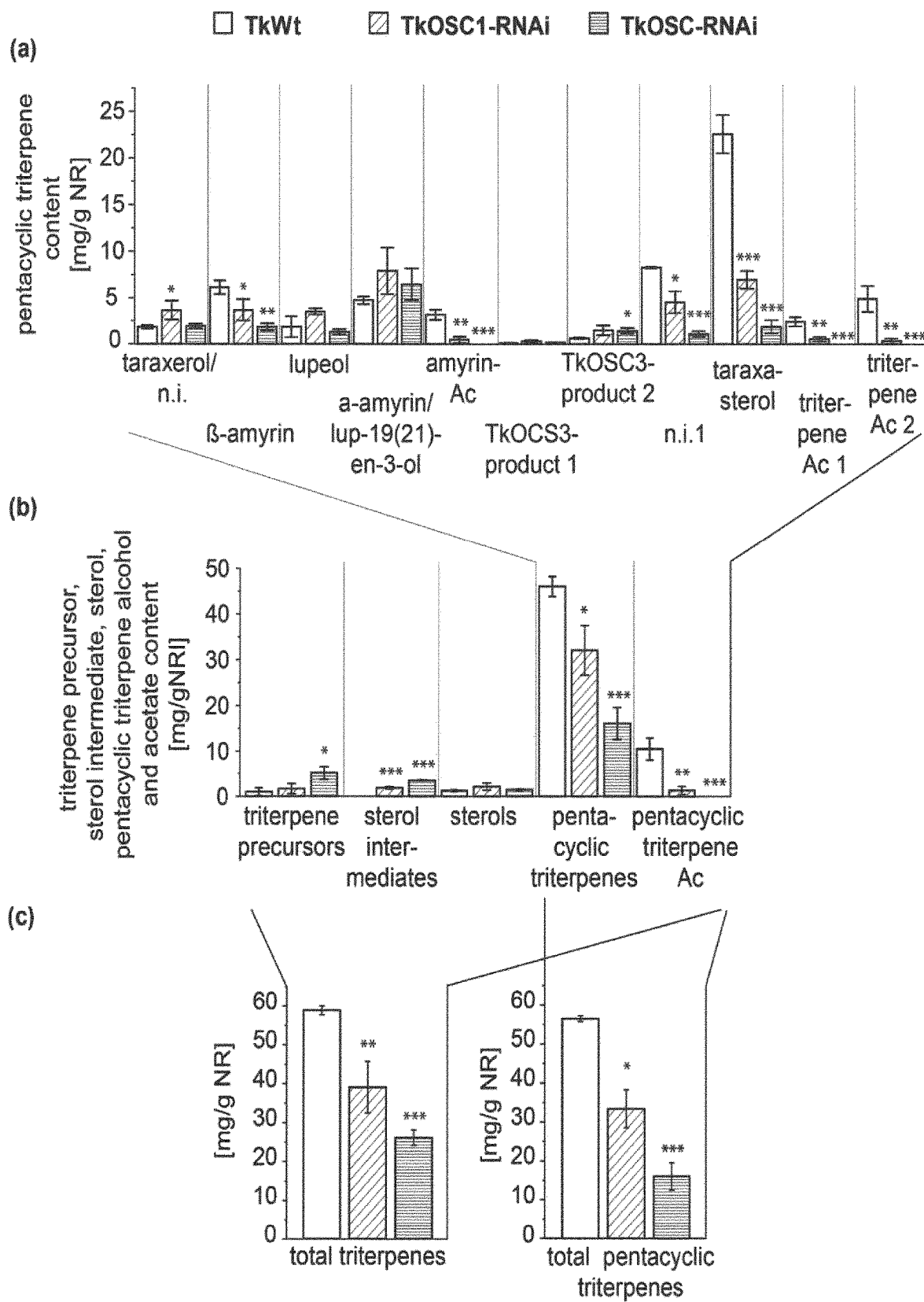


Figure 3

TKOSC1	MWKLRIGEKNGKFNIGDNGDDYLHSTNNFVGRQTWEFFDPA-GTQKERDEIERLREQFL	59
TKOSC2	MWKLRIGEKNGKFNIGDNGDEYLYSTNNFVGRQTWEFFDPA-GTQEERDEVERLREQFL	59
TkOSC8	MWKLKIAEGN-----DPYLFSTNNFVGRQIWEFFDPA-GTPEERQEVENARQNFL	49
TKOSC3	MWELKIAEGD-----GPYLYSTNNFVGRQFWEFNPDA-GTPEEKKEIEKVRQKFK	49
TkOSC7	MWKLKIAQGN-----DPYLFSTNNFVGRQIWEFFDPA-GTPAERQEVEDARQYFR	49
TklUP	MWKLKIAEGS-----DDEWLTSTNNHVGRQHWQFDPDA-GTEEEERAQIEKIRLNFK	50
TKOSC4	MWRLRIGQGN-----KDDPYLFSTNNFVGRQIWEYDENYKATPEELEEVKQARSSF	52
TKOSC5	MWKLKIAEGG-----NDPYLHSTNNFIGRQIWEFFDSNH-GTPEDRAEAEHARINFW	50
TKOSC6	MWKLKIAEGG-----NDPYLHSTNNFIGRQIWEFFDPNH-GTTEDRAEVEQARLDFW	50
	.*:*:*:* . . :* :*.*:*:* :* :* :* :*	
TKOSC1	INKKKLDIRCCADLLMRNQLIKESGIDLISVPPVRLGDDEVDVNYEAVTTAVRKAVRLNRA	119
TKOSC2	INKKKLDISCCADLLMRNQLIKESGIDLISEAPVKLGHDEVDVNYKAVTTAVRKAVRLNRA	119
TkOSC8	N-RQKEGFQTCGDLLMRIQLIRENGIDLISIPPARLGEKEDVDYEAVTTSIRKAVRYS	108
TKOSC3	DNRKGGFHACGDLLMRMQLMKENAIDLTSILPVRISEGEQVNYEATTIAVRKAVRLHRG	109
TkOSC7	NNRR-EGVHPCGDLLMRAQLIKESGIDVSSIPTTRLEENEENVNYDALTTAVKKALRLNRA	108
TklUP	LNRF--QFKQSADLLMRTQLRKENPINK-IPDAIKLNETEEVTNDAVSTTLKRAISFYST	107
TKOSC4	NNRH--KQRPCDDRLWRFQFLREKNFKQ-TIPREIIVDESEIKYETVDNALKRAVRYWTA	109
TKOSC5	NLRH--RVKPSDDVLWRMQFLKEKQFKQ-TIAQVKIEEFEEISYEKATVTLKRCVNLFAA	107
TKOSC6	NHRH--QVKPSSDVLWRMQFLKEKEFKQ-TIAQVKIEDSEEEISYEKTTTLRLRCVSEFAA	107
	: . * * * * : * : : * : * : * : *	

Figure 3 (cont.)

TKOSC1	FTKGAWTFSDQDQGWVVS	<u>DCTAE</u>	AVKCLLALSQMPQEISGEKVDVERLYDAINVLLYLQS	533
TKOSC2	FTKGAWTFDQDQGWVVS	DCTAE	AVKCLLAMSQMPQEIAGEKVEVERLYDAINVLLYLQS	533
TkOSC8	FSKGSWTFDQDHGWPVS	DCTGE	ALKCLLLSQMPPEEISGEKVDNQRLYDAVNFLHVQS	525
TKOSC3	FTKGSWTFSDQDHGWTVS	DCTAE	ALKCLLLSNMPKNIAGEKDDTARLYEAVNVLLYMQS	526
TkOSC7	FTKGAWTFSDQDQGWVVS	DCTAE	ALKCLLLSQMPPEEISGEKADNERLYEAVNVLLYLQS	525
TkLUP	TSKGAWTFSDQDHGQVVS	DCTAE	GLKVALLYSQMSPELVGEKLETEHLYDAVNVILSLQS	526
TKOSC4	ISKGGWTFSDQDHGQVVS	DCTAH	GLMSCLLLSKMPLEIVGEKMETERLFDCLNLLSLQY	528
TKOSC5	ISKGSWTFSDQDHGQVVS	DTTAE	ALKCCLLFSTMPPEIVGEKMKPEQLNDAVNVILSLQS	525
TKOSC6	ISKGSWTFSDQDHGQVVS	DCTAE	GLKCCLLFLTMPAEIVGEHMKPEQFNDAVNVILSLQS	525
	:**.*::: **:*	** *	* * * * * : : : * : * : *	
TKOSC1	PETGGFAIWEAPVPKPYLEKLNPS	ELFADIVVEREHVECTGSI	IQTLQTFKTLHPGHRQK	593
TKOSC2	PETGGFAVWEVPVPKPYLEKLNPS	ELFADITVEREHVECTGSI	IQALQTFKTLHPGHRK	593
TkOSC8	PTTGGFAVWEKPIPHPYLQTLNP	SEMFADIVVEREHVECTT	SVMQALIAFQHFHSGHREK	585
TKOSC3	PVSGGFAVWEPPIPKPFLLQLLNP	SEIFADIVVEKEHVECTSS	IIGALIEFKRVHPNHRKE	586
TkOSC7	PISGGFAIWEPPVPQPYLQMLNP	SEIFADIVVEKEHVECTSS	IIGALLAFKRLHPGHRK	585
TkLUP	E-NGGFPAWEPQRAYAWLEKFN	PTEFFEDVLIEREYVECTSS	AIQGLTLFKKLHPGHRK	585
TKOSC4	K-NGGFSGWEPAGAPKWLEMLNP	SEMFADIMIEIQYVECTSS	ALQALILFKKLYPEHRSQ	587
TKOSC5	K-NGGLASWEPAGSSEWLEIFNP	TEFFADIVIEHEYVECTSS	AIQAIVLFNKLYPQHRKK	584
TKOSC6	K-NGGLAAWEPAGSSEWLEVLNP	TEFFADIVIEHEYVECTSS	ASQALVLFKKLYPGHRRK	584
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Figure 3 (cont.)

TKOSC1
AKLLINGQMDNGDFPQQEITGVYMKNCMLHYAEYRNTFPLWALGEYRKRVLAKQET-
- 770

TKOSC2
SKLLINGQTDNGDFPQQEITGVYNKNCMLHYAEYRNTFPLWALGEYRKRVLAKQET-
- 770

TkOSC8
ARILINAQMDNGDFPQQEMTGAAMRNCMLHYPLYSNIFPLLALSKYRQQFWDK-----
- 758

TKOSC3
AKLLINAQMDNGDFPQQEITGVYMKNCMLLYAQYRNIFPLWALGEYRKRVSIIKKGIQ
N 765

TkOSC7
AKILINSQMDNGDFPQQEITGVYMKNCMLHYAEYRNIFPLWALGEYRKRVVVN-----
- 758

TkLUP
VRLINSQMEEGDFPQQEITGVFMKNCNLNYSSYRNIFPIWALGEYRRIVQNI-----
- 758

TKOSC4
AKFLINSQMEDGDFPQKEITGSFRKNCMLHYACHRNIFPMWALAEYKNRVSKII----
- 761

TKOSC5
ARLLINSQLENGDFPQQEITGVFMKNCMLHYALYRNIIYPMWALADYRKKVLSTCKK--
- 760

TKOSC6
AKLLINSQLENGDFPQQKIAGVFKMNCMLHYALYRNIFPMWALADYRKHVLKGI----
- 758

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METHOD TO OBTAIN LOW TRITERPENE/TRITERPENOID-CONTAINING NATURAL RUBBER LATEX

PRIORITY CLAIM

This application claims priority to International Application No. PCT/EP2019/080668, filed Nov. 8, 2019, which claims priority to European Application No. 18205121.9, filed Nov. 8, 2018, wherein the contents of said applications are incorporated herein by reference in their entireties. Also, the entire contents of the ASCII text file entitled “IPM0119US Sequence Listing.txt” created on May 2, 2021, and having a size of 107 kilobytes, is incorporated herein by reference.

TECHNICAL FIELD OF THE INVENTION

The present invention provides a method of reducing or eliminating the content of triterpenes and/or triterpenoids of latex or rubber from a laticiferous plant or part thereof, comprising genetically modifying the laticiferous plant or part thereof, such that the activity of at least one oxidosqualene cyclase is reduced or eliminated. Further, the present invention provides a laticiferous plant or part thereof being genetically modified such that the activity of at least one oxidosqualene cyclase is reduced or eliminated, compared to the respective wild-type or untreated laticiferous plant or part thereof. Additionally, the present invention also includes an RNAi being specifically directed against a mRNA transcript of at least one oxidosqualene cyclase, for use in said method of reducing or eliminating the content of triterpenes and/or triterpenoids of latex or rubber from a laticiferous plant or part thereof.

BACKGROUND ART

Supplied by the mevalonic acid (MVA) pathway oxidosqualene cyclases (OSCs) catalyse the cyclisation of 2,3-oxidosqualene (Abe, 2007; Thimmappa et al., 2014). This leads to the production of triterpenes, 30-carbon compounds, which comprise nearly 200 different scaffolds in plants (Xu et al., 2004). Tailoring enzymes can further modify triterpenes via acetylation, oxidation or by adding sugar chains, resulting in a huge diversity with more than 20,000 known triterpenoids (Xu et al., 2004; Thimmappa et al., 2014).

Due to the cyclisation reaction type, a differentiation between two classes of triterpenoids exist: In the first class chair-boat-chair conformation occurs and results initially in the production of cycloartenol, which is a phytosterol intermediate (Abe, 2007; Thimmappa et al., 2014). The resulting sterols function e.g. as membrane compounds or signalling molecules. Therefore, they are important for plant development and growth (Hartmann, 1998; Clouse, 2002). In the second class chair-chair-chair conformation results among others in the formation of pentacyclic triterpenes such as lupeol and β -amyrin (Thimmappa et al., 2014). A large number of pentacyclic triterpenes show antimicrobial and antifungal characteristics and play a role in plant defence.

Due to this characteristics, they are interesting for broad applications in food, cosmetics and pharmaceutical industries (Augustin et al., 2011; Osbourn et al., 2011; Moses et al., 2014). In dandelion, a high amount of pentacyclic triterpenes mainly occurs in roots, more precisely in laticifers, a specialised cell type, which produces a milky sap containing high concentrations of secondary metabolites

2

(Furuno et al., 1993; Akashi et al., 1994). This cell type is also the place for poly(cis-1,4-isoprene) biosynthesis, the main component of natural rubber (NR) (Hagel et al., 2008). NR is an important biopolymer that is used for the production of more than 40,000 consumables, including e.g. tyres and medical devices (Davis, 1997; Mooibroek and Cornish, 2000; Hagel et al., 2008). High quality NR for commercial use usually derives from the rubber tree *Hevea brasiliensis*; however, it can also be extracted from the roots of the Russian dandelion, *T. koksaghyz*. NR derived from dandelion incorporates small amounts of additives like fatty acids, proteins and triterpenes, which have impact on the properties of NR (Nor and Ebdon, 1998; Sakdapipanich, 2007; Schulze Gronover et al., 2011).

Since *T. koksaghyz* is a self-incompatible, sexually reproductive plant, it produces heterogenic offspring (Krotkov, 1945; Schulze Gronover et al., 2011). Consequently, the amount of pentacyclic triterpenes varies between single plants, which is not desirable for industrial applications.

Moreover, Jungkong et al. and Ikeda et al. have shown that substances such as triterpenes generally have an important influence on the properties of natural rubber or latex.

Accordingly, there is not only the need for a consistent level of triterpenes and/or triterpenoids, but also for a reduced content of triterpenes and/or triterpenoids of latex or rubber from laticiferous plants. The present invention aims to solve this problem.

SUMMARY OF THE INVENTION

The problem is solved by the subject-matter as defined in the claims and as defined herein.

In one embodiment, the present invention provides a method of reducing or eliminating the content of triterpenes and/or triterpenoids of latex or rubber from a laticiferous plant or part thereof, comprising genetically modifying the laticiferous plant or part thereof, such that the activity of at least one oxidosqualene cyclase is reduced or eliminated, compared to the respective wild-type or untreated laticiferous plant or part thereof. The laticiferous plant or part thereof can also be a progeny of the respective laticiferous plant.

In a preferred embodiment thereof, the laticiferous plant or part thereof is genetically modified by at least one heterologous nucleotide sequence that is capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase, and/or wherein the laticiferous plant or part thereof is genetically modified by subjecting it to and/or incubating it with agents, which are capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase. The term “is/are capable of reducing or eliminating” is also to be considered compared to the respective wild-type or untreated laticiferous plant, progeny thereof or part thereof.

In a further embodiment, the at least one heterologous nucleotide sequence encodes at least one RNAi that interferes with the mRNA encoding the at least one oxidosqualene cyclase, with the proviso that the at least one RNAi is capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase. The term “is/are capable of reducing or eliminating” is also to be considered compared to the respective wild-type or untreated laticiferous plant, progeny thereof or part thereof.

In one embodiment of the method of the present invention, the at least one RNAi is specifically directed against the mRNA transcript of the at least one oxidosqualene cyclase, preferably wherein the at least one RNAi comprises or consists of a nucleotide sequence according to SEQ ID NO:

1 and/or comprises or consists of a nucleotide sequence according to SEQ ID NO: 2 and/or comprises or consists of a nucleotide sequence being at least 60%, 70%, 80%, 90%, 95% or 99% identical to SEQ ID NO: 1 or SEQ ID NO: 2.

In a further embodiment of the method of the present invention, the laticiferous plant or part thereof is selected from the group of plant families consisting of Apocynaceae, Asteraceae, Euphorbiaceae and Moraceae. The laticiferous plant or part thereof can also be the progeny of the respective laticiferous plant in this embodiment.

In a specific embodiment of the method of the present invention, the at least one oxidosqualene cyclase is selected from the group consisting of oxidosqualene cyclase 1 according to SEQ ID NO: 4, the oxidosqualene cyclase 2 according to SEQ ID NO: 6, the oxidosqualene cyclase 3 according to SEQ ID NO: 8, the oxidosqualene cyclase 4 according to SEQ ID NO: 10, the oxidosqualene cyclase 5 according to SEQ ID NO: 12, the oxidosqualene cyclase 6 according to SEQ ID NO: 14, the oxidosqualene cyclase 7 according to SEQ ID NO: 16, the oxidosqualene cyclase 8 according to SEQ ID NO: 18, the lupeol synthase according to SEQ ID NO: 20 and an amino acid sequence being at least 60%, 70%, 80%, 90%, 95% or 99% identical to SEQ ID NOs: 4, 6, 8, 10, 12, 14, 16, 18 or 20.

In one embodiment of the method of the present invention, the part of the laticiferous plant is selected from the group consisting of cells, tissues, organs, roots, stems, branches, leaves, exudates, latex, and mixtures thereof.

The present invention also includes a laticiferous plant or part thereof being genetically modified such that the activity of at least one oxidosqualene cyclase is reduced or eliminated, compared to the respective wild-type or untreated laticiferous plant or part thereof. The term "is/are capable of reducing or eliminating" is also to be considered compared to the respective wild-type or untreated laticiferous plant, progeny thereof or part thereof.

In one embodiment, such a laticiferous plant or part thereof is genetically modified by at least one heterologous nucleotide sequence that is capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase, or wherein the laticiferous plant or part thereof is genetically modified by subjecting it to and/or incubating it with agents, which are capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase. The term "is/are capable of reducing or eliminating" is also to be considered compared to the respective wild-type or untreated laticiferous plant, progeny thereof or part thereof.

In another embodiment, in the laticiferous plant or part thereof of the present invention, the at least one heterologous nucleotide sequence encodes at least one RNAi that interferes with the mRNA encoding the at least one oxidosqualene cyclase with the proviso that the at least one RNAi is capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase. The laticiferous plant or part thereof can also be the progeny of the respective laticiferous plant. The term "is/are capable of reducing or eliminating" is also to be considered compared to the respective wild-type or untreated laticiferous plant, progeny thereof or part thereof.

In a further embodiment, in the laticiferous plant or part thereof of the present invention, the at least one RNAi is specifically directed against the mRNA transcript of the at least one oxidosqualene cyclase, preferably wherein the at least one RNAi comprises or consists of a nucleotide sequence according to SEQ ID NO: 1 and/or comprises or consists of a nucleotide sequence according to SEQ ID NO: 2 and/or comprises or consists of a nucleotide sequence

being at least 60%, 70%, 80%, 90%, 95% or 99% identical to SEQ ID NO: 1 or SEQ ID NO: 2.

In a further embodiment, the laticiferous plant or part thereof is selected from the group of plant families consisting of Apocynaceae, Asteraceae, Euphorbiaceae, Moraceae and crossings thereof, preferably wherein the laticiferous plant or part thereof is selected from the group consisting of *Agoseris glauca*, *Apocynum venetum*, *Chrysanthamnus graveolens*, *Chrysanthamnus nauseosus*, *Chrysanthamnus viscidiflorus*, *Funtumia elastica*, *Parthenium argentatum*, *Scorzonera acanthoclada*, *Scorzonera albicaulis*, *Scorzonera divaricate*, *Scorzonera hissaricata*, *Scorzonera racemosa*, *Scorzonera tau-saghyz*, *Scorzonera trapapogonoides*, *Scorzonera turkestanica*, *Scorzonera virgate*, *Solidago canadensis scabra*, *Solidago fistulosa*, *Solidago leavenworthii*, *Solidago rigida*, *Sonchus oleraceus*, *Taraxacum brevicomiculatum*, *Taraxacum hybernum*, *Taraxacum koksaghyz*, *Taraxacum megalorrhizon*, *Taraxacum officinale* and crossings thereof. In a further embodiment, the laticiferous plant or part thereof is selected from the group of plant families consisting of Apocynaceae, Asteraceae, Euphorbiaceae, Moraceae and crossings thereof, preferably wherein the laticiferous plant or part thereof is selected from the group consisting of *Agoseris glauca*, *Apocynum venetum*, *Chrysanthamnus graveolens*, *Chrysanthamnus nauseosus*, *Chrysanthamnus viscidiflorus*, *Cichorium intybus* L., *Funtumia elastica*, *Parthenium argentatum*, *Scorzonera acanthoclada*, *Scorzonera albicaulis*, *Scorzonera divaricate*, *Scorzonera hissaricata*, *Scorzonera racemosa*, *Scorzonera tau-saghyz*, *Scorzonera trapapogonoides*, *Scorzonera turkestanica*, *Scorzonera virgate*, *Solidago canadensis scabra*, *Solidago fistulosa*, *Solidago leavenworthii*, *Solidago rigida*, *Sonchus oleraceus*, *Taraxacum brevicomiculatum*, *Taraxacum hybernum*, *Taraxacum koksaghyz*, *Taraxacum megalorrhizon*, *Taraxacum officinale* and crossings thereof. In a more preferred embodiment, the laticiferous plant or part thereof is *Cichorium intybus* L.

In a further embodiment of the present invention, in the laticiferous plant or part thereof the at least one oxidosqualene cyclase is selected from the group consisting of oxidosqualene cyclase 1 according to SEQ ID NO: 4, the oxidosqualene cyclase 2 according to SEQ ID NO: 6, the oxidosqualene cyclase 3 according to SEQ ID NO: 8, the oxidosqualene cyclase 4 according to SEQ ID NO: 10, the oxidosqualene cyclase 5 according to SEQ ID NO: 12, the oxidosqualene cyclase 6 according to SEQ ID NO: 14, the oxidosqualene cyclase 7 according to SEQ ID NO: 16, the oxidosqualene cyclase 8 according to SEQ ID NO: 18, the lupeol synthase according to SEQ ID NO: 20 and an amino acid sequence being at least 60%, 70%, 80%, 90%, 95% or 99% identical to SEQ ID NOs: 4, 6, 8, 10, 12, 14, 16, 18 or 20.

The present invention is also directed to a RNAi being specifically directed against a mRNA transcript of at least one oxidosqualene cyclase, preferably wherein the RNAi comprises or consists of a nucleotide sequence according to SEQ ID NO: 1 and/or comprises or consists of a nucleotide sequence according to SEQ ID NO: 2 and/or comprises or consists of a nucleotide sequence being at least 60%, 70%, 80%, 90%, 95% or 99% identical to SEQ ID NO: 1 or SEQ ID NO: 2, for use in a method of reducing or eliminating the content of triterpenes and/or triterpenoids of latex or rubber from a laticiferous plant or part thereof.

The invention will be better understood with reference to the detailed description when considered in conjunction with the drawings, the non-limiting examples and the accompanying claims.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1: Analysis of TkOSC1- and TkOSC-RNAi lines. FIG. 1(a) shows normalised TkOSC1 expression of 9 to 16 weeks old wild-type (empty boxes, n=4), TkOSC1-RNAi (diagonally striped boxes, n=7) and TkOSC-RNAi plants (horizontally striped boxes, n=12) was determined by qRT-PCR using the constitutively expressed genes elongation factor 1a (TkEF1a) and ribosomal protein L27 (TkRP) from *T. koksaghyz*. FIG. 1(b) shows the expression analysis of TkOSC2-3, TkOSC5-6 and TkLUP, and FIG. 1(c) shows the one of TkSQS1 and TkSQE2. Therefore, three plants of each group were analysed. Quantification by GC/MS of single pentacyclic triterpenes shown in FIG. 1(d) and triterpene subclasses shown in FIG. 1(e) was performed on triterpene extracts from freeze-dried root material harvested from 5- to 7-month-old plants. Precursors=squalene and 2,3-oxidosqualene; sterol intermediates=cycloartenol and 24-methylene cycloartanol; pentacyclic triterpenes=taraxerol/n.i. (not identified pentacyclic triterpene), β -amyrin, lupeol, α -amyrin/lup-19(21)-en-3-ol, TkOSC3-product 1, TkOSC3-product 2, n.i.1 and taraxasterol; sterols=campesterol, stigmasterol and sitosterol; bar charts with appropriate standard deviations; boxplots: boxes=25%-75%, bar=median, square=average, hash=outlier, whisker=1.5 interquartile range; asterisks=statistical significance compared to *T. koksaghyz* wild-type (two-tailed t-test was applied for n=3, Mann-Whitney U test was applied for n>3, *= $p<0.05$, **= $p<0.01$).

FIG. 2: Analysis of triterpenes in NR extracts from TkOSC1- and TkOSC-RNAi lines. NR was extracted from independent wild-type (empty boxes, n=3), TkOSC1-RNAi (diagonally striped boxes, n=3) and TkOSC-RNAi plants (horizontally striped boxes, n=3), the corresponding triterpene extracts were measured via GC/MS. Pentacyclic triterpenes (shown in FIG. 2(a)), triterpene subclasses (shown in FIG. 2(b)), total triterpenoids and total pentacyclic triterpenes (shown in FIG. 2(c)) were depicted in bar charts with appropriate standard deviations; asterisks indicate statistical significant differences between root and NR (two-tailed t-test, *= $p<0.05$, **= $p<0.01$, ***= $p<0.001$). Precursors=squalene and 2,3-oxidosqualene; sterol intermediates=cycloartenol and 24-methylene cycloartanol; pentacyclic triterpenes=taraxerol, β -amyrin, lupeol, α -amyrin, lup-19(21)-en-3-ol, taraxasterol, OSC3-products and not identified (n.i.) triterpenes; sterols=campesterol, stigmasterol and sitosterol; acetate=Ac.

FIG. 3: Alignment of TkOSC1-8 and TkLUP amino acid sequences. The alignment was created with Clustal Omega (ebi.ac.uk/Tools/msa/clustalo/). Dashes represent gaps in aligned sequences. Conserved sites include a domain for product determination (M (W/Y) CY (C/S) R-motif, dashed box; e.g., SEQ ID NO: 6, amino acids 264-269), a region involved in substrate binding and cyclization (DCTAE-motif, solid box; e.g., SEQ ID NO: 8, amino acids 485-489) and repeated QW-motifs (underlined) involved in stabilization of carbocationic intermediates, which are localized near the 5' and 3'-end of the proteins. TkOSC1 relates to SEQ ID NO: 4, TkOSC2 relates to SEQ ID NO: 6, TkOSC3 relates to SEQ ID NO: 8, TkOSC4 relates to SEQ ID NO: 10, TkOSC5 relates to SEQ ID NO: 12, TkOSC6 relates to SEQ ID NO: 14, TkOSC7 relates to SEQ ID NO: 16, TkOSC8 relates to SEQ ID NO: 18 and TkLUP relates to SEQ ID NO: 20.

DETAILED DESCRIPTION OF THE INVENTION

The invention is based on the surprising finding that the triterpene and/or triterpenoid content of latex or rubber can

be reduced or even completely eliminated by the method according to the present invention. This allows the provision of a laticiferous plant, progeny thereof or part thereof, which is characterized by a low or even absent triterpene and/or triterpenoid content. This provides advantages for the products or materials, which are produced with rubber or latex received from these laticiferous plants, progeny thereof or parts thereof according to the present invention, for example with regard to durability, robustness and longevity.

Example 2 shows that a knockdown of oxidosqualene cyclases (OSC) is possible by RNAi (RNA interference)-mediated knockdown. Example 3 shows that the knockdown of OSCs results in the reduction of the pentacyclic triterpenoid content (triterpenes and triterpenoids) in root and rubber of a laticiferous plant such as *Taraxacum koksaghyz*. For example, with regard to the root, the following observations were made: The pentacyclic triterpenoids were strongly affected in the transgenic lines (FIG. 1(d) and FIG. 1(e)). As a result, the total amount was reduced from 4.6 mg/g dry root weight in wild-type plants to 2.8 mg/g dry root weight in TkOSC1-RNAi plants comprising a nucleotide sequence of SEQ ID NO: 2 and to 2.6 mg/g in TkOSC-RNAi plants comprising a nucleotide sequence of SEQ ID NO: 1. Especially, the most abundant triterpenes, taraxasterol, β -amyrin and one unidentified triterpene (n.i.1), were remarkably reduced with a higher level of significance in TkOSC-RNAi plants. Besides the pentacyclic triterpenes, further compounds were measured via GC/MS (FIG. 1(e), Table 3). It was found that precursors like squalene and 2,3-oxidosqualene accumulated significantly in the transgenic lines, with a higher level of significance in TkOSC-RNAi plants. Additionally, cycloartenol and 24-methylene cycloartanol, two sterol intermediates, were detected in both transgenic lines, with an average of 0.1 mg/g dry root weight in TkOSC1-RNAi plants and 0.9 mg/g dry root weight in TkOSC-RNAi plants. Sterols including campesterol, stigmasterol and sitosterol were detected to equal amounts of about 0.5 mg/g dry root weight in all plants and were not affected by the knock-down. In sum, the total triterpene content including precursors, sterol intermediates, sterols and pentacyclic triterpenes was significantly reduced in transgenic RNAi lines to about two-thirds with a higher level of significance in TkOSC-RNAi plants (FIG. 1(e)). In conclusion, it can be claimed that an OSC knockdown strongly reduces triterpenes in dandelion root and NR. For example, it could be observed that in dandelion root the total content of triterpenes could be reduced to a total content of 70%, while in dandelion NR the total content of triterpenes could be reduced to a total content of 44%.

For example, with regard to the NR, the following observations were made: After rubber extraction from root material of wild-type, TkOSC1-RNAi and TkOSC-RNAi plants, the triterpenes in NR extracts were isolated by acetone extraction and quantified in relation to NR weight (FIG. 2, Table 4). The crude acetone extracts contained acetate derivatives of different triterpenes. Those acetates as well as the predominant triterpenes in latex (taraxasterol, β -amyrin and n.i.1) were strongly reduced compared to wild-type plants with a higher level of significance in TkOSC-RNAi plants compared to TkOSC1-RNAi plants (FIGS. 2(a) and 2(b)). The average amount of total pentacyclic triterpenes in wild-type plants with 56.6 mg/g NR was reduced to 33.4 mg/g NR in TkOSC1-RNAi lines or even to 16.1 mg/g NR in TkOSC-RNAi lines (FIG. 2(c)). In contrast to the reduced pentacyclic triterpene content, increasing amounts of triterpene precursors as well as sterol intermediates were again detected in RNAi-lines compared to wild-type, while equal

amounts of sterols were measured in all three groups. However, total triterpenoid amount was still significantly decreased to two thirds in TkOSC1-RNAi NR and to less than the half in TkOSC-RNAi NR (FIG. 2(b)). In conclusion, it can be stated that an OSC knockdown strongly

The present invention provides a method of reducing or eliminating the content of triterpenes and/or triterpenoids of latex or rubber from a laticiferous plant or part thereof, comprising genetically modifying the laticiferous plant or part thereof, such that the activity of at least one oxidosqualene cyclase is reduced or eliminated, compared to the respective wild-type or untreated laticiferous plant or part thereof.

In another embodiment, the present invention provides a method of reducing the content of triterpenes and/or triterpenoids of latex or rubber from a laticiferous plant or part thereof, comprising genetically modifying the laticiferous plant or part thereof, such that the activity of at least one oxidosqualene cyclase is reduced, compared to the respective wild-type or untreated laticiferous plant or part thereof.

In a further embodiment, the present invention provides a method of eliminating the content of triterpenes and/or triterpenoids of latex or rubber from a laticiferous plant or part thereof, comprising genetically modifying the laticiferous plant or part thereof, such that the activity of at least one oxidosqualene cyclase is eliminated, compared to the respective wild-type or untreated laticiferous plant or part thereof.

In another embodiment, the present invention provides a method of reducing or eliminating the content of triterpenes and/or triterpenoids of rubber from a laticiferous plant or part thereof, comprising genetically modifying the laticiferous plant or part thereof, such that the activity of at least one oxidosqualene cyclase is reduced or eliminated, compared to the respective wild-type or untreated laticiferous plant or part thereof.

In a further embodiment, the present invention provides a method of reducing or eliminating the content of triterpenes and/or triterpenoids of latex from a laticiferous plant or part thereof, comprising genetically modifying the laticiferous plant or part thereof, such that the activity of at least one oxidosqualene cyclase is reduced or eliminated, compared to the respective wild-type or untreated laticiferous plant or part thereof.

In another embodiment, the present invention provides a method of reducing the content of triterpenes and/or triterpenoids of rubber from a laticiferous plant or part thereof, comprising genetically modifying the laticiferous plant or part thereof, such that the activity of at least one oxidosqualene cyclase is reduced, compared to the respective wild-type or untreated laticiferous plant or part thereof.

In another embodiment, the present invention provides a method of reducing the content of triterpenes and/or triterpenoids of latex from a laticiferous plant or part thereof, comprising genetically modifying the laticiferous plant or part thereof, such that the activity of at least one oxidosqualene cyclase is reduced, compared to the respective wild-type or untreated laticiferous plant or part thereof.

In a further embodiment, the present invention provides a method of eliminating the content of triterpenes and/or triterpenoids of rubber from a laticiferous plant or part thereof, comprising genetically modifying the laticiferous plant or part thereof, such that the activity of at least one oxidosqualene cyclase is eliminated, compared to the respective wild-type or untreated laticiferous plant or part thereof.

In a further embodiment, the present invention provides a method of eliminating the content of triterpenes and/or triterpenoids of latex from a laticiferous plant or part thereof, comprising genetically modifying the laticiferous plant or part thereof, such that the activity of at least one oxidosqualene cyclase is eliminated, compared to the respective wild-type or untreated laticiferous plant or part thereof.

In a further embodiment, the present invention provides a method of reducing or eliminating the content of triterpenes and/or triterpenoids of latex or rubber from a laticiferous plant, progeny thereof or part thereof, comprising genetically modifying the laticiferous plant or part thereof, such that the activity of at least one oxidosqualene cyclase is reduced or eliminated, compared to the respective wild-type or untreated laticiferous plant or part thereof.

"Triterpenes" as used herein relate to a class of chemical compounds composed of three terpene units with the molecular formula $C_{30}H_{48}$; they may also be thought of as comprising of six isoprene units. Animals, plants and fungi all produce triterpenes, including squalene, the precursor to all steroids. Triterpenes as such do not comprise heteroatoms.

"Triterpenoids" as used herein relate to triterpenes, which possess at least one heteroatom. Examples of triterpenoids include, but are not limited to, taraxerol, β -amyrin, lupeol, α -amyrin, lup-19(21)-en-3-ol or taraxasterol. Triterpenes and triterpenoids may also be summarized as pentacyclic triterpene content of the laticiferous plant or part thereof. Triterpenes and triterpenoids may also be summarized as pentacyclic triterpene content of the method of the present invention.

As used herein, the term "progeny" or "progenies" refers in a non limiting manner to offspring or descendant plants. According to certain embodiments, the term "progeny" or "progenies" refers to plants developed or grown or produced from the respective laticiferous plants. Those plants grown under consideration of the methods according to the present invention as described herein are characterized by a reduced or eliminated content of triterpenes and/or triterpenoids of the latex or rubber, which is gained from these laticiferous plant progeny or progenies.

For the method of the present invention, it is necessary that the activity of the oxidosqualene cyclase is reduced or eliminated compared to the respective wild-type or untreated laticiferous plant, progeny thereof or part thereof. There are many possible ways to achieve this reduction or elimination of oxidosqualene cyclase activity including reducing/eliminating the amount of the enzyme and/or reducing/eliminating the activity of the enzyme oxidosqualene cyclase. Ways to reduce the oxidosqualene cyclase amount, i.e. reduction of the expression of oxidosqualene cyclase, are known to a person skilled in the art and include the use gene knockdown technologies such as RNA interference (RNAi) or modification of regulatory elements. The expression of the oxidosqualene cyclase can be eliminated e.g. by knocking out the gene(s) encoding an oxidosqualene cyclase. Examples of technologies for knocking out genes include homologous recombination or site-specific nucleases such as Zinc-Finger nucleases, TALEN or CRISPR.

However, a "reduction of the content of triterpenes and/or triterpenoids" or "the content of triterpenes and/or triterpenoids is reduced" or "being capable of reducing the content of triterpenes and/or triterpenoids" means in the context of the present invention and as used herein that the content of triterpenes and/or triterpenoids contained in a laticiferous plant, progeny thereof or part thereof is reduced for at least 1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 11%, 12%,

13%, 14%, 15%, 16%, 17%, 18%, 19%, 20%, 21%, 22%, 23%, 24%, 25%, 26%, 27%, 28%, 29%, 30%, 31%, 32%, 33%, 34%, 35%, 36%, 37%, 38%, 39%, 40%, 41%, 42%, 43%, 44%, 45%, 46%, 47%, 48%, 49%, 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94% or 95% compared to the wild-type or untreated laticiferous plant, progeny thereof or part thereof, meaning without being genetically modified or before being genetically modified.

An “elimination of the content of triterpenes and/or triterpenoids” or “the content of triterpenes and/or triterpenoids is eliminated” or “being capable of eliminating the content of triterpenes and/or triterpenoids” means in the context of the present invention and as used herein that the content of triterpenes and/or triterpenoids contained in a laticiferous plant, progeny thereof or part thereof is reduced for at least 95.1%, 95.2%, 95.3%, 95.4%, 95.5%, 95.6%, 95.7%, 95.8%, 95.9%, 96.0%, 96.1%, 96.2%, 96.3%, 96.4%, 96.5%, 96.6%, 96.7%, 96.8%, 96.9%, 97.0%, 97.1%, 97.2%, 97.3%, 97.4%, 97.5%, 97.6%, 97.7%, 97.8%, 97.9%, 98.0%, 98.1%, 98.2%, 98.3%, 98.4%, 98.5%, 98.6%, 98.7%, 98.8%, 98.9%, 99.0%, 99.1%, 99.2%, 99.3%, 99.4%, 99.5%, 99.6%, 99.7%, 99.8%, 99.9% or even 100.00% compared to the wild-type or untreated laticiferous plant, progeny thereof or part thereof, meaning without being genetically modified or before being genetically modified.

The “activity of at least one oxidosqualene cyclase is reduced” or the “reduction of the activity of the at least one oxidosqualene cyclase” or “being capable of reducing the activity of at least one oxidosqualene cyclase” means in the context of the present invention and as used herein that the activity of the at least one oxidosqualene cyclase contained in a laticiferous plant, progeny thereof or part thereof is reduced for at least 1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 11%, 12%, 13%, 14%, 15%, 16%, 17%, 18%, 19%, 20%, 21%, 22%, 23%, 24%, 25%, 26%, 27%, 28%, 29%, 30%, 31%, 32%, 33%, 34%, 35%, 36%, 37%, 38%, 39%, 40%, 41%, 42%, 43%, 44%, 45%, 46%, 47%, 48%, 49%, 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94% or 95%, compared to the wild-type or untreated laticiferous plant, progeny thereof or part thereof, meaning without being genetically modified or before being genetically modified.

The “activity of at least one oxidosqualene cyclase is eliminated” or an “elimination of the activity of the at least one oxidosqualene cyclase” or “being capable of eliminating the activity of at least one oxidosqualene cyclase” means in the context of the present invention and as used herein that the activity of at least one oxidosqualene cyclase of the laticiferous plant, progeny thereof or part thereof is reduced for at least 95.1%, 95.2%, 95.3%, 95.4%, 95.5%, 95.6%, 95.7%, 95.8%, 95.9%, 96.0%, 96.1%, 96.2%, 96.3%, 96.4%, 96.5%, 96.6%, 96.7%, 96.8%, 96.9%, 97.0%, 97.1%, 97.2%, 97.3%, 97.4%, 97.5%, 97.6%, 97.7%, 97.8%, 97.9%, 98.0%, 98.1%, 98.2%, 98.3%, 98.4%, 98.5%, 98.6%, 98.7%, 98.8%, 98.9%, 99.0%, 99.1%, 99.2%, 99.3%, 99.4%, 99.5%, 99.6%, 99.7%, 99.8%, 99.9% or even 100.00%, compared to the wild-type or untreated

laticiferous plant, progeny thereof or part thereof, meaning without being genetically modified or before being genetically modified.

“Latex” or “latexes” as used herein relate to products well known to those skilled in the art. They consist of aqueous dispersions of water-insoluble polymers. These fluid systems contain, as dispersed phase, particles of polymers consisting of several entangled polymer chains in an aqueous dispersion medium. The diameter of the polymer particles within the dispersion can range between 5 nm and 10 μ m. These latexes find many applications, in particular as additives in formulations for paints, for paper (coating mixtures, bulk paper) or in formulations intended to be applied in the building industry (adhesive, bonding agents, smoothing coatings, etc.). They impart important properties on these formulations by virtue, for example, of their binding power and their film-forming power.

“Rubber” as used herein relates to natural rubber, component of the latex of the rubber tree *Hevea brasiliensis* and of other rubber-supplying plants. It contains polyisoprene, which is present very finely dispersed in the aqueous phase and represents the actual natural rubber, and also, as secondary components, proteins, sugars, polysaccharides and inorganic salts. These secondary components, in particular, form a culture medium for bacteria, which multiply quickly in the latex and cause it to perish, which finally results in coagulation, i.e., in the precipitation of the polymer phase. There are often long distances, and possibly relatively long storage times, between extraction at the rubber plantations and further processing. If processing to solid rubber does not take place on site, the latex is normally treated with suitable agents immediately after extraction. This treatment is to provide protection against bacterial attack of the latex and for stabilization of the physico-chemical properties of the latex.

The term “laticiferous” or “laticifers” as used in the context of the present invention and as used herein is related to or is a type of elongated secretory cells found in the leaves and/or stems of plants that produce latex and rubber as secondary metabolites. Laticifers may be divided into: Articulated laticifers, i.e., composed of a series of cells joined together, or non-articulated laticifers, consisting of one long coenocytic cell. Non-articulated laticifers begin their growth from the meristematic tissue of the embryo, termed the laticifer initial, and can exhibit continual growth throughout the lifetime of the plant. Laticifer tubes have irregularly edged walls and a larger inner diameter than the surrounding parenchyma cells. In the development of the cell, elongation occurs via karyokinesis and no cell plate develops resulting in coenocytic cells, which extend throughout the plant. These cells can reach up to tens of centimeters long and can be branched or unbranched. They are thought to have a role in wound healing and as defense against herbivory, as well as pathogen defense, and are often used for taxonomy.

The term “wild-type” plant, respectively laticiferous plant, or “untreated” plant, respectively laticiferous plant, refers in the context of the present invention and as used herein to plants, which serve as starting material for the production of the plants described herein and according to the present invention, i.e. plants of which the genetic information—apart from the genetic modification introduced—is identical to the genetic information of the plant of the present invention. This specifically means herein that the “wild-type” plant, respectively laticiferous plant, or

“untreated” plant, respectively laticiferous plant, is without being genetically modified or before being genetically modified as described herein.

In an embodiment of the method of the present invention, the laticiferous plant or part thereof is genetically modified by at least one heterologous nucleotide sequence that is capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase. In one further embodiment of the method of the present invention, the laticiferous plant, progeny thereof or part thereof is genetically modified by at least one heterologous nucleotide sequence that is capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase. The term “genetically modified” means, as used herein and in the context of the present invention, that the laticiferous plant, progeny thereof or part thereof comprises one or more heterologous nucleotide sequence(s) that is/are capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase. The term “is/are capable of reducing or eliminating” is also to be considered compared to the respective wild-type or untreated laticiferous plant, progeny thereof or part thereof as defined above. A “heterologous nucleotide sequence” or “heterologous nucleic acid” is a sequence that is not naturally occurring in the respective nucleic acid. Generally, the heterologous nucleic acid comprises an open reading frame that encodes a polypeptide or non-translated RNA of interest (e.g., for delivery to a cell or subject). In another further embodiment, the laticiferous plant or part thereof is genetically modified by at least one heterologous nucleotide sequence that is capable of reducing the activity of the at least one oxidosqualene cyclase. In a further embodiment, the laticiferous plant, progeny thereof or part thereof is genetically modified by at least one heterologous nucleotide sequence that is capable of reducing the activity of the at least one oxidosqualene cyclase. Preferably, the laticiferous plant or part thereof is genetically modified by at least one heterologous nucleotide sequence that is capable of reducing or eliminating the expression of the gene of the at least one oxidosqualene cyclase. Also preferred, the laticiferous plant, progeny thereof or part thereof is genetically modified by at least one heterologous nucleotide sequence that is capable of reducing or eliminating the expression of the gene of the at least one oxidosqualene cyclase. The term “is capable of reducing or eliminating” is also to be considered compared to the respective wild-type or untreated laticiferous plant, progeny thereof or part thereof as defined above. Examples for such heterologous nucleotide sequences include nucleotides that can induce RNA interference, such as siRNA or shRNA.

Accordingly, in one embodiment of the method of the present invention, the at least one heterologous nucleotide sequence encodes at least one RNAi that interferes with the mRNA encoding the at least one oxidosqualene cyclase, with the proviso that the at least one RNAi is capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase. Activity in this context may relate to a reduced or eliminated amount of triterpenes or triterpenoids in the resulting rubber or latex of the laticiferous plant, progeny thereof or part thereof. In this context “capable of reducing the activity of the at least one oxidosqualene cyclase” means that the respective RNAi can reduce the activity of the at least one oxidosqualene cyclase contained in the laticiferous plant, progeny thereof or part thereof for at least 1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 11%, 12%, 13%, 14%, 15%, 16%, 17%, 18%, 19%, 20%, 21%, 22%, 23%, 24%, 25%, 26%, 27%, 28%, 29%, 30%, 31%, 32%, 33%, 34%, 35%, 36%, 37%, 38%, 39%, 40%, 41%,

42%, 43%, 44%, 45%, 46%, 47%, 48%, 49%, 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94% or 95%, compared to the wild-type or untreated laticiferous plant, progeny thereof or part thereof, meaning without or in the absence of the respective RNAi. In this context, “capable of eliminating the activity of the at least one oxidosqualene cyclase” means that the respective RNAi can reduce the activity of the at least one oxidosqualene cyclase contained in the laticiferous plant, progeny thereof or part thereof for at least 95.1%, 95.2%, 95.3%, 95.4%, 95.5%, 95.6%, 95.7%, 95.8%, 95.9%, 96.0%, 96.1%, 96.2%, 96.3%, 96.4%, 96.5%, 96.6%, 96.7%, 96.8%, 96.9%, 97.0%, 97.1%, 97.2%, 97.3%, 97.4%, 97.5%, 97.6%, 97.7%, 97.8%, 97.9%, 98.0%, 98.1%, 98.2%, 98.3%, 98.4%, 98.5%, 98.6%, 98.7%, 98.8%, 98.9%, 99.0%, 99.1%, 99.2%, 99.3%, 99.4%, 99.5%, 99.6%, 99.7%, 99.8%, 99.9% or even 100.00%, compared to the wild-type or untreated laticiferous plant, progeny thereof or part thereof, meaning without or in the absence of the respective RNAi.

The amount or content of triterpenes or triterpenoids can be determined by methods known to a person skilled in the art. Such methods e.g. include, but are not limited to, mass spectrometry, chromatography such as RP-HPLC or gas chromatography, or NMR. An exemplary method for the quantification of triterpenes and/or triterpenoids in natural rubber is described in Example 1.

In another embodiment of the method of the present invention, the at least one RNAi is specifically directed against the mRNA transcript of the at least one oxidosqualene cyclase. Preferably, the at least one RNAi comprises or consists of a nucleotide sequence according to SEQ ID NO: 1, and/or comprises or consists of a nucleotide sequence according to SEQ ID NO: 2, and/or comprises or consists of a nucleotide sequence being at least 60%, 70%, 80%, 90%, 95% or 99% identical to SEQ ID NO: 1 or SEQ ID NO: 2.

The nucleotide sequence according to SEQ ID NO: 1 of the present invention is given as follows:

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TCTCCAAGCGCCCATAGCGGAAAAGTGTCTCTATATTCTGCATAATGTAAC
ATGCAATTCTTCATGTACACTCCTGTTATTTCTGTTGAGGAAAGTCTCCA
TTATCCATTGTCCATTTATGAGCAATTTGCTGCCTTGTGTAATGGTGTCT
GGATCTCTTTCAGCCTGTCCACCACATAGAAGACCAAGCATGGCCCATGAA
GTTTGCACAAAATTGTTCTGTTCCCTTCCAAAGGTATAAATTTCTCTTGT
GGGCATGACTCAAAGTTCTCTCCCAACCACCTTCTGAGTTTGTGTCGAG
AGGAAAACTTGACAGCTTTCGAACTGCTTCACTATTTTCATATGTTTGT
CCACAAGCTACCAATCTTGTAAACACGAAATAGGTTCCATAAGATAACAT
ATACCCCAATAACCATACCAAGAACCATTTTCTTGT.
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13

The complementary sequence thereto, which is, for example, also present in a vector construct comprising the SEQ ID NO: 1 if the vector comprises a hairpin-structure, is given as follows:

(SEQ ID NO: 47)
 AACAAAGAAAATGGTTCTTGGTATGGTTATTGGGGTATATGTTATCTTTATG
 GAACCTATTTCTGTGTACAAGATTGGTAGCTTGTGGACAACATATGAAA
 ATAGTGAAGCAGTTTCGGAAGCTGTCAAGTTTTTCCTCTCGACACAAAAC
 CAGAAGGTGGTTGGGAGAGAACTTTGAGTCATGCCACAAGAGAAATTTA
 TACCTTTGGAAGGGAACAGAACAAATTTTGTGCAAACTTCATGGGCCATGC
 TTGGTCTTCTATGTGGTGGACAGGCTGAAAGAGATCCGACACCATTACACA
 AGGCAGCAAAATTGCTCATAAATGGACAAATGGATAATGGAGACTTTCTCT
 AACAGGAAATAACAGGAGTGTACATGAAGAATTGCATGTTACATTATGCAG
 AATATAGGAACACTTTTCCGCTATGGGCGCTTGGAGA.

The nucleotide sequence according to SEQ ID NO: 2 of the present invention is given as follows:

ATCCCAAGCTTGAATCGCACGGTTCAGTCTAACTGCCTTTCTACGGCTGT
 GTGCACCGCTTCATAGTTCACATCTTCGTCGCTCCAAGTCTCACTGGAGG
 TACGCTTATAAGATCAATTCGCG.

The complementary sequence thereto, which is, for example, also present in a vector construct comprising the SEQ ID NO: 2 if the vector comprises a hairpin-structure, is given as follows:

(SEQ ID NO: 48)
 GCGGAATTGATCTTATAAGCGTACCTCCAGTGAGACTTGGAGACGACGAAG
 ATGTGAACATATGAAGCGGTGACAACAGCGGTGAGAAAGGCAGTTAGACTGA
 ACCGTGCGATTCAAGCTTGGGAT.

More preferably, the at least one RNAi comprises or consists of a nucleotide sequence being at least 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or even 100% identical to SEQ ID NO: 1.

Even more preferably, the at least one RNAi consists of a nucleotide sequence being at least 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or even 100% identical to SEQ ID NO: 1.

More preferably, the at least one RNAi comprises or consists of a nucleotide sequence being at least 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or even 100% identical to SEQ ID NO: 2.

Even more preferably, the at least one RNAi consists of a nucleotide sequence being at least 50%, 51%, 52%, 53%,

14

54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or even 100% identical to SEQ ID NO: 2.

More preferably, the at least one RNAi comprises or consists of a nucleotide sequence according to SEQ ID NO: 1, even more preferably, the at least one RNAi consists of a nucleotide sequence according to SEQ ID NO: 1. More preferably, the at least one RNAi comprises or consists of a nucleotide sequence according to SEQ ID NO: 2, even more preferably, the at least one RNAi consists of a nucleotide sequence according to SEQ ID NO: 2.

Two amino-acid or nucleotide sequences are said to be “identical” within the context of the present invention, if the sequence of amino-acid or nucleotide residues in the two sequences is the same when aligned for maximum correspondence as described below. Sequence comparisons between two (or more) peptides or polynucleotides are typically performed by comparing sequences of two optimally aligned sequences over a segment or “comparison window” to identify and compare local regions of sequence similarity. Optimal alignment of sequences for comparison may be conducted by the local homology algorithm of Smith and Waterman, *Ad. App. Math* 2: 482 (1981), by the homology alignment algorithm of Neddleman and Wunsch, *J. Mol. Biol.* 48: 443 (1970), by the search for similarity method of Pearson and Lipman, *Proc. Natl. Acad. Sci. (U.S.A.)* 85: 2444 (1988), by computerized implementation of these algorithms (GAP, BESTFIT, FASTA, and TFASTA in the Wisconsin Genetics Software Package, Genetics Computer Group (GCG), 575 Science Dr., Madison, Wis.), or by visual inspection.

“Percentage of sequence identity” (or “degree” or “identity”) is determined by comparing two optimally aligned sequences over a comparison window, where the portion of the peptide or polynucleotide sequence in the comparison window may comprise additions or deletions (i.e., gaps) as compared to the reference sequence (which does not comprise additions or deletions) for optimal alignment of the two sequences. The percentage is calculated by determining the number of positions at which the identical amino-acid residue or nucleic acid base occurs in both sequences to yield the number of matched positions, dividing the number of matched positions by the total number of positions in the window of comparison and multiplying the result by 100 to yield the percentage of sequence identity.

The definition of sequence identity given above is the definition that would use one of skill in the art. The definition by itself does not need the help of any algorithm, said algorithms being helpful only to achieve the optimal alignments of sequences, rather than the calculation of sequence identity.

From the definition given above, it follows that there is a well-defined and only one value for the sequence identity between two compared sequences, which value corresponds to the value obtained for the best or optimal alignment.

Instead of reducing the amount of the enzyme oxidosqualene cyclase, it is also within the context of the present invention that the reduction or elimination of the activity of the oxidosqualene cyclase may be inhibited by inhibitors. Accordingly, in an embodiment of the method of the present invention, the laticiferous plant or part thereof is genetically modified by subjecting it to and/or incubating it with agents, which are capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase.

15

Further, in one embodiment of the method of the present invention, the laticiferous plant, progeny thereof or part thereof is genetically modified by subjecting it to and/or incubating it with agents, which are capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase. This means, that the term “genetically modified” or “genetically modification” may also comprise subjecting the laticiferous plant, progeny thereof or part thereof of the present invention or the laticiferous plant, progeny thereof or part thereof used in any embodiments of the method of the present invention to and/or incubating it with agents, which are capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase. The term “is/are capable of reducing or eliminating” is also to be considered compared to the respective wild-type or untreated laticiferous plant, progeny thereof or part thereof as defined above. Preferably, such an agent is an inhibitor of the activity of the at least one oxidosqualene cyclase. More preferably, the inhibitor of the activity of the at least one oxidosqualene cyclase is selected from the group consisting of 2,3-sulfidosqualene, derivatives of oxidosqualene, even more preferably 10- and 19-azasqualene derivatives, 4,4,10 β -trimethyl-trans-decal-3- β -ol (TMD) or derivatives thereof, aminoketones, azadecaline and derivatives of imidazoline, epohelmins, lanopylins, isoquinoline derivatives, 4-piperidinopyridines and 4-piperidinopyrimidines, even more preferably BIBB 515 (1-(4-chlorobenzoyl)-4-((4-2-oxazolin-2-yl)benzylidene)) piperidine), vinyl sulfide and ketene dithioacetal derivatives, umbelliferone aminoalkyl derivatives and digalactosylglycerols.

In another embodiment of the method of the present invention, the laticiferous plant or part thereof is genetically modified by at least one heterologous nucleotide sequence that is capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase, preferably that is capable of reducing or eliminating the expression of the gene of the at least one oxidosqualene cyclase, and/or wherein the laticiferous plant or part thereof is genetically modified by subjecting it to and/or incubating it with agents, which are capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase. In another embodiment of the method of the present invention, the laticiferous plant, progeny thereof or part thereof is genetically modified by at least one heterologous nucleotide sequence that is capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase, preferably that is capable of reducing or eliminating the expression of the gene of the at least one oxidosqualene cyclase, and/or wherein the laticiferous plant, progeny thereof or part thereof is genetically modified by subjecting it to and/or incubating it with agents, which are capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase. The above given definitions also apply for this embodiment, e.g. the term “is/are capable of reducing or eliminating” is also to be considered compared to the respective wild-type or untreated laticiferous plant, progeny thereof or part thereof as defined above.

In a further embodiment of the method of the present invention, the laticiferous plant or part thereof is selected from the group of plant families consisting of Apocynaceae, Asteraceae, Euphorbiaceae and Moraceae. In a further embodiment of the method of the present invention, the laticiferous plant, progeny thereof or part thereof is selected from the group of plant families consisting of Apocynaceae, Asteraceae, Euphorbiaceae and Moraceae.

In a further embodiment of the method of the present invention, it is preferred that the laticiferous plant or part thereof is selected from the group consisting of *Agoseris*

16

glauca, *Apocynum venetum*, *Chrysanthamnus graveolens*, *Chrysanthamnus nauseosus*, *Chrysanthamnus viscidiflorus*, *Funtumia elastica*, *Parthenium argentatum*, *Scorzonera acanthoclada*, *Scorzonera albicaulis*, *Scorzonera divaricate*, *Scorzonera hissaricata*, *Scorzonera racemosa*, *Scorzonera tau-saghyz*, *Scorzonera tragapogonoides*, *Scorzonera turkestanica*, *Scorzonera virgate*, *Solidago canadensis scabra*, *Solidago fistulosa*, *Solidago leavenworthii*, *Solidago rigida*, *Sonchus oleraceus*, *Taraxacum brevicomiculatum*, *Taraxacum hybemum*, *Taraxacum koksaghyz*, *Taraxacum megalorrhizon*, *Taraxacum officinale* and crossings thereof. In a further embodiment of the method of the present invention, it is preferred that the laticiferous plant or part thereof is selected from the group consisting of *Agoseris glauca*, *Apocynum venetum*, *Chrysanthamnus graveolens*, *Chrysanthamnus nauseosus*, *Chrysanthamnus viscidiflorus*, *Cichorium intybus* L., *Funtumia elastica*, *Parthenium argentatum*, *Scorzonera acanthoclada*, *Scorzonera albicaulis*, *Scorzonera divaricate*, *Scorzonera hissaricata*, *Scorzonera racemosa*, *Scorzonera tau-saghyz*, *Scorzonera tragapogonoides*, *Scorzonera turkestanica*, *Scorzonera virgate*, *Solidago canadensis scabra*, *Solidago fistulosa*, *Solidago leavenworthii*, *Solidago rigida*, *Sonchus oleraceus*, *Taraxacum brevicomiculatum*, *Taraxacum hybemum*, *Taraxacum koksaghyz*, *Taraxacum megalorrhizon*, *Taraxacum officinale* and crossings thereof. In a more preferred embodiment, the laticiferous plant or part thereof is *Cichorium intybus* L. The term “crossings thereof” means in the context of the present invention plant varieties or lines that have been created by the deliberate interbreeding of closely or distantly related plant species, e.g. to introduce genes from one variety or line into a new genetic background. The laticiferous plant or part thereof can also be the progeny of the respective laticiferous plant. It is even more preferred that the laticiferous plant or part thereof is selected from the group consisting of *Parthenium argentatum*, *Chrysanthamnus nauseosus*, *Taraxacum koksaghyz*, *Taraxacum officinale* and crossings thereof. However, most preferably the laticiferous plant or part thereof is *Taraxacum koksaghyz*. In one embodiment of the method of the present invention, the laticiferous plant or part thereof is not a member of the family Euphorbiaceae. This means the present invention is also directed to a method of reducing or eliminating the content of triterpenes and/or triterpenoids of latex or rubber from a laticiferous plant or part thereof, comprising genetically modifying the laticiferous plant or part thereof, such that the activity of at least one oxidosqualene cyclase is reduced or eliminated, compared to the respective wild-type or untreated laticiferous plant or part thereof, wherein the laticiferous plant or part thereof is not a member of the family Euphorbiaceae. In another related embodiment of the method of the present invention, the laticiferous plant or part thereof is not *Hevea brasiliensis*. This means the present invention is also directed to a method of reducing or eliminating the content of triterpenes and/or triterpenoids of latex or rubber from a laticiferous plant or part thereof, comprising genetically modifying the laticiferous plant or part thereof, such that the activity of at least one oxidosqualene cyclase is reduced or eliminated, compared to the respective wild-type or untreated laticiferous plant or part thereof, wherein the laticiferous plant or part thereof is not *Hevea brasiliensis*. In a further related embodiment of the method of the present invention, the laticiferous plant or part thereof is not from the genus *Landolphia*. This means, the present invention is also directed to a method of reducing or eliminating the content of triterpenes and/or triterpenoids of latex or rubber from a laticiferous plant or part thereof, comprising genetically

modifying the laticiferous plant or part thereof, such that the activity of at least one oxidosqualene cyclase is reduced or eliminated, compared to the respective wild-type or untreated laticiferous plant or part thereof, wherein the laticiferous plant or part thereof is not from the genus *Landolphia*.

In another embodiment of the method of the present invention, the at least one oxidosqualene cyclase is selected from the group consisting of oxidosqualene cyclase 1 according to SEQ ID NO: 4, the oxidosqualene cyclase 2 according to SEQ ID NO: 6, the oxidosqualene cyclase 3 according to SEQ ID NO: 8, the oxidosqualene cyclase 4 according to SEQ ID NO: 10, the oxidosqualene cyclase 5 according to SEQ ID NO: 12, the oxidosqualene cyclase 6 according to SEQ ID NO: 14, the oxidosqualene cyclase 7 according to SEQ ID NO: 16, the oxidosqualene cyclase 8 according to SEQ ID NO: 18, the lupeol synthase according to SEQ ID NO: 20 and an amino acid sequence being at least 60%, 70%, 80%, 90%, 95% or 99% identical to SEQ ID NOs: 4, 6, 8, 10, 12, 14, 16, 18 or 20.

In another embodiment of the method of the present invention, the at least one oxidosqualene cyclase is selected from the group consisting of an amino acid sequence being at least 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or even 100% identical to SEQ ID NOs: 4, 6, 8, 10, 12, 14, 16, 18 or 20.

It is further preferred that the at least one oxidosqualene cyclase is selected from the group consisting of the oxidosqualene cyclase 1 according to SEQ ID NO: 4, the oxidosqualene cyclase 2 according to SEQ ID NO: 6, the oxidosqualene cyclase 3 according to SEQ ID NO: 8 and the oxidosqualene cyclase 5 according to SEQ ID NO: 12 and an amino acid sequence being at least 60%, 70%, 80%, 90%, 95% or 99% identical to SEQ ID NOs: 4, 6, 8 or 12.

In another embodiment of the method of the present invention, the at least one oxidosqualene cyclase is selected from the group consisting of an amino acid sequence being at least 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or even 100% identical to SEQ ID NOs: 4, 6, 8 or 12.

It is even more preferred that the at least one oxidosqualene cyclase is the oxidosqualene cyclase 1 according to SEQ ID NO: 4 or an amino acid sequence being at least 60%, 70%, 80%, 90%, 95% or 99% identical thereto.

In another embodiment of the method of the present invention, the at least one oxidosqualene cyclase is selected from the group consisting of an amino acid sequence being at least 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or even 100% identical to SEQ ID NO: 4. It is most preferred that at least one oxidosqualene cyclase is the oxidosqualene cyclase 1 according to SEQ ID NO: 4.

The nucleotide sequence of the oxidosqualene cyclase 1 is further given herein as SEQ ID NO: 3. The nucleotide sequence of the oxidosqualene cyclase 2 is further given herein as SEQ ID NO: 5. The nucleotide sequence of the oxidosqualene cyclase 3 is further given herein as SEQ ID

NO: 7. The nucleotide sequence of the oxidosqualene cyclase 4 is further given herein as SEQ ID NO: 9. The nucleotide sequence of the oxidosqualene cyclase 5 is further given herein as SEQ ID NO: 11. The nucleotide sequence of the oxidosqualene cyclase 6 is further given herein as SEQ ID NO: 13. The nucleotide sequence of the oxidosqualene cyclase 7 is further given herein as SEQ ID NO: 15. The nucleotide sequence of the oxidosqualene cyclase 8 is further given herein as SEQ ID NO: 17. The nucleotide sequence of the lupeol synthase is further given herein as SEQ ID NO: 19.

“Oxidosqualene cyclases (OSC)” as used herein are enzymes involved in the cyclization reactions of 2,3-oxidosqualene to form sterols or triterpenes. It can be differentiated between two different types of cyclisation reaction: The chair-boat-chair conformation results in the formation of cycloartenol, a phytosterol intermediate, while the chair-chair-chair conformation results in the formation of triterpenes (Abe, 2007; Thimmappa et al., 2014). The lupeol synthase also belongs to the oxidosqualene cyclases, wherein the lupeol synthase catalyses the cyclisation of 2,3-oxidosqualene to lupeol.

The polypeptides of TkOSC1-8 and TkLUP according to SEQ ID NOs: 4, 6, 8, 10, 12, 14, 16, 18 and 20 of the present invention comprise 758 to 770 amino acids, whereas TkOSC1 compared to TkOSC2 displays the highest (94%) sequence identity among the newly identified sequences. Analysis of the amino acid sequence reveals that conserved domains reported for OSC are present with only slight modifications for TkOSC4, TkOSC5, TkOSC6 and TkOSC8. Those include a domain essential for product determination (M(W/Y)CY(C/S)R-motif), one region involved in substrate binding and polycyclization initiation (DCTAE-motif) and six repeated QW motifs involved in stabilization of carbocationic intermediates and localized near the 5' and 3'-ends of the proteins (see FIG. 3).

In a further embodiment of the method of the present invention, the part of the laticiferous plant is selected from the group consisting of cells, tissues, organs, roots, stems, branches, leaves, exudates, latex and mixtures thereof. In the Russian dandelion *Taraxacum koksaghyz*, the part of the laticiferous plant with the highest concentration of latex is the root. Accordingly, the part of the laticiferous plant preferably is root, if the plant or part thereof is *Taraxacum koksaghyz*. In the context of the present invention, the term “exudate(s)” comprises saps, gums, resins or nectar of the respective plant.

The present invention also provides a laticiferous plant or part thereof being genetically modified such that the activity of the at least one oxidosqualene cyclase is reduced or eliminated, compared to the respective wild-type or untreated laticiferous plant or part thereof. The present invention also provides a laticiferous plant, progeny thereof or part thereof being genetically modified such that the activity of the at least one oxidosqualene cyclase is reduced or eliminated, compared to the respective wild-type or untreated laticiferous plant, progeny thereof or part thereof.

In one embodiment of the laticiferous plant or part thereof according to the present invention, the laticiferous plant or part thereof is genetically modified by at least one heterologous nucleotide sequence that is capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase, preferably that is capable of reducing or eliminating the expression of the gene of the at least one oxidosqualene cyclase. In one embodiment of the laticiferous plant, progeny thereof or part thereof according to the present invention, the laticiferous plant, progeny thereof or part thereof is

genetically modified by at least one heterologous nucleotide sequence that is capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase, preferably that is capable of reducing or eliminating the expression of the gene of the at least one oxidosqualene cyclase. The term “is/are capable of reducing or eliminating” is also to be considered compared to the respective wild-type or untreated laticiferous plant, progeny thereof or part thereof as defined above.

The “activity of at least one oxidosqualene cyclase is reduced” or “reducing the activity of the at least one oxidosqualene cyclase” or “being capable of reducing the activity of at least one oxidosqualene cyclase” means in the context of the present invention and as used herein that the activity of the at least one oxidosqualene cyclase contained in a laticiferous plant, progeny thereof or part thereof is reduced for at least 1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 11%, 12%, 13%, 14%, 15%, 16%, 17%, 18%, 19%, 20%, 21%, 22%, 23%, 24%, 25%, 26%, 27%, 28%, 29%, 30%, 31%, 32%, 33%, 34%, 35%, 36%, 37%, 38%, 39%, 40%, 41%, 42%, 43%, 44%, 45%, 46%, 47%, 48%, 49%, 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94% or 95%, compared to the wild-type or untreated laticiferous plant, progeny thereof or part thereof, meaning without being genetically modified or before being genetically modified.

The “activity of at least one oxidosqualene cyclase is eliminated” or “eliminating the activity of the at least one oxidosqualene cyclase” or “being capable of eliminating the activity of at least one oxidosqualene cyclase” means in the context of the present invention and as used herein that the activity of at least one oxidosqualene cyclase of the laticiferous plant, progeny thereof or part thereof is reduced for at least 95.1%, 95.2%, 95.3%, 95.4%, 95.5%, 95.6%, 95.7%, 95.8%, 95.9%, 96.0%, 96.1%, 96.2%, 96.3%, 96.4%, 96.5%, 96.6%, 96.7%, 96.8%, 96.9%, 97.0%, 97.1%, 97.2%, 97.3%, 97.4%, 97.5%, 97.6%, 97.7%, 97.8%, 97.9%, 98.0%, 98.1%, 98.2%, 98.3%, 98.4%, 98.5%, 98.6%, 98.7%, 98.8%, 98.9%, 99.0%, 99.1%, 99.2%, 99.3%, 99.4%, 99.5%, 99.6%, 99.7%, 99.8%, 99.9% or even 100.00%, compared to the wild-type or untreated laticiferous plant, progeny thereof or part thereof, meaning without being genetically modified or before being genetically modified.

In another embodiment, the laticiferous plant or part thereof according to the present invention, is genetically modified by subjecting it to and/or incubating it with agents, which are capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase. In another embodiment, the laticiferous plant, progeny thereof or part thereof according to the present invention, is genetically modified by subjecting it to and/or incubating it with agents, which are capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase. The term “is/are capable of reducing or eliminating” is also to be considered compared to the respective wild-type or untreated laticiferous plant, progeny thereof or part thereof as defined above. Such an agent may be an inhibitor of the activity of the at least one oxidosqualene cyclase, preferably an inhibitor of the activity of the at least one oxidosqualene cyclase selected from the group consisting of 2,3-sulfidosqualene, derivatives of oxidosqualene, more preferably 10- and 19-azasqualene derivatives, 4,4,10 β -trimethyl-trans-decal-3- β -ol (TMD) or derivatives thereof, aminoketones, azadecaline and deriva-

tives of imidazoline, epohelmins, lanopylins, isoquinoline derivatives, 4-piperidinopyridines and 4-piperidinopyrimidines, even more preferably BIBB 515 (1-(4-chlorobenzoyl)-4-((4-2-oxazolin-2-yl)benzylidene))piperidine), vinyl sulfide and ketene dithioacetal derivatives, umbelliferone aminoalkyl derivatives and digalactosylglycerols. In a further embodiment, such an agent may be an inhibitor of the activity of the at least one oxidosqualene cyclase and preferably an inhibitor of the activity of the at least one oxidosqualene cyclase selected from the group consisting of 2,3-sulfidosqualene, derivatives of oxidosqualene, more preferably 10- and 19-azasqualene derivatives, 4,4,10 β -trimethyl-trans-decal-3- β -ol (TMD) or derivatives thereof, aminoketones, ethyl methanesulfonates, azadecaline and derivatives of imidazoline, epohelmins, lanopylins, isoquinoline derivatives, 4-piperidinopyridines and 4-piperidinopyrimidines, even more preferably BIBB 515 (1-(4-chlorobenzoyl)-4-((4-2-oxazolin-2-yl)benzylidene))piperidine), vinyl sulfide and ketene dithioacetal derivatives, umbelliferone aminoalkyl derivatives and digalactosylglycerols. In a preferred embodiment, an inhibitor of the activity of the at least one oxidosqualene cyclase is ethyl methanesulfonates.

In a further embodiment of the laticiferous plant or part thereof according to the present invention, the laticiferous plant or part thereof is genetically modified by at least one heterologous nucleotide sequence that is capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase, preferably that is capable of reducing or eliminating the expression of the gene of the at least one oxidosqualene cyclase, and/or wherein the laticiferous plant or part thereof is genetically modified by subjecting it to and/or incubating it with agents, which are capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase. In a further embodiment of the laticiferous plant, progeny thereof or part thereof according to the present invention, the laticiferous plant, progeny thereof or part thereof is genetically modified by at least one heterologous nucleotide sequence that is capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase, preferably that is capable of reducing or eliminating the expression of the gene of the at least one oxidosqualene cyclase, and/or wherein the laticiferous plant, progeny thereof or part thereof is genetically modified by subjecting it to and/or incubating it with agents, which are capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase. The “activity of at least one oxidosqualene cyclase is reduced” or “reducing the activity of the at least one oxidosqualene cyclase” or “being capable of reducing the activity of at least one oxidosqualene cyclase” means in this context and as used herein that the activity of the at least one oxidosqualene cyclase contained in a laticiferous plant, progeny thereof or part thereof is reduced for at least 1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 11%, 12%, 13%, 14%, 15%, 16%, 17%, 18%, 19%, 20%, 21%, 22%, 23%, 24%, 25%, 26%, 27%, 28%, 29%, 30%, 31%, 32%, 33%, 34%, 35%, 36%, 37%, 38%, 39%, 40%, 41%, 42%, 43%, 44%, 45%, 46%, 47%, 48%, 49%, 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94% or 95%, compared to the wild-type or untreated laticiferous plant, progeny thereof or part thereof, meaning without being genetically modified or before being genetically modified. The “activity of at least one oxidosqualene cyclase is eliminated” or “eliminating the activity of the at least one oxidosqualene cyclase” or “being capable of

eliminating the activity of at least one oxidosqualene cyclase” means in this context that the activity of at least one oxidosqualene cyclase of the laticiferous plant, progeny thereof or part thereof is reduced for at least 95.1%, 95.2%, 95.3%, 95.4%, 95.5%, 95.6%, 95.7%, 95.8%, 95.9%, 96.0%, 96.1%, 96.2%, 96.3%, 96.4%, 96.5%, 96.6%, 96.7%, 96.8%, 96.9%, 97.0%, 97.1%, 97.2%, 97.3%, 97.4%, 97.5%, 97.6%, 97.7%, 97.8%, 97.9%, 98.0%, 98.1%, 98.2%, 98.3%, 98.4%, 98.5%, 98.6%, 98.7%, 98.8%, 98.9%, 99.0%, 99.1%, 99.2%, 99.3%, 99.4%, 99.5%, 99.6%, 99.7%, 99.8%, 99.9% or even 100.00%, compared to the wild-type or untreated laticiferous plant, progeny thereof or part thereof, meaning without being genetically modified or before being genetically modified.

Thus, in one embodiment of the present invention, the laticiferous plant or part thereof may be a transgenic plant. In one embodiment of the present invention, the laticiferous plant, progeny thereof or part thereof may be a transgenic plant. A “transgenic plant” as used herein refers to a plant that contains genetic material not found in a wild-type plant of the same species, variety or cultivar. The genetic material may include a transgene, the product of a mutagenesis event (such as by transposon or T-DNA insertional mutagenesis), including insertion, deletion or substitution, an activation tagging sequence, a mutated sequence, the product of a homologous recombination event or a sequence modified by chimera-plasty. Typically, the foreign genetic material has been introduced into the plant by human manipulation, but any method can be used as one of skill in the art recognizes. A transgenic plant may contain an expression vector or cassette. The expression cassette typically comprises a polypeptide-encoding sequence operably linked (i.e., under regulatory control of) to appropriate inducible or constitutive regulatory sequences that allow for the expression of polypeptide. The expression cassette can be introduced into a plant by transformation or by breeding after transformation of a parent plant. The term “plant” as used herein refers to a whole plant, including seedlings and mature plants, as well as to a plant part, such as seed, fruit, leaf, or root, plant tissue, plant cells or any other plant material, e.g., a plant explant, as well as to progeny thereof, and to in vitro systems that mimic biochemical or cellular components or processes in a cell.

In one embodiment of the laticiferous plant or part thereof according to the present invention, the at least one heterologous nucleotide sequence encodes at least one RNAi that interferes with the mRNA encoding the at least one oxidosqualene cyclase, with the proviso that the at least one RNAi is capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase. In one embodiment of the laticiferous plant, progeny thereof or part thereof according to the present invention, the at least one heterologous nucleotide sequence encodes at least one RNAi that interferes with the mRNA encoding the at least one oxidosqualene cyclase, with the proviso that the at least one RNAi is capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase. In this context, “capable of reducing the activity of the at least one oxidosqualene cyclase” means that the respective RNAi can reduce the activity of the at least one oxidosqualene cyclase contained in the laticiferous plant, progeny thereof or part thereof for at least 1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 11%, 12%, 13%, 14%, 15%, 16%, 17%, 18%, 19%, 20%, 21%, 22%, 23%, 24%, 25%, 26%, 27%, 28%, 29%, 30%, 31%, 32%, 33%, 34%, 35%, 36%, 37%, 38%, 39%, 40%, 41%, 42%, 43%, 44%, 45%, 46%, 47%, 48%, 49%, 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%,

60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94% or 95%, compared to the wild-type or untreated laticiferous plant, progeny thereof or part thereof, meaning without or in the absence of the respective RNAi. In this context, “capable of eliminating the activity of the at least one oxidosqualene cyclase” means that the respective RNAi can reduce the activity of the at least one oxidosqualene cyclase contained in the laticiferous plant, progeny thereof or part thereof for at least 95.1%, 95.2%, 95.3%, 95.4%, 95.5%, 95.6%, 95.7%, 95.8%, 95.9%, 96.0%, 96.1%, 96.2%, 96.3%, 96.4%, 96.5%, 96.6%, 96.7%, 96.8%, 96.9%, 97.0%, 97.1%, 97.2%, 97.3%, 97.4%, 97.5%, 97.6%, 97.7%, 97.8%, 97.9%, 98.0%, 98.1%, 98.2%, 98.3%, 98.4%, 98.5%, 98.6%, 98.7%, 98.8%, 98.9%, 99.0%, 99.1%, 99.2%, 99.3%, 99.4%, 99.5%, 99.6%, 99.7%, 99.8%, 99.9% or even 100.00%, compared to the wild-type or untreated laticiferous plant, progeny thereof or part thereof, meaning without or in the absence of the respective RNAi.

In a further embodiment of the laticiferous plant or part thereof according to the present invention, the at least one RNAi is specifically directed against the mRNA transcript of the at least one oxidosqualene cyclase, preferably wherein the at least one RNAi comprises or consists of a nucleotide sequence according to SEQ ID NO: 1 and/or comprises or consists of a nucleotide sequence according to SEQ ID NO: 2 and/or comprises or consists of a nucleotide sequence being at least 60%, 70%, 80%, 90%, 95% or 99% identical to SEQ ID NO: 1 or SEQ ID NO: 2. In a further embodiment of the laticiferous plant, progeny thereof or part thereof according to the present invention, the at least one RNAi is specifically directed against the mRNA transcript of the at least one oxidosqualene cyclase, preferably wherein the at least one RNAi comprises or consists of a nucleotide sequence according to SEQ ID NO: 1 and/or comprises or consists of a nucleotide sequence according to SEQ ID NO: 2 and/or comprises or consists of a nucleotide sequence being at least 60%, 70%, 80%, 90%, 95% or 99% identical to SEQ ID NO: 1 or SEQ ID NO: 2.

Also for all embodiments of the laticiferous plant, progeny thereof or part thereof according to the present invention, the nucleotide sequence according to SEQ ID NO: 1 is given as follows:

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TCTCCAAGCGCCCATAGCGGAAAAGTGTTCTATATTCTGCATAATGTAAC
ATGCAATTCTTCATGTACACTCCTGTTATTTCTGTTGAGGAAAGTCTCCA
TTATCCATTGTCCATTATGAGCAATTTTGTGCTTGTGTAATGGTGTCTC
GGATCTCTTTGAGCTGTCCACCACATAGAAGACCAAGCATGGCCCATGAA
GTTTGCACAAAATTTGTTCTGTTCCCTTCCAAAGGTATAAATTTCTCTTGT
GGGCATGACTCAAAGTTCTCTCCCAACCACCTTCTGAGTTTGTGTCGAG
AGGAAAAACTTGACAGCTTTCGCAACTGCTTCACTATTTTCATATGTTTGT
CCACAAGCTACCAATCCTTGTAACACGAAATAGGTTCCATAAAGATAACAT
ATACCCCAATAACCATACCAAGAACCATTTTCTTGT.
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23

The complementary sequence thereto, which is, for example, also present in a vector construct comprising the SEQ ID NO: 1 if the vector comprises a hairpin-structure, is given as follows:

(SEQ ID NO: 47)
 AACAGAAATGGTCTTGGTATGGTTATTGGGGTATATGTTATCTTTATG
 GAACCTATTTCTGTGTACAAGATTGGTAGCTTGTGGACAAACATATGAAA
 ATAGTGAAGCAGTTCGGAAGCTGTCAAGTTTTCTCTCGACACAAAAC
 CAGAAGGTGGTGGGAGAGAACTTTGAGTCATGCCACAAGAGAAATTTA
 TACCTTTGGAAGGGAACAGAACAAATTTTGTGCAAACTTCATGGGCCATGC
 TTGGTCTTCTATGTGGTGGACAGGCTGAAAGAGATCCGACACCATTACACA
 AGGCAGCAAAATGCTCATAAATGGACAAATGGATAATGGAGACTTTCTCTC
 AACAGGAAATAACAGGAGTGTACATGAAGAATTGCATGTTACATTATGCAG
 AATATAGGAACACTTTTCCGCTATGGGCGCTTGGAGA.

Also for all embodiments of the laticiferous plant, progeny thereof or part thereof according to the present invention, the nucleotide sequence according to SEQ ID NO: 2 of the present invention is given as follows:

ATCCCAAGCTTGAATCGCACGGTTCAGTCTAACTGCCTTTCTCACGGCTGT
 TGTCACCGCTTCATAGTTCACATCTTCGTCGTCTCCAAGTCTCACTGGAGG
 TACGCTTATAAGATCAATTCGCT.

The complementary sequence thereto, which is, for example, also present in a vector construct comprising the SEQ ID NO: 2 if the vector comprises a hairpin-structure, is given as follows:

(SEQ ID NO: 48)
 GCGGAATTGATCTTATAAGCGTACCTCCAGTGAGACTTGGAGACGACGAAG
 ATGTGAACACTGAAGCGGTGACAACAGCCGTGAGAAAGGCAGTTAGACTGA
 ACCGTGCGATTCAAGCTTGGGAT.

More preferably, the at least one RNAi comprises or consists of a nucleotide sequence being at least 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or even 100% identical to SEQ ID NO: 1.

Even more preferably, the at least one RNAi consists of a nucleotide sequence being at least 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or even 100% identical to SEQ ID NO: 1.

More preferably, the at least one RNAi comprises or consists of a nucleotide sequence being at least 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or even 100% identical to SEQ ID NO: 2.

24

Even more preferably, the at least one RNAi consists of a nucleotide sequence being at least 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or even 100% identical to SEQ ID NO: 2.

More preferably, the at least one RNAi comprises or consists of a nucleotide sequence according to SEQ ID NO: 1, even more preferably, the at least one RNAi consists of a nucleotide sequence according to SEQ ID NO: 1. More preferably, the at least one RNAi comprises or consists of a nucleotide sequence according to SEQ ID NO: 2, even more preferably, the at least one RNAi consists of a nucleotide sequence according to SEQ ID NO: 2.

In a further embodiment of the laticiferous plant or part thereof according to the present invention, the laticiferous plant or part thereof is selected from the group of plant families consisting of Apocynaceae, Asteraceae, Euphorbiaceae and Moraceae. Preferably, the laticiferous plant or part thereof according to the present invention is selected from the group consisting of *Agoseris glauca*, *Apocynum venetum*, *Chrysanthamnus graveolens*, *Chrysanthamnus nauseosus*, *Chrysanthamnus viscidiflorus*, *Funtumia elastica*, *Parthenium argentatum*, *Scorzonera acanthoclada*, *Scorzonera albicaulis*, *Scorzonera divaricate*, *Scorzonera hissaricata*, *Scorzonera racemosa*, *Scorzonera tau-saghyz*, *Scorzonera tragapogonoides*, *Scorzonera turkestanica*, *Scorzonera virgate*, *Solidago canadensis scabra*, *Solidago fistulosa*, *Solidago leavenworthii*, *Solidago rigida*, *Sonchus oleraceus*, *Taraxacum brevicomiculatum*, *Taraxacum hybernium*, *Taraxacum koksaghyz*, *Taraxacum megalorrhizon*, *Taraxacum officinale* and crossings thereof. In a further embodiment of the laticiferous plant or part thereof according to the present invention, the laticiferous plant or part thereof is selected from the group of plant families consisting of Apocynaceae, Asteraceae, Euphorbiaceae and Moraceae. Preferably, the laticiferous plant or part thereof according to the present invention is selected from the group consisting of *Agoseris glauca*, *Apocynum venetum*, *Chrysanthamnus graveolens*, *Chrysanthamnus nauseosus*, *Chrysanthamnus viscidiflorus*, *Cichorium intybus* L., *Funtumia elastica*, *Parthenium argentatum*, *Scorzonera acanthoclada*, *Scorzonera albicaulis*, *Scorzonera divaricate*, *Scorzonera hissaricata*, *Scorzonera racemosa*, *Scorzonera tau-saghyz*, *Scorzonera tragapogonoides*, *Scorzonera turkestanica*, *Scorzonera virgate*, *Solidago canadensis scabra*, *Solidago fistulosa*, *Solidago leavenworthii*, *Solidago rigida*, *Sonchus oleraceus*, *Taraxacum brevicomiculatum*, *Taraxacum hybernium*, *Taraxacum koksaghyz*, *Taraxacum megalorrhizon*, *Taraxacum officinale* and crossings thereof. In a more preferred embodiment, the laticiferous plant or part thereof is *Cichorium intybus* L. It is even more preferred that the laticiferous plant or part thereof is selected from the group consisting of *Parthenium argentatum*, *Chrysanthamnus nauseosus*, *Taraxacum koksaghyz*, *Taraxacum officinale* and crossings thereof. Most preferably, the laticiferous plant or part thereof is *Taraxacum koksaghyz*. In one embodiment of the laticiferous plant or part thereof according to the present invention, the laticiferous plant or part thereof is not a member of the family Euphorbiaceae. This means the present invention also provides a laticiferous plant or part thereof being genetically modified such that the activity of the at least one oxidosqualene cyclase is reduced or eliminated, compared to the respective wild-type or untreated laticiferous plant or part thereof, wherein the laticiferous

plant or part thereof is not a member of the family Euphorbiaceae. In another related embodiment of the laticiferous plant or part thereof according to the present invention, the laticiferous plant or part thereof is not *Hevea brasiliensis*. This means, the present invention also provides a laticiferous plant or part thereof being genetically modified such that the activity of the at least one oxidosqualene cyclase is reduced or eliminated, compared to the respective wild-type or untreated laticiferous plant or part thereof, wherein the laticiferous plant or part thereof is not *Hevea brasiliensis*. In a further related embodiment of the laticiferous plant or part thereof according to the present invention, the laticiferous plant or part thereof is not from the genus *Landolphia*. This means the present invention also provides a laticiferous plant or part thereof being genetically modified such that the activity of the at least one oxidosqualene cyclase is reduced or eliminated, compared to the respective wild-type or untreated laticiferous plant or part thereof, wherein the laticiferous plant or part thereof is not from the genus *Landolphia*. The laticiferous plant or part thereof can also be the progeny of the respective laticiferous plant as mentioned in this paragraph.

In another embodiment of the laticiferous plant or part thereof according to the present invention, the at least one oxidosqualene cyclase is selected from the group consisting of oxidosqualene cyclase 1 according to SEQ ID NO: 4, the oxidosqualene cyclase 2 according to SEQ ID NO: 6, the oxidosqualene cyclase 3 according to SEQ ID NO: 8, the oxidosqualene cyclase 4 according to SEQ ID NO: 10, the oxidosqualene cyclase 5 according to SEQ ID NO: 12, the oxidosqualene cyclase 6 according to SEQ ID NO: 14, the oxidosqualene cyclase 7 according to SEQ ID NO: 16, the oxidosqualene cyclase 8 according to SEQ ID NO: 18, the lupeol synthase according to SEQ ID NO: 20 and an amino acid sequence being at least 60%, 70%, 80%, 90%, 95% or 99% identical to SEQ ID NOs: 4, 6, 8, 10, 12, 14, 16, 18 or 20. Preferably, the at least one oxidosqualene cyclase is selected from the group consisting of the oxidosqualene cyclase 1 according to SEQ ID NO: 4, the oxidosqualene cyclase 2 according to SEQ ID NO: 6, the oxidosqualene cyclase 3 according to SEQ ID NO: 8, the oxidosqualene cyclase 5 according to SEQ ID NO: 12 and an amino acid sequence being at least 60%, 70%, 80%, 90%, 95% or 99% identical to SEQ ID NOs: 4, 6, 8 or 12. More preferably, the at least one oxidosqualene cyclase is the oxidosqualene cyclase 1 according to SEQ ID NO: 4 or an amino acid sequence being at least 60%, 70%, 80%, 90%, 95% or 99% identical thereto. Most preferably, the at least one oxidosqualene cyclase is the oxidosqualene cyclase 1 according to SEQ ID NO: 4.

In another embodiment of the laticiferous plant or part thereof according to the present invention, the at least one oxidosqualene cyclase is selected from the group consisting of an amino acid sequence being at least 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or even 100% identical to SEQ ID NOs: 4, 6, 8, 10, 12, 14, 16, 18 or 20.

In another embodiment of the laticiferous plant or part thereof according to the present invention, it is further preferred that the at least one oxidosqualene cyclase is selected from the group consisting of the oxidosqualene cyclase 1 according to SEQ ID NO: 4, the oxidosqualene cyclase 2 according to SEQ ID NO: 6, the oxidosqualene cyclase 3 according to SEQ ID NO: 8 and the oxidosqualene

cyclase 5 according to SEQ ID NO: 12 and an amino acid sequence being at least 60%, 70%, 80%, 90%, 95% or 99% identical to SEQ ID NOs: 4, 6, 8 or 12. In a further embodiment of the laticiferous plant, progeny thereof or part thereof according to the present invention, the at least one RNAi is specifically directed against the mRNA transcript of the at least one oxidosqualene cyclase, preferably wherein the at least one RNAi comprises or consists of a nucleotide sequence according to SEQ ID NO: 1 and/or comprises or consists of a nucleotide sequence according to SEQ ID NO: 2 and/or comprises or consists of a nucleotide sequence being at least 60%, 70%, 80%, 90%, 95% or 99% identical to SEQ ID NO: 1 or SEQ ID NO: 2.

In another embodiment of the laticiferous plant or part thereof according to the present invention, the at least one oxidosqualene cyclase is selected from the group consisting of an amino acid sequence being at least 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or even 100% identical to SEQ ID NOs: 4, 6, 8 or 12.

It is even more preferred for the laticiferous plant or part thereof according to the present invention that the at least one oxidosqualene cyclase is the oxidosqualene cyclase 1 according to SEQ ID NO: 4 or an amino acid sequence being at least 60%, 70%, 80%, 90%, 95% or 99% identical thereto.

In another embodiment of the laticiferous plant or part thereof according to the present invention, the at least one oxidosqualene cyclase is selected from the group consisting of an amino acid sequence being at least 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or even 100% identical to SEQ ID NO: 4. It is most preferred that at least one oxidosqualene cyclase is the oxidosqualene cyclase 1 according to SEQ ID NO: 4.

The nucleotide sequence of the oxidosqualene cyclase 1 is further given herein as SEQ ID NO: 3. The nucleotide sequence of the oxidosqualene cyclase 2 is further given herein as SEQ ID NO: 5. The nucleotide sequence of the oxidosqualene cyclase 3 is further given herein as SEQ ID NO: 7. The nucleotide sequence of the oxidosqualene cyclase 4 is further given herein as SEQ ID NO: 9. The nucleotide sequence of the oxidosqualene cyclase 5 is further given herein as SEQ ID NO: 11. The nucleotide sequence of the oxidosqualene cyclase 6 is further given herein as SEQ ID NO: 13. The nucleotide sequence of the oxidosqualene cyclase 7 is further given herein as SEQ ID NO: 15. The nucleotide sequence of the oxidosqualene cyclase 8 is further given herein as SEQ ID NO: 17. The nucleotide sequence of the lupeol synthase is further given herein as SEQ ID NO: 19.

The definitions as well as the preferred embodiments provided herein above with regard to the method of the present invention apply mutatis mutandis also to the embodiments relating to the laticiferous plant, progeny thereof or part thereof of the present invention. Further, the definitions as well as the embodiments and preferred embodiments provided herein above with regard to the laticiferous plant or part thereof apply mutatis mutandis also to the progeny of the respective laticiferous plant or part thereof.

The present invention also provides an RNAi being specifically directed against a mRNA transcript of at least one oxidosqualene cyclase, for use in a method of reducing or eliminating the content of triterpenes and/or triterpenoids of latex or rubber from a laticiferous plant or part thereof. The present invention also provides an RNAi being specifically directed against a mRNA transcript of at least one oxidosqualene cyclase, for use in a method of reducing or eliminating the content of triterpenes and/or triterpenoids of latex or rubber from a laticiferous plant, progeny thereof or part thereof. In this context, the term “reducing the content of triterpenes and/or triterpenoids” or the term “the content of triterpenes and/or triterpenoids is reduced” or the term “being capable of reducing the content of triterpenes and/or triterpenoids” means, as used herein, that the content of triterpenes and/or triterpenoids contained in a laticiferous plant, progeny thereof or part thereof is reduced for at least 1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 11%, 12%, 13%, 14%, 15%, 16%, 17%, 18%, 19%, 20%, 21%, 22%, 23%, 24%, 25%, 26%, 27%, 28%, 29%, 30%, 31%, 32%, 33%, 34%, 35%, 36%, 37%, 38%, 39%, 40%, 41%, 42%, 43%, 44%, 45%, 46%, 47%, 48%, 49%, 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94% or 95% compared to the wild-type or untreated laticiferous plant, progeny thereof or part thereof, meaning without being genetically modified or before being genetically modified. Further, the term “eliminating the content of triterpenes and/or triterpenoids” or the term “the content of triterpenes and/or triterpenoids is eliminated” or the term “being capable of eliminating the content of triterpenes and/or triterpenoids” means that the content of triterpenes and/or triterpenoids contained in a laticiferous plant, progeny thereof or part thereof is reduced for at least 95.1%, 95.2%, 95.3%, 95.4%, 95.5%, 95.6%, 95.7%, 95.8%, 95.9%, 96.0%, 96.1%, 96.2%, 96.3%, 96.4%, 96.5%, 96.6%, 96.7%, 96.8%, 96.9%, 97.0%, 97.1%, 97.2%, 97.3%, 97.4%, 97.5%, 97.6%, 97.7%, 97.8%, 97.9%, 98.0%, 98.1%, 98.2%, 98.3%, 98.4%, 98.5%, 98.6%, 98.7%, 98.8%, 98.9%, 99.0%, 99.1%, 99.2%, 99.3%, 99.4%, 99.5%, 99.6%, 99.7%, 99.8%, 99.9% or even 100.00% compared to the wild-type or untreated laticiferous plant, progeny thereof or part thereof, meaning without being genetically modified or before being genetically modified.

Preferably, the RNAi according to the present invention comprises or consists of a nucleotide sequence according to SEQ ID NO: 1 and/or comprises or consists of a nucleotide sequence according to SEQ ID NO: 2 and/or comprises or consists of a nucleotide sequence being at least 60%, 70%, 80%, 90%, 95% or 99% identical to SEQ ID NO: 1 or SEQ ID NO: 2, for use in a method of reducing or eliminating the content of triterpenes and/or triterpenoids of latex or rubber from a laticiferous plant or part thereof. The laticiferous plant or part thereof can also be a progeny of the respective laticiferous plant as mentioned herein.

In a further embodiment of the present invention, the RNAi according to the present invention comprises or consists of a nucleotide sequence according to SEQ ID NO: 1 and/or comprises or consists of a nucleotide sequence being at least 60%, 70%, 80%, 90%, 95% or 99% identical to SEQ ID NO: 1. More preferably, the RNAi according to the present invention consists of a nucleotide sequence according to SEQ ID NO: 1 and/or consists of a nucleotide sequence being at least 60%, 70%, 80%, 90%, 95% or 99%

identical to SEQ ID NO: 1. Most preferably, the RNAi according to the present invention consists of a nucleotide sequence according to SEQ ID NO: 1.

In a further embodiment of the present invention, the RNAi according to the present invention comprises or consists of a nucleotide sequence according to SEQ ID NO: 2 and/or comprises or consists of a nucleotide sequence being at least 60%, 70%, 80%, 90%, 95% or 99% identical to SEQ ID NO: 2. More preferably, the RNAi according to the present invention consists of a nucleotide sequence according to SEQ ID NO: 2 and/or consists of a nucleotide sequence being at least 60%, 70%, 80%, 90%, 95% or 99% identical to SEQ ID NO: 2. Most preferably, the RNAi according to the present invention consists of a nucleotide sequence according to SEQ ID NO: 2.

Also for all embodiments of the RNAi according to the present invention, the nucleotide sequence according to SEQ ID NO: 1 is given as follows:

```
TCTCCAAGCGCCCATAGCGAAAAGTGTCTCTATATTCTGCATAATGTAAC
ATGCAATTCTTCATGTACACTCCTGTTATTTCTGTTGAGGAAAAGTCTCCA
TTATCCATTGTGTCATTATGAGCAATTTGCTGCCTTGTAATGGTGTCTC
GGATCTCTTTCAGCCTGTCCACCACATAGAAGACCAAGCATGGCCCATGAA
GTTTGACAAAAATTGTTCTGTTCCCTTCCAAAGGTATAAATTTCTCTTGT
GGGCATGACTCAAAGTTCTCTCCCAACCACTTCTGAGTTTGTGTGTCGAG
AGGAAAAACTTGACAGCTTTCGAACTGCTTCACTATTTTCATATGTTTGT
CCACAAGCTACCAATCCTTGTAACACGAAATAGGTTCCATAAGATAACAT
ATACCCCAATAACCATAACCAAGAACCATTTTCTTGT.
```

The complementary sequence thereto, which is, for example, also present in a vector construct comprising the SEQ ID NO: 1 if the vector comprises a hairpin-structure, is given as follows:

```
(SEQ ID NO: 47)
AACAGAAAATGGTTCTTGGTATGGTTATTGGGTATATGTTATCTTTATG
GAACCTATTTCTGTGTACAAGGATTGGTAGCTTGTGGACAAACATATGAAA
ATAGTGAAGCAGTTCGGAAAGCTGTCAAGTTTTCTCTCGACACAAAACT
CAGAAGGTGGTTGGGGAGAGAACTTTGAGTCATGCCCAAGAGAAATTTA
TACCTTTGGAAGGGAACAGAACAAATTTGTGCAAACTTCATGGCCATGC
TTGGTCTTCTATGTGGTGGACAGGCTGAAAGAGATCCGACACCATTACACA
AGGCAGCAAAATGCTCATAAATGGACAAATGGATAATGGAGACTTTCCTC
AACAGGAAATAACAGGAGTGTACATGAAGAATGCATGTTACATTATGCAG
AATATAGGAACACTTTTCCGCTATGGCGCTTGAGAA.
```

Also for all embodiments of the RNAi according to the present invention, the nucleotide sequence according to SEQ ID NO: 2 of the present invention is given as follows:

```
ATCCCAAGCTTGAATCGCACGGTTCAGTCTAACTGCCTTTCTCAGGCTGT
TGTCACCGCTTCATAGTTCACATCTTCGTCGTCTCCAAGTCTCACTGGAGG
TACGCTTATAAGATCAATTCCGC.
```

The complementary sequence thereto, which is, for example, also present in a vector construct comprising the SEQ ID NO: 2 if the vector comprises a hairpin-structure, is given as follows:

(SEQ ID NO: 48)

CGCGAATTGATCTTATAAGCGTACCTCCAGTGAGACTTGGAGACGACGAAG
ATGTGAACATATGAAGCGGTGACAACAGCCGTGAGAAAGGCAGTTAGACTGA
ACCGTGCATTCAAGCTTGGGAT.

Preferably, the RNAi according to the present invention comprises or consists of a nucleotide sequence being at least 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or even 100% identical to SEQ ID NO: 1.

More preferably, the at least one RNAi consists of a nucleotide sequence being at least 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or even 100% identical to SEQ ID NO: 1.

Preferably, the RNAi according to the present invention comprises or consists of a nucleotide sequence being at least 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or even 100% identical to SEQ ID NO: 2.

More preferably, the RNAi according to the present invention consists of a nucleotide sequence being at least 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or even 100% identical to SEQ ID NO: 2.

More preferably, the RNAi according to the present invention comprises or consists of a nucleotide sequence according to SEQ ID NO: 1. Even more preferably, the RNAi according to the present invention consists of a nucleotide sequence according to SEQ ID NO: 1.

More preferably, the RNAi according to the present invention comprises or consists of a nucleotide sequence according to SEQ ID NO: 2. Even more preferably, the at least one RNAi consists of a nucleotide sequence according to SEQ ID NO: 2.

The term "RNA interference" refers generally and in the context of the present invention as used herein to a process in which a double-stranded RNA molecule changes the expression of a nucleic acid sequence with which the double-stranded or short hairpin RNA molecule shares substantial or total homology.

The term "RNAi" refers generally and in the context of the present invention as used herein to an RNA sequence that elicits RNA interference, and which is transcribed from a vector. The terms "short hairpin RNA" or "shRNA" refer to an RNA structure having a duplex region and a loop region. This term should also be understood to specifically include RNA molecules with stem-loop or panhandle secondary

structures. In some embodiments of the present invention, RNAi's are expressed initially as shRNAs.

A better understanding of the present invention and of its advantages will be available from the following examples, offered for illustrative purposes only. The examples are not intended to limit the scope of the present invention in any way.

The invention is further characterized by the following items:

1. Method of reducing or eliminating the content of triterpenes and/or triterpenoids of latex or rubber from a laticiferous plant or part thereof, comprising genetically modifying the laticiferous plant or part thereof, such that the activity of at least one oxidosqualene cyclase is reduced or eliminated, compared to the respective wild-type or untreated laticiferous plant or part thereof.

2. Method according to item 1, wherein the laticiferous plant or part thereof is genetically modified by at least one heterologous nucleotide sequence that is capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase, preferably that is capable of reducing or eliminating the expression of the gene of the at least one oxidosqualene cyclase, and/or wherein the laticiferous plant or part thereof is genetically modified by subjecting it to and/or incubating it with agents, which are capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase.

3. Method according to item 2, wherein the at least one heterologous nucleotide sequence encodes at least one RNAi that interferes with the mRNA encoding the at least one oxidosqualene cyclase, with the proviso that the at least one RNAi is capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase.

4. Method according to item 3, wherein the at least one RNAi is specifically directed against the mRNA transcript of the at least one oxidosqualene cyclase, preferably wherein the at least one RNAi comprises or consists of a nucleotide sequence according to SEQ ID NO: 1 and/or comprises or consists of a nucleotide sequence according to SEQ ID NO: 2 and/or comprises or consists of a nucleotide sequence being at least 60%, 70%, 80%, 90%, 95% or 99% identical to SEQ ID NO: 1 or SEQ ID NO: 2.

5. Method according to any one of the preceding items, wherein the laticiferous plant or part thereof is selected from the group of plant families consisting of Apocynaceae, Asteraceae, Euphorbiaceae and Moraceae.

6. Method according to any one of the preceding items, wherein the laticiferous plant or part thereof is selected from the group consisting of *Agoseris glauca*, *Apocynum venetum*, *Chrysothamnus graveolens*, *Chrysothamnus nauseosus*, *Chrysothamnus viscidiflorus*, *Funtumia elastica*, *Parthenium argentatum*, *Scorzonera acanthoclada*, *Scorzonera albicaulis*, *Scorzonera divaricate*, *Scorzonera hissaricata*, *Scorzonera racemosa*, *Scorzonera tau-saghyz*, *Scorzonera tragapogonoides*, *Scorzonera turkestanica*, *Scorzonera virgate*, *Solidago canadensis scabra*, *Solidago fistulosa*, *Solidago leavenworthii*, *Solidago rigida*, *Sonchus oleraceus*, *Taraxacum brevicomiculatum*, *Taraxacum hybemale*, *Taraxacum koksaghyz*, *Taraxacum megalorrhizon*, *Taraxacum officinale* and crossings thereof, preferably wherein the laticiferous plant or part thereof is selected from the group consisting of *Parthenium argentatum*, *Chrysothamnus nauseosus*, *Taraxacum koksaghyz*, *Taraxacum officinale* and crossings thereof, more preferably wherein the laticiferous plant or part thereof is *Taraxacum koksaghyz*.

7. Method according to any one of the preceding items, wherein the at least one oxidosqualene cyclase is selected from the group consisting of oxidosqualene cyclase 1

according to SEQ ID NO: 4, the oxidosqualene cyclase 2 according to SEQ ID NO: 6, the oxidosqualene cyclase 3 according to SEQ ID NO: 8, the oxidosqualene cyclase 4 according to SEQ ID NO: 10, the oxidosqualene cyclase 5 according to SEQ ID NO: 12, the oxidosqualene cyclase 6 according to SEQ ID NO: 14, the oxidosqualene cyclase 7 according to SEQ ID NO: 16, the oxidosqualene cyclase 8 according to SEQ ID NO: 18, the lupeol synthase according to SEQ ID NO: 20 and an amino acid sequence being at least 60%, 70%, 80%, 90%, 95% or 99% identical to SEQ ID NOs: 4, 6, 8, 10, 12, 14, 16, 18 or 20, preferably wherein the at least one oxidosqualene cyclase is selected from the group consisting of the oxidosqualene cyclase 1 according to SEQ ID NO: 4, the oxidosqualene cyclase 2 according to SEQ ID NO: 6, the oxidosqualene cyclase 3 according to SEQ ID NO: 8 and the oxidosqualene cyclase 5 according to SEQ ID NO: 12 and an amino acid sequence being at least 60%, 70%, 80%, 90%, 95% or 99% identical to SEQ ID NOs: 4, 6, 8 or 12, more preferably wherein the at least one oxidosqualene cyclase is the oxidosqualene cyclase 1 according to SEQ ID NO: 4 or an amino acid sequence being at least 60%, 70%, 80%, 90%, 95% or 99% identical thereto.

8. Method according to any one of the preceding items, wherein the part of the laticiferous plant is selected from the group consisting of cells, tissues, organs, roots, stems, branches, leaves, exudates, latex and mixtures thereof.

9. Laticiferous plant or part thereof being genetically modified such that the activity of at least one oxidosqualene cyclase is reduced or eliminated, compared to the respective wild-type or untreated laticiferous plant or part thereof.

10. Laticiferous plant or part thereof according to item 9, wherein the laticiferous plant or part thereof is genetically modified by at least one heterologous nucleotide sequence that is capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase, preferably that is capable of reducing or eliminating the expression of the gene of the at least one oxidosqualene cyclase, and/or wherein the laticiferous plant or part thereof is genetically modified by subjecting it to and/or incubating it with agents, which are capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase.

11. Laticiferous plant or part thereof according to item 9 or 10, wherein the at least one heterologous nucleotide sequence encodes at least one RNAi that interferes with the mRNA encoding the at least one oxidosqualene cyclase with the proviso that the at least one RNAi is capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase.

12. Laticiferous plant or part thereof according to item 11, wherein the at least one RNAi is specifically directed against the mRNA transcript of the at least one oxidosqualene cyclase, preferably wherein the at least one RNAi comprises or consists of a nucleotide sequence according to SEQ ID NO: 1 and/or comprises or consists of a nucleotide sequence according to SEQ ID NO: 2 and/or comprises or consists of a nucleotide sequence being at least 60%, 70%, 80%, 90%, 95% or 99% identical to SEQ ID NO: 1 or SEQ ID NO: 2.

13. Laticiferous plant or part thereof according to any one of items 9 to 12, wherein the laticiferous plant or part thereof is selected from the group of plant families consisting of Apocynaceae, Asteraceae, Euphorbiaceae and Moraceae, preferably wherein the laticiferous plant or part thereof is selected from the group consisting of *Agoseris glauca*, *Apocynum venetum*, *Chrysothamnus graveolens*, *Chrysothamnus nauseosus*, *Chrysothamnus viscidiflorus*, *Funtumia elastica*, *Parthenium argentatum*, *Scorzonera acanthoclada*, *Scorzonera albicaulis*, *Scorzonera divaricate*, *Scorzonera*

hissaricata, *Scorzonera racemosa*, *Scorzonera tau-saghyz*, *Scorzonera tragapogonoides*, *Scorzonera turkestanica*, *Scorzonera virgate*, *Solidago canadensis scabra*, *Solidago fistulosa*, *Solidago leavenworthii*, *Solidago rigida*, *Sonchus oleraceus*, *Taraxacum brevicomiculatum*, *Taraxacum hybridum*, *Taraxacum koksaghyz*, *Taraxacum megalorrhizon*, *Taraxacum officinale* and crossings thereof, preferably wherein the laticiferous plant or part thereof is selected from the group consisting of *Parthenium argentatum*, *Chrysothamnus nauseosus*, *Taraxacum koksaghyz*, *Taraxacum officinale* and crossings thereof, more preferably wherein the laticiferous plant or part thereof is *Taraxacum koksaghyz*.

14. Laticiferous plant or part thereof according to any one of items 9 to 13, wherein the at least one oxidosqualene cyclase is selected from the group consisting of oxidosqualene cyclase 1 according to SEQ ID NO: 4, the oxidosqualene cyclase 2 according to SEQ ID NO: 6, the oxidosqualene cyclase 3 according to SEQ ID NO: 8, the oxidosqualene cyclase 4 according to SEQ ID NO: 10, the oxidosqualene cyclase 5 according to SEQ ID NO: 12, the oxidosqualene cyclase 6 according to SEQ ID NO: 14, the oxidosqualene cyclase 7 according to SEQ ID NO: 16, the oxidosqualene cyclase 8 according to SEQ ID NO: 18, the lupeol synthase according to SEQ ID NO: 20, and an amino acid sequence being at least 60%, 70%, 80%, 90%, 95% or 99% identical to SEQ ID NOs: 4, 6, 8, 10, 12, 14, 16, 18 or 20, preferably wherein the at least one oxidosqualene cyclase is selected from the group consisting of the oxidosqualene cyclase 1 according to SEQ ID NO: 4, the oxidosqualene cyclase 2 according to SEQ ID NO: 6, the oxidosqualene cyclase 3 according to SEQ ID NO: 8, the oxidosqualene cyclase 5 according to SEQ ID NO: 12, and an amino acid sequence being at least 60%, 70%, 80%, 90%, 95% or 99% identical to SEQ ID NOs: 4, 6, 8 or 12, more preferably wherein the at least one oxidosqualene cyclase is the oxidosqualene cyclase 1 according to SEQ ID NO: 4 or an amino acid sequence being at least 60%, 70%, 80%, 90%, 95% or 99% identical thereto.

15. RNAi being specifically directed against a mRNA transcript of at least one oxidosqualene cyclase, preferably wherein the RNAi comprises or consists of a nucleotide sequence according to SEQ ID NO: 1 and/or comprises or consists of a nucleotide sequence according to SEQ ID NO: 2 and/or comprises or consists of a nucleotide sequence being at least 60%, 70%, 80%, 90%, 95% or 99% identical to SEQ ID NO: 1 or SEQ ID NO: 2, for use in a method of reducing or eliminating the content of triterpenes and/or triterpenoids of latex or rubber from a laticiferous plant or part thereof.

EXAMPLES

Example 1: Materials and Methods

Plant Material and Cultivation Conditions

T. koksaghyz wild-type and transgenic plants were grown in standard soil (ED73 Einheitserde, Fröndenberger, Germany). Every four weeks a commercial fertilizer was supplemented according to the manufacturer's recommendations (Hakaphos Plus, Compo GmbH, Munster, Germany). The plants were cultivated in controlled growth chambers or in the greenhouse at 18° C. and 20 klux (high pressure sodium lamp, HPS 600 Watts, Greenbud, enhanced yellow and red spectrum) with a 16-h photoperiod.

Total RNA Extraction and cDNA Synthesis

Latex was harvested from wounded *T. koksaghyz* root in 90 µl REB buffer (Schmidt et al., 2010). Subsequently, total

RNA was extracted using the innuPREP RNA Mini Kit (Analytik Jena, Jena, Germany) according to manufacturer's instructions.

Full-length cDNA was synthesized from 500 ng total RNA using PrimeScript RT Master Mix (TaKaRa, Clontech, Saint-Germain-en-Laye, France) according to manufacturer's instructions.

Quantitative RT-PCR

Quantitative RT-PCR analysis was performed as previously described (Putter et al., 2017). The *T. kokzaghlyz* TkOSC1- and TkOSC-RNAi-plants were 9 to 16 weeks old at the time of expression analysis. All oligonucleotide sequences for expression analysis are listed in Table 1. Primer efficiencies and amplification factors are summarized in Table 2.

TABLE 1

Sequences of oligonucleotides used.	
Oligo	Sequence (5'43')
cloning procedures	
TkOSC1-RNAi-fwd-NcoI	AAACCATGGGCGGAATTGATCTTATAAGCG (SEQ ID NO: 21)
TkOSC1-RNAi-rev-XhoI	AAACTCGAGATCCCAAGCTTGAATCGCAC (SEQ ID NO: 22)
TkOSC-RNAi-fwd-NcoI	AAACCATGGAACAAGAAAATGGTTCTTGG (SEQ ID NO: 23)
TkOSC-RNAi-rev-XhoI	AAACTCGAGTCTCCAAGCGCCCATAGCGG (SEQ ID NO: 24)
qRT-PCR	
TkEFlalpha-fw-realtime	CGAGAGATTCGAGAAGGAAGC (SEQ ID NO: 25)
TkEFlalpha-rv-realtime	CTGTGCAGTAGTACTTGGTGG (SEQ ID NO: 26)
TkLUP-fw-realtime	GCTGACCACCACCAACAACCAC (SEQ ID NO: 27)
TkLUP-rv-realtime	AGCACGTTCTCTTCGGTTCCAG (SEQ ID NO: 28)
TkOSC1-fw-realtime	ACTCCTCCCTTGATAATTGCCC (SEQ ID NO: 29)
TkOSC1-rv-realtime	TTGTGCTTCTGCCTGATATATAGAAC (SEQ ID NO: 30)
TkOSC2-fw-realtime	CCGGTGAGAAGGTGGAAGTT (SEQ ID NO: 31)
TkOSC2-rv-realtime	GGAACCGGTACCTCCCAAC (SEQ ID NO: 32)
TkOSC3-fw-realtime	TCCATCCAAACCACAGAAAAG (SEQ ID NO: 33)
TkOSC3-rv-realtime	ATGAAGCATACTCCCCAATAAC (SEQ ID NO: 34)
TkOSC4-fw-realtime	CCGACAATTGTAAGGCCACTG (SEQ ID NO: 35)
TkOSC4-rv-realtime	TGTTTGCAACCGTTCGACC (SEQ ID NO: 36)
TkOSC5-fw-realtime	GAAACACAAC TAGAAGATGGCGGT (SEQ ID NO: 37)
TkOSC5-rv-realtime	CATAGCCCATGAAGTGTGCACT (SEQ ID NO: 38)
TkOSC6-fw-realtime	GGTCATAGCACCATGTTTGGG (SEQ ID NO: 39)
TkOSC6-rv-realtime	GGTGACTGAGCCATGATCCAGG (SEQ ID NO: 40)
TkOSC7-fw-realtime	CGATGCTGGAACACCCGC (SEQ ID NO: 41)
TkOSC7-rv-realtime	GTGGTGGTATGCTCGATACATC (SEQ ID NO: 42)
TkOSC8-fw-realtime	CTGAGCATACCACCTGCGAG (SEQ ID NO: 43)
TkOSC8-rv-realtime	CAGCTTTTCTTATTGATGTCGCTACTG (SEQ ID NO: 44)
TkRP-fw-realtime	CGTCGATCTCAAGGATGTTGTC (SEQ ID NO: 45)
TkRP-rv-realtime	GGAGCTTTGAGAAGAACCAACG (SEQ ID NO: 46)

TABLE 2

Primer efficiency and amplification factors of qRT-PCR oligos. All values are obtained from cDNA synthesized from *T. koksaghyz* mRNA. The values were calculated using the Bio-Rad CFX Manager v3.1 software (Bio-Rad Laboratories Inc., Hercules, CA, USA) and the qPCR primer efficiency calculator provided by Thermo Fisher Scientific (thermoscientificbio.com/webtools/qpcr/efficiency/).

Oligo pair	Efficiency	Amplification factor (66° C.)
TkLUP-realtime	99.17%	1.99
TkOSC1-realtime	100.16%	2.00
TkOSC2-realtime	101.87%	2.02
TkOSC3-realtime	109.67%	2.10
TkOSC4-realtime	108.54%	2.09
TkOSC5-realtime	106.95%	2.07
TkOSC6-realtime	94.39%	1.94
TkOSC7-realtime	91.29%	1.91
TkOSC8-realtime	98.6%	1.99
TkEF1alpha-realtime	104.48%	2.04
TkRP-realtime	105.44%	2.05
TkSQE1-realtime	95.53%	1.96
TkSQS1-realtime	102.56%	2.03

Cloning and Stable Transformation Procedures

For cloning of RNAi-vectors for stable transformation, cDNA fragments were inserted initially, into the Gateway-compatible entry vector pBluescript II KS (+) (Addgene, Cambridge, MA, USA). Therefore, cDNA fragments TkOSC1-RNAi (125 bp) (SEQ ID NO: 48) and TkOSC-RNAi (445 bp) (SEQ ID NO: 47) were amplified by using the oligos TkOSC1-RNAi-fwd-NcoI (SEQ ID NO: 21) and TkOSC1-RNAi-rev-XhoI (SEQ ID NO: 22) or TkOSC-RNAi-fwd-NcoI (SEQ ID NO: 23) and TkOSC-RNAi-rev-XhoI (SEQ ID NO: 24), respectively. The insertion of the RNAi-fragments by using the NcoI/XhoI restriction sites of pBluescript II KS (+) resulted in the entry vectors pBluescript-TkOSC1 and pBluescript-TkOSC. Subsequently, the entry vectors were used for recombination into the expression vector pLab12.5-pREF (Epping et al., 2015) resulting in the expression vectors pLab12.5-pREF-TkOSC1-RNAi (SEQ ID NO: 50) and pLab12.5-pREF-TkOSC-RNAi (SEQ ID NO: 49) vectors. The fragment according to SEQ ID NO: 48 (GCGGAATTGATCTTATAAGCGTACCTCCAGTGA-GACTTGGAGACGACGAAGATGTGAACT ATGAAGCGGTGACAACAGCCGTGAGAAAGGCAGT-TAGACTGAACCGTGCGATTCAAGCTT GGGAT) was used to construct pLab12.5-pREF-TkOSC1-RNAi (SEQ ID NO: 50).

The fragment according to SEQ ID NO: 47 (AACAAGAAAATGGTTCTTGGTATGGTTATTGGG GTATATGTTATCTTTATGGAACCTATTTTCGTGTTA-CAAGGATTGGTAGCTTGTGGACAAACA TAT-50 GAAAATAGTGAAGCAGTTCCGAAAGCTGT-CAAGTTTTTCTCTCGACACAAAACCTCAGA AGGTGGTTGGGAGAGAACTTTGAGTCATGCC-CACAAGAGAAAATTATACCTTTGGAAGGG AACAGAACAAATTTGTGCAAACTTCATGGGC-CATGCTTGGTCTTCTATGTGGTGGACAGG CTGAAAGAGATCCGACACCATTA-CACAAGGCAGCAAAATGGCTCAT-AAATGGACAAATGGA TAATGGAGACTTTCCT-CAACAGGAAATAACAGGAGTGATGAAGAATTG-60 CATGTTACATT ATGCAGAATATAGGAACACTTTTCCGC-TATGGGCGTGTGGAGA) was used to construct pLab12.5-pREF-TkOSC-RNAi (SEQ ID NO: 49).

All vector sequences were verified by sequencing. The complete vector sequence or construct for pLab12.5-pREF-TkOSC1-RNAi is given as SEQ ID NO: 50. The complete

vector sequence or construct for pLab12.5-pREF-TkOSC-RNAi is given as SEQ ID NO: 49. Oligonucleotide sequences are shown in Table 1.

The transformation of *T. koksaghyz* by *A. tumefaciens* strain EHA105 with OE- and RNAi-constructs was carried out as previously described (Stolze et al., 2017).

Triterpene Extraction and Analysis

Whole *T. koksaghyz* roots were frozen in liquid nitrogen, freeze-dried, and grinded to a fine powder. Root material was harvested from 5- to 7-month old RNAi-lines. For triterpene analysis, 100 mg of dry root material were used for saponification by adding 20 ml of methanol containing 6% potassium hydroxide and heating to 80° C. for 2 h. 100 µl of betuline (2.5 mg/ml stock solutions in acetone) were added as internal standard. Samples were refilled with water to a volume of 15 ml and extracted three times with one volume of hexane by vortexing and subsequent centrifugation at 4000 rpm for 5 min. The pooled hexane phases were evaporated in the Rocket Evaporator 4D (Genvac Aerospace Inc., Cleveland, USA) and redissolved in 1 ml acetone.

To extract NR from root material, 2 g of the dry root powder as mentioned above and 20 ml of water were transferred into a 50 ml reaction tube and a mixture of steal beads (six 7 g-beads and nine 4 g-beads) were added. The sample was vortexed for 30 min and centrifuged for 15 min at 5000 g. The floating rubber flakes were removed, washed in fresh water and dried over night at 40° C. For triterpene extraction, 100 mg of the resulting NR were filled up with 10 ml acetone and 100 µl internal betuline standard (2.5 mg/ml stock solution in acetone) and were incubated by rotating at room temperature for one week. The resulting extract was dried and redissolved in 1 ml acetone.

Triterpene analysis was performed with a GC-MS-QP 210 Ultra system (Shimadzu, Duisburg, Germany) equipped with a Rxi®-5 ms column (Restek GmbH, Bad Homburg, Germany). 0.5 µl of the extract were injected using split modus (1:10) at an injector and interface temperature of 260° C. The GC temperature program was as follows: 120° C. for 3 min, temperature gradient of 15° C. per minute up to 330° C., 330° C. for 10 min. Electron ionization (EI) in the MS was set to 70 eV. Peak integration and identification was performed with the LabSolution software (Shimadzu, Duisburg, Germany) using a NIST library (NIST=National Institute of Standards and Technology) or by analyzing the corresponding standards obtained from Extrasynthese (Genay, France). Quantification was performed in relation to the internal standard. Retention indices for the triterpene compounds were determined using the LabSolution software in relation to a C8-C40 alkane calibration standard (Sigma Aldrich, Taufkirchen, Germany) using the same operating conditions.

Statistical Analysis

For statistical analysis with more than three samples, the Mann-Whitney U test was applied. This test is a nonparametric test of the null hypothesis that does not require the assumption of normal distributions. The corresponding results were displayed in boxplots. All other statistics were performed using the two-tailed t-test. If variances were assumed to be unequal Welch's correction was applied. The corresponding results were displayed in bar charts.

Example 2: Generation of *T. koksaghyz* Plants Comprising RNAi Targeting OSCs

In order to gain *T. koksaghyz* plants with reduced triterpene levels in latex, two different RNAi constructs driven by the laticifer predominant REF-promoter were transformed in

T. koksaghyz plants. The fragment (SEQ ID NO: 48) used to construct plab12.5-pREF-TkOSC1-RNAi (SEQ ID NO: 50) (comprising SEQ ID NO: 2 as the RNAi that is specifically directed against the mRNA transcript) comprises a 125 bp long not conserved region and was supposed to knock-down only TkOSC1, whereas the fragment (SEQ ID NO: 47) of pLab12.5-pREF-TkOSC-RNAi (SEQ ID NO: 49) (comprising SEQ ID NO: 1 as the RNAi that is specifically directed against the mRNA transcript) contains a 445 bp long sequence derived from a more conserved region and was expected to knock-down several TkOSCs. After stable transformation, the inventors received two individual TkOSC1-RNAi lines comprising seven plants and five TkOSC-RNAi lines consisting of 12 plants.

In 9- to 16-week-old plants, the expression of TkOSC1 in latex was analysed via qPCR (FIG. 1(a)). Compared to *T. koksaghyz* wild-type plants (TkWt; n=4) the expression of TkOSC1 was reduced significantly in all lines. For the TkOSC1-RNAi plants, the expression was reduced to one-seventh. For the TkOSC-RNAi plants, the effect was even stronger, resulting in a reduction to one-eighteenth.

To analyse the expression of further triterpene biosynthesis related genes, three wild-type, three TkOSC1-RNAi plants and four TkOSC-RNAi plants were exemplarily used for further qPCR analysis (FIG. 1(b) and FIG. 1(c)). TkOSC-RNAi plants show a remarkable reduction of TkOSC2, TkOSC3, TkOSC5 and TkLUP expression. Only TkOSC6 expression was slightly enhanced. Surprisingly, the plants with the TkOSC1 specific construct showed also a reduced expression of TkOSC2, TkOSC3 and TkOSC5, but the effect was weaker compared to the TkOSC-RNAi plants. TkOSC6 expression was not affected and TkLUP expression was slightly upregulated. TkOSC4 expression was not detectable in latex of any plants. Earlier studies revealed the co-regulation of TkOSC1 with the upstream genes TkSQS1 and TkSQE1 (Unland et al., 2018). Therefore, the expression of TkSQS1 and TkSQE1 was also analysed in the RNAi-lines (FIG. 3(c)). Mean expression of TkSQS1 in TkOSC1-RNAi plants was reduced to one-half, whereas the TkSQE1 expression was comparable to the expression in wild-type

plants. In the TkOSC-RNAi lines, TkSQS1 expression was even further reduced to one-quarter and TkSQE1 expression was reduced to one-half compared to the wild-type, confirming the co-regulation of TkSQS1, TkSQE1 and TkOSC1. It can be noted that the expression of several triterpene biosynthesis related genes are affected in both RNAi lines with a general stronger effect in the plants carrying the TkOSC-RNAi fragment.

Example 3: Effect of Oxidosqualene Cyclase (OSC) Knockdown on Triterpene Composition

To analyse if alterations in expression have an effect on triterpene composition in the RNAi lines, triterpene extracts of dry root material from 5- to 7-month-old plants were measured by GC/MS (FIG. 1). The pentacyclic triterpenes were strongly affected in the transgenic lines (FIG. 1(d) and FIG. 1(e)). As a result, the total amount was reduced from 4.6 mg/g in wild-type plants to 2.8 mg/g in TkOSC1-RNAi plants and to 2.6 mg/g in TkOSC-RNAi plants. Especially the most abundant triterpenes, taraxasterol, β -amyrin and one unidentified triterpene (n.i.1), were remarkably reduced with a higher level of significance in TkOSC-RNAi plants.

Besides the pentacyclic triterpenes, further compounds were measured via GC/MS (FIG. 1(e), Table 3). It was found that precursors like squalene and 2,3-oxidosqualene accumulated significantly in the transgenic lines, with a higher level of significance in TkOSC-RNAi plants. Additionally, cycloartenol and 24-methylene cycloartenol, two sterol intermediates, were detected in both transgenic lines, with an average of 0.1 mg/g dry root weight in TkOSC1-RNAi plants and 0.9 mg/g dry root weight in TkOSC-RNAi plants. Sterols including campesterol, stigmasterol and sitosterol were detected to equal amounts of about 0.5 mg/g dry root weight in all plants and were not affected by the knock-down. In sum, the total triterpene content including precursors, sterol intermediates, sterols and pentacyclic triterpenes was significantly reduced in transgenic RNAi lines to about two-thirds with a higher level of significance in TkOSC-RNAi plants (FIG. 1(e)).

TABLE 3

Triterpene content of root material from <i>T. koksaghyz</i> TkOSC1-/TkOSC-RNAi lines.				
Triterpene compounds were identified and quantified by GC/MS [mg/g root dry weight], corresponding retention indices (RI) were determined in relation to a C8-C40 alkane calibration standard. The mean values represent independent extractions from wild-type plants (n = 4), TkOSC1-RNAi plants (n = 7) and TkOSC-RNAi plants (n = 12) conducted in triplicates and are indicated with the corresponding standard deviations.				
	RI	TkWt	TkOSC1-RNAi	TkOSC-RNAi
precursors		0.05 ± 0.01	0.08 ± 0.04	0.17 ± 0.08
squalene	2843	0.01 ± 0.00	0.01 ± 0.00	0.03 ± 0.02
2,3-oxidosqualene	2965	0.04 ± 0.01	0.06 ± 0.03	0.14 ± 0.08
sterol intermediates		0.00 ± 0.00	0.14 ± 0.06	0.27 ± 0.07
cycloartenol	3478	0.00 ± 0.00	0.09 ± 0.03	0.15 ± 0.04
24-methylene cycloartenol	3536	0.00 ± 0.00	0.05 ± 0.04	0.13 ± 0.03
sterols		0.50 ± 0.06	0.52 ± 0.05	0.54 ± 0.05
campesterol	3313	0.07 ± 0.01	0.07 ± 0.01	0.07 ± 0.01
stigmasterol	3343	0.22 ± 0.04	0.21 ± 0.02	0.24 ± 0.03
sitosterol	3405	0.21 ± 0.01	0.24 ± 0.03	0.23 ± 0.02
pentacyclic triterpenes		4.57 ± 0.38	2.79 ± 0.58	2.60 ± 0.42
taraxerol and n.i.	3448	0.70 ± 0.09	0.81 ± 0.14	0.79 ± 0.10
β -amyrin	3465	0.73 ± 0.06	0.25 ± 0.08	0.23 ± 0.05
lupeol	3510	0.12 ± 0.02	0.18 ± 0.07	0.11 ± 0.02
α -amyrin and lup-19(21)-en-3-ol	3516	0.70 ± 0.06	0.52 ± 0.14	0.56 ± 0.17
TkOSC3-product-1	3550	0.06 ± 0.01	0.05 ± 0.01	0.07 ± 0.01

TABLE 3-continued

Triterpene content of root material from <i>T. koksaghyz</i> TkOSC1-/TkOSC-RNAi lines. Triterpene compounds were identified and quantified by GC/MS [mg/g root dry weight], corresponding retention indices (RI) were determined in relation to a C8-C40 alkane calibration standard. The mean values represent independent extractions from wild-type plants (n = 4), TkOSC1-RNAi plants (n = 7) and TkOSC-RNAi plants (n = 12) conducted in triplicates and are indicated with the corresponding standard deviations.				
	RI	TkWt	TkOSC1-RNAi	TkOSC-RNAi
TkOSC3-product-2	3590	0.48 ± 0.05	0.45 ± 0.10	0.58 ± 0.12
n.i.1	3603	0.43 ± 0.04	0.19 ± 0.06	0.12 ± 0.02
taraxasterol	3615	1.37 ± 0.12	0.33 ± 0.13	0.14 ± 0.05

15

For triterpene quantification in NR, root material of three wild-type, TkOSC1-RNAi and TkOSC-RNAi plants, respectively, were used for rubber extraction. The triterpenes in NR extracts were isolated by acetone extraction and quantified in relation to NR weight (FIG. 2, Table 4). The crude acetone extracts contained acetate derivatives of different triterpenes. Those acetates as well as the predominant triterpenes in latex (taraxasterol, β -amyirin and n.i.1) were strongly reduced compared to wild-type plants with a higher level of significance in TkOSC-RNAi plants compared to TkOSC1-RNAi plants (FIGS. 2(a) and 2(b)). The average amount of total pentacyclic triterpenes in wild-type plants with 56.6 mg/g NR was reduced to 33.4 mg/g NR in TkOSC1-RNAi lines or even to 16.1 mg/g NR in TkOSC-RNAi lines (FIG. 2(c)).

still significantly decreased to two thirds in TkOSC1-RNAi NR and to less than the half in TkOSC-RNAi NR (FIG. 2(b)).

In conclusion, it can be claimed that an OSC knockdown strongly reduces triterpenes/triterpenoids in dandelion root and NR.

The listing or discussion of a previously published document in this specification should not necessarily be taken as an acknowledgement that the document is part of the state of the art or is common general knowledge.

The invention illustratively described herein may suitably be practiced in the absence of any element or elements, limitation or limitations, not specifically disclosed herein. Thus, for example, the terms “comprising”, “including”, “containing”, etc. shall be read expansively and without

TABLE 4

Triterpene content in NR extracts from <i>T. koksaghyz</i> TkOSC1-/TkOSC-RNAi lines. Triterpene compounds were identified and quantified by GC/MS [mg/g NR dry weight], corresponding retention indices (RI) were determined in relation to a C8-C40 alkane calibration standard. The mean values represent independent extractions from wild-type plants (WtTk; n = 3), TkOSC1-RNAi plants (n = 3) and TkOSC-RNAi plants (n = 3) and are indicated with the corresponding standard deviations.				
	RI	TkWt	TkOSC1-RNAi	TkOSC-RNAi
precursors		1.02 ± 0.69	1.69 ± 0.92	5.12 ± 1.14
squalene	2843	0.59 ± 0.53	0.54 ± 0.39	0.58 ± 0.23
2,3-oxidosqualene	2965	0.44 ± 0.17	1.16 ± 0.53	4.55 ± 0.95
sterol precursors		0.00 ± 0.00	1.86 ± 0.23	3.51 ± 0.06
cycloartenol	3478	0.00 ± 0.00	1.48 ± 0.46	2.06 ± 0.11
24-methylene cycloartenol	3536	0.00 ± 0.00	0.37 ± 0.30	1.45 ± 0.06
sterols		1.25 ± 0.15	2.14 ± 0.63	1.45 ± 0.09
campesterol	3313	0.15 ± 0.08	0.23 ± 0.07	0.16 ± 0.02
stigmasterol	3343	0.43 ± 0.04	0.59 ± 0.19	0.51 ± 0.05
sitosterol	3405	0.67 ± 0.14	1.32 ± 0.38	0.78 ± 0.02
pentacyclic triterpenes		56.60 ± 0.62	33.37 ± 3.99	16.05 ± 2.86
taraxerol and n.i.	3448	1.84 ± 0.14	3.65 ± 0.85	1.93 ± 0.21
β -amyirin	3465	6.14 ± 0.60	3.69 ± 0.95	1.83 ± 0.32
lupeol	3510	1.85 ± 0.91	3.53 ± 0.29	1.30 ± 0.24
α -amyirin and lup-19(21)-en-3-ol	3516	4.74 ± 0.34	7.89 ± 2.03	6.46 ± 1.40
β -amyirin acetate	3562	3.17 ± 0.47	0.47 ± 0.27	0.00 ± 0.00
TkOSC3-product-1	3550	0.08 ± 0.01	0.28 ± 0.12	0.17 ± 0.02
TkOSC3-product-2	3590	0.62 ± 0.06	1.49 ± 0.43	1.43 ± 0.24
n.i.1	3603	8.27 ± 0.07	4.54 ± 0.97	1.09 ± 0.26
taraxasterol	3615	22.58 ± 1.68	7.00 ± 0.77	1.85 ± 0.59
triterpene acetate 1	3695	2.41 ± 0.40	0.54 ± 0.20	0.00 ± 0.00
triterpene acetate 2	3707	4.90 ± 1.14	0.33 ± 0.24	0.00 ± 0.00

In contrast to the reduced pentacyclic triterpene content, increasing amounts of triterpene precursors as well as sterol intermediates were again detected in RNAi-lines compared to wild-type, while equal amounts of sterols were measured in all three groups. However, total triterpenoid amount was

limitation. Additionally, the terms and expressions employed herein have been used as terms of description and not of limitation, and there is no intention in the use of such terms and expressions of excluding any equivalents of the features shown and described or portions thereof, but it is

recognized that various modifications are possible within the scope of the invention claimed. Thus, it should be understood that although the present invention has been specifically disclosed by exemplary embodiments and optional features, modifications and variations of the inventions disclosed herein are also considered to be within the scope of this invention.

The invention has been described broadly and generically herein. Each of the narrower species and sub-generic groupings falling within the generic disclosure also form part of the invention. This includes the generic description of the invention with a proviso or negative limitation removing any subject matter from the genus, regardless of whether or not the excised material is specifically recited herein.

Other embodiments are within the following claims. In addition, where features or aspects of the invention are described in terms of Markush groups, those skilled in the art will recognize that the invention is also thereby described in terms of any individual member or subgroup of members of the Markush group.

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Leu Tyr Ile Ser Gly Thr Leu Asp Thr Ile Leu Thr Gln Glu His Lys
145    150    155    160
Arg Glu Met Ile Arg Tyr Met Tyr Ile His Gln Asn Glu Asp Gly Gly
165    170    175
Trp Gly Phe Tyr Ile Ser Gly Arg Ser Thr Met Ile Gly Thr Ala Leu
180    185    190
Asn Tyr Val Gly Leu Arg Leu Leu Gly Glu Asp Asp Asn Asp Ala Ile
195    200    205
Ala Lys Gly Arg Lys Trp Ile Leu Asp His Gly Gly Ala Thr Ser Ile
210    215    220

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Pro 225	Ser	Trp	Gly	Lys	Val 230	Tyr	Leu	Ser	Val	Leu 235	Gly	Val	Tyr	Glu	Trp 240
Ala	Gly	Cys	Asn	Pro 245	Leu	Pro	Pro	Glu	Phe 250	Trp	Leu	Phe	Pro	Ser 255	Phe
Leu	Pro	Tyr	His 260	Pro	Ala	Lys	Met	Trp 265	Cys	Tyr	Cys	Arg	Thr	Thr	Tyr
Met	Pro	Met	Ser	Tyr	Leu	Tyr	Gly 280	Thr	Gly	Phe	Gln	Gly 285	Pro	Ile	Thr
Asp	Leu	Val	Lys	Ser	Leu	Arg 295	Lys	Glu	Ile	His	Val 300	Ile	Pro	Tyr	His
Gln 305	Ile	Asp	Trp	Asn	Lys 310	Gln	Arg	His	Asn	Cys 315	Cys	Lys	Glu	Asp	Leu 320
Tyr	Tyr	Pro	His 325	Thr	Tyr	Ile	Gln	Asp	Leu 330	Leu	Trp	Asp	Gly	Leu 335	His
Tyr	Phe	Ser	Glu 340	Pro	Leu	Val	Ser	Lys 345	Trp	Pro	Leu	Lys	Lys	Leu	Arg
Glu	Lys	Gly 355	Leu	Lys	Arg	Val	Leu	Asp 360	Leu	Met	Gln	Tyr 365	Asn	Ala	Glu
Glu	Gly 370	Arg	Tyr	Ile	Thr	Leu 375	Gly	Cys	Val	Glu	Lys 380	Ser	Leu	Gln	Met
Met 385	Cys	Phe	Ser	Ala	Leu 390	Asp	Pro	Asn	Gly 395	Ile	Asp	Phe	Lys	Arg	His 400
Leu	Ala	Arg	Val 405	Pro	Asp	Tyr	Leu	Trp	Val 410	Ala	Glu	Asp	Gly	Met 415	Lys
Met	Gln	Ser	Phe 420	Gly	Ser	Gln	Leu	Trp 425	Asp	Cys	Thr	Leu 430	Val	Thr	Gln
Ala	Ile 435	Ile	Ala	Ser	Asp	Met	Val 440	Glu	Glu	Tyr	Gly 445	Asp	Ser	Leu	Lys
Lys 450	Ala	Asn	Phe	Tyr	Leu	Lys 455	Glu	Ser	Gln	Ile	Lys 460	Gln	Asn	Pro	Lys
Gly 465	Asp	Phe	Glu	Asn	Met 470	Cys	Arg	Gln	Phe	Thr	Lys 475	Gly	Ala	Trp	Thr 480
Phe	Ser	Asp	Gln 485	Asp	Gln	Gly	Trp	Val	Val 490	Ser	Asp	Cys	Thr	Ala 495	Glu
Ala	Val	Lys	Cys 500	Leu	Leu	Ala	Leu	Ser	Gln 505	Met	Pro	Gln	Glu	Ile	Ser
Gly	Glu 515	Lys	Val	Asp	Val	Glu	Arg 520	Leu	Tyr	Asp	Ala	Ile 525	Asn	Val	Leu
Leu 530	Tyr	Leu	Gln	Ser	Pro	Glu 535	Thr	Gly	Gly	Phe	Ala 540	Ile	Trp	Glu	Ala
Pro 545	Val	Pro	Lys	Pro	Tyr 550	Leu	Glu	Lys	Leu	Asn 555	Pro	Ser	Glu	Leu	Phe 560
Ala	Asp	Ile	Val 565	Val	Glu	Arg	Glu	His	Val 570	Glu	Cys	Thr	Gly	Ser	Ile 575
Ile	Gln	Thr	Leu 580	Gln	Thr	Phe	Lys	Thr 585	Leu	His	Pro	Gly 590	His	Arg	Gln
Lys	Glu 595	Ile	Glu	Val	Ala	Ile	Glu 600	Lys	Gly	Ile	Arg 605	Phe	Leu	Glu	Asn
Arg 610	Gln	Gln	Glu	Asn	Gly	Ser 615	Trp	Tyr	Gly	Tyr	Trp 620	Gly	Ile	Cys	Tyr
Leu 625	Tyr	Gly	Thr	Tyr	Phe 630	Val	Leu	Gln	Gly	Leu 635	Val	Ala	Cys	Gly	Gln 640

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Thr Tyr Glu Asn Ser Glu Ala Val Arg Lys Ala Val Lys Phe Phe Leu
645 650 655

Ser Thr Gln Asn Ser Glu Gly Gly Trp Gly Glu Asn Phe Glu Ser Cys
660 665 670

Pro Gln Glu Lys Phe Ile Pro Leu Glu Gly Asn Arg Thr Asn Phe Val
675 680 685

Gln Thr Ser Trp Ala Met Leu Gly Leu Leu Cys Gly Gly Gln Ala Glu
690 695 700

Arg Asp Pro Thr Pro Leu His Lys Ala Ala Lys Leu Leu Ile Asn Gly
705 710 715 720

Gln Met Asp Asn Gly Asp Phe Pro Gln Gln Glu Ile Thr Gly Val Tyr
725 730 735

Met Lys Asn Cys Met Leu His Tyr Ala Glu Tyr Arg Asn Thr Phe Pro
740 745 750

Leu Trp Ala Leu Gly Glu Tyr Arg Lys Arg Val Trp Leu Ala Lys Gln
755 760 765

Glu Thr
770

<210> SEQ ID NO 5
<211> LENGTH: 2313
<212> TYPE: DNA
<213> ORGANISM: Taraxacum koksaghyz

<400> SEQUENCE: 5

atgtggaagc tgagaatagg tgaaaagaat ggaaagtcca acattggtga tggaaacggc	60
gatgaatatt tgtatagcac caacaacttt gtggggagac aaacatggga gttcgaccct	120
gatgcaggca cacaggaaga gcgtgatgaa gttgaaaggc ttcgagaaca atttctaatac	180
aataagaaga agcttgacat cagttgttgt gcagacttgc tcatgcaaaa tcagcttatt	240
aaggaaagcg gaattgatct tataagcgaa gctccagtga aacttggaac cgatgaggat	300
gtgaactata aagcggtaac aacagcagta agaaaggcag tcagactgaa ccgtgcaatc	360
caagcttggg atggctcattg gccggctgaa aatgcgggtc cgctcttttt cactcctcct	420
ttgataatcg tcttgtatat tagtgggtaca ttggataatg tccttacaca agaccacaag	480
aaggagatga ttcgttatat gtacatccat caaaatgaag atggaggggtg gggattctat	540
atatcaggca gaagcacaat gatagggaact gcgctgaatt atgtgggtct aagacttctt	600
ggagaagatg ataatgatgc aattgctaaa ggcgctaaat ggataactga ccacggtggc	660
gccacctcta ttccttcgtg gggaaagggtg tatctttcgg tacttgaggt gtacgaatgg	720
gcaggctgta accctcttcc tccagaattc tggctttttc cttccttttt cccttatcat	780
ccagcaaaaa tgtgggtgta ttgcagaacc acatacatgc caatgtcata cttgtatggt	840
agaggcatcc aagggcctat tacggacctc gttaagtctt tgagaaaaga gatccatgtc	900
attccttacc atcagattga ttggaataaa caacgacata attgttgcaa ggaggatctg	960
tactaccctc atacctacat ccaagatctc ttatgggatg gtcttcatta ttttagtgag	1020
ccacttatca caaaatggcc cttcgagaag ttaagagcga agggtttgaa aagagtattg	1080
gatctaatagc aatataatgc tgaggaagga cgttacataa ccctaggttc cgttgagaag	1140
tctttacaaa tgatgtgttt ttatgcccta gatccaaatg gaattgactt taaacgacac	1200
cttgctagag tocccogatta cttatgggtg gcggaagatg gtatgaaaat gcaaagtttt	1260
ggtagtcagt tatgggattg tactcttgta actcaagcta ttatcgctag tgatatgggt	1320

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gaagaatatg gagattcact taaaaaagcc aacttttact taaaagaatc acagatcaaa 1380
gaaaacccaa aaggcgattt tgaaaacatg tgtcgctcagt ttacaaaagg ggcatggact 1440
ttcactgata aagatcaagg gtgggtcgtc tcagattgca ctgctgaagc tgtgaagtgt 1500
ttattggcaa tgtcacaat gccacaagaa attgccggtg agaagggtga agttgagcga 1560
ttgtatgatg ccattaacgt ccttctttac ctacagagtc cagaaaccgg tggttttgca 1620
gtttgggagg taccggttcc aaaaccatat ttagagaaat tgaatccttc ggaacttttt 1680
gcagacataa ccgtcgagag agagcatgtt gaatgtacag gctccataat tcaagcgttg 1740
caaacattta aaactctgca tccagggcat cgtgaaaaag aaatagaagt tgctattgaa 1800
aaaggcatac acttttttga aaataggcaa caagaaaatg gttcttggtg tggtttttgg 1860
ggatatgtt atatttatgg aacgtatttt gtgctacaag gattggtagc ttgtggaaaa 1920
acatatgaaa attgtgaagc tgtcaggaaa gctgttaagt ttttctctc gacacaaaac 1980
tcagaaggtg gttggggaga gcactttgag tcatgccac aagagaaatt tataccttta 2040
gagggcaaca gaacacattt tgtgcatact tcatgggcca tgcttggtct tctatatggt 2100
ggacaggttg aaagagatcc gacaccatta cacaaggcat caaaattgct cataaatgga 2160
caaacggata atggagattt tctcaacag gaaataacag gagtgtacaa caagaattgc 2220
atgttacatt atgcagaata taggaacact ttcccgtat gggcacttgg agagtaccgc 2280
aagcgtgttt ggtagcaaa acaagaaacc taa 2313

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<210> SEQ ID NO 6

<211> LENGTH: 770

<212> TYPE: PRT

<213> ORGANISM: Taraxacum koksaghyz

<400> SEQUENCE: 6

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Met Trp Lys Leu Arg Ile Gly Glu Lys Asn Gly Lys Phe Asn Ile Gly
1           5           10           15
Asp Gly Asn Gly Asp Glu Tyr Leu Tyr Ser Thr Asn Asn Phe Val Gly
20          25          30
Arg Gln Thr Trp Glu Phe Asp Pro Asp Ala Gly Thr Gln Glu Glu Arg
35          40          45
Asp Glu Val Glu Arg Leu Arg Glu Gln Phe Leu Ile Asn Lys Lys Lys
50          55          60
Leu Asp Ile Ser Cys Cys Ala Asp Leu Leu Met Arg Asn Gln Leu Ile
65          70          75          80
Lys Glu Ser Gly Ile Asp Leu Ile Ser Glu Ala Pro Val Lys Leu Gly
85          90          95
His Asp Glu Asp Val Asn Tyr Lys Ala Val Thr Thr Ala Val Arg Lys
100         105         110
Ala Val Arg Leu Asn Arg Ala Ile Gln Ala Trp Asp Gly His Trp Pro
115         120         125
Ala Glu Asn Ala Gly Pro Leu Phe Phe Thr Pro Pro Leu Ile Ile Val
130         135         140
Leu Tyr Ile Ser Gly Thr Leu Asp Asn Val Leu Thr Gln Asp His Lys
145         150         155         160
Lys Glu Met Ile Arg Tyr Met Tyr Ile His Gln Asn Glu Asp Gly Gly
165         170         175
Trp Gly Phe Tyr Ile Ser Gly Arg Ser Thr Met Ile Gly Thr Ala Leu
180         185         190
Asn Tyr Val Gly Leu Arg Leu Leu Gly Glu Asp Asp Asn Asp Ala Ile

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195	200	205
Ala Lys Gly Arg Lys Trp Ile Leu Asp His Gly Gly Ala Thr Ser Ile 210 215 220		
Pro Ser Trp Gly Lys Val Tyr Leu Ser Val Leu Gly Val Tyr Glu Trp 225 230 235 240		
Ala Gly Cys Asn Pro Leu Pro Pro Glu Phe Trp Leu Phe Pro Ser Phe 245 250 255		
Phe Pro Tyr His Pro Ala Lys Met Trp Cys Tyr Cys Arg Thr Thr Tyr 260 265 270		
Met Pro Met Ser Tyr Leu Tyr Gly Arg Gly Ile Gln Gly Pro Ile Thr 275 280 285		
Asp Leu Val Lys Ser Leu Arg Lys Glu Ile His Val Ile Pro Tyr His 290 295 300		
Gln Ile Asp Trp Asn Lys Gln Arg His Asn Cys Cys Lys Glu Asp Leu 305 310 315 320		
Tyr Tyr Pro His Thr Tyr Ile Gln Asp Leu Leu Trp Asp Gly Leu His 325 330 335		
Tyr Phe Ser Glu Pro Leu Ile Thr Lys Trp Pro Phe Glu Lys Leu Arg 340 345 350		
Ala Lys Gly Leu Lys Arg Val Leu Asp Leu Met Gln Tyr Asn Ala Glu 355 360 365		
Glu Gly Arg Tyr Ile Thr Leu Gly Ser Val Glu Lys Ser Leu Gln Met 370 375 380		
Met Cys Phe Tyr Ala Leu Asp Pro Asn Gly Ile Asp Phe Lys Arg His 385 390 395 400		
Leu Ala Arg Val Pro Asp Tyr Leu Trp Val Ala Glu Asp Gly Met Lys 405 410 415		
Met Gln Ser Phe Gly Ser Gln Leu Trp Asp Cys Thr Leu Val Thr Gln 420 425 430		
Ala Ile Ile Ala Ser Asp Met Val Glu Glu Tyr Gly Asp Ser Leu Lys 435 440 445		
Lys Ala Asn Phe Tyr Leu Lys Glu Ser Gln Ile Lys Glu Asn Pro Lys 450 455 460		
Gly Asp Phe Glu Asn Met Cys Arg Gln Phe Thr Lys Gly Ala Trp Thr 465 470 475 480		
Phe Thr Asp Gln Asp Gln Gly Trp Val Val Ser Asp Cys Thr Ala Glu 485 490 495		
Ala Val Lys Cys Leu Leu Ala Met Ser Gln Met Pro Gln Glu Ile Ala 500 505 510		
Gly Glu Lys Val Glu Val Glu Arg Leu Tyr Asp Ala Ile Asn Val Leu 515 520 525		
Leu Tyr Leu Gln Ser Pro Glu Thr Gly Gly Phe Ala Val Trp Glu Val 530 535 540		
Pro Val Pro Lys Pro Tyr Leu Glu Lys Leu Asn Pro Ser Glu Leu Phe 545 550 555 560		
Ala Asp Ile Thr Val Glu Arg Glu His Val Glu Cys Thr Gly Ser Ile 565 570 575		
Ile Gln Ala Leu Gln Thr Phe Lys Thr Leu His Pro Gly His Arg Glu 580 585 590		
Lys Glu Ile Glu Val Ala Ile Glu Lys Gly Ile His Phe Leu Glu Asn 595 600 605		
Arg Gln Gln Glu Asn Gly Ser Trp Tyr Gly Phe Trp Gly Ile Cys Tyr 610 615 620		

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Ile Tyr Gly Thr Tyr Phe Val Leu Gln Gly Leu Val Ala Cys Gly Lys
 625 630 635 640

Thr Tyr Glu Asn Cys Glu Ala Val Arg Lys Ala Val Lys Phe Phe Leu
 645 650 655

Ser Thr Gln Asn Ser Glu Gly Gly Trp Gly Glu His Phe Glu Ser Cys
 660 665 670

Pro Gln Glu Lys Phe Ile Pro Leu Glu Gly Asn Arg Thr His Phe Val
 675 680 685

His Thr Ser Trp Ala Met Leu Gly Leu Leu Tyr Gly Gly Gln Val Glu
 690 695 700

Arg Asp Pro Thr Pro Leu His Lys Ala Ser Lys Leu Leu Ile Asn Gly
 705 710 715 720

Gln Thr Asp Asn Gly Asp Phe Pro Gln Gln Glu Ile Thr Gly Val Tyr
 725 730 735

Asn Lys Asn Cys Met Leu His Tyr Ala Glu Tyr Arg Asn Thr Phe Pro
 740 745 750

Leu Trp Ala Leu Gly Glu Tyr Arg Lys Arg Val Trp Leu Ala Lys Gln
 755 760 765

Glu Thr
 770

<210> SEQ ID NO 7

<211> LENGTH: 2298

<212> TYPE: DNA

<213> ORGANISM: Taraxacum koksaghyz

<400> SEQUENCE: 7

atgtgggagt taaagatagc ggaaggggat ggcccttatt tgtacagtac caacaacttt	60
gttggttagac aattttggga gtttaacca gatgctggaa ctccagaaga gaaagaggaa	120
atcgaaaagg ttgcacagaa atttaagat aatcgtaaaa aaggaggatt tcatgcttgt	180
ggcgatctac tcatgctgat gcagctaag aaagaaaatg caatcgatct tacgagcata	240
cttcgggtga gaataagtga gggcgagcaa gttaaattatg aagcaacaac aattgcagta	300
cgaaaagcag ttcgactaca tcgtggaatc caagcgaaag atggacattg gcctgctgaa	360
aatgctggtc ccttgttttt taccctcca cttgttattg ctttatatat cagtggtagc	420
attaataaccg tottaagtga agaacacaaa aaagagatga tacgatactt ctacaatcat	480
caaaatgaag atgggggatg ggggttctat atcgagggtc atagtacaat gatcggatca	540
gtactgagtt atgtagccct acgtctactc ggagaaggag aagatgatgg agatggtgca	600
atcgcccag cagcaagtg gatacttgac catggtggtg ctgcctctat tccctcatgg	660
ggaaagggtt atctttcggg gcttgagtg tacgaatggg agggctgcaa tccattgcca	720
ccagaatttt ggctctttcc ttgcacattt ccttttcac ctgctgagat gtggtgetac	780
tgccggacaa cctacatgcc tatgtcatat ttatatggga aaagaatcca aggacctctc	840
acccctctcg tttctcatt gagaaaagaa attcatthaa ccccttttga ggatattaat	900
tgaataaac aaaggaataa ttgttgtaag aaggacttct actatccaca ttcatttctt	960
caagatgcat tgtggcatag ccttcactac cttactgagc ctgtectcaa gtattggcca	1020
ttttcaaac tacgagggag atctcttgat agagttgtag aactaatgag ctatgaatct	1080
gaagagacca gatatatgac cataggggtg gtcgaaaaaa gtctacaaat gatgtgttgg	1140
tgggcagaaa atcctaagtg tgatgagttc aaatatcatc tagcaagagt tccagattac	1200

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ttatggattg cagaagatgg aatgacaatg catagtttcg gtagtcaagt gtgggattgt 1260
tctcttgcaa ctcaagcaat tatagcaagt aacatggtag aggaatatga tgattgtctg 1320
gaaaaggcac acttctatctt aagagaatca caggtaaaag aaaatccttc aggagatttc 1380
actcgcatgt gtcggcagtt cactaaggga tcgtggacct tctctgatca agatcatgga 1440
tggacagtct ctgactgtac agctgaagca ctaaagtgtc tattgttact atcaaacatg 1500
cctaaaaata tcgctggaga gaaagatgac actgccagac tatatgaagc agttaatgtg 1560
cttctttaca tgcaaagtcc tgtaagtggg ggatttgctg tttgggagcc accaattcca 1620
aaaccgttct tacagcttct taatccttca gagatttttg cagatatcgt cgttgagaaa 1680
gagcatgtgg agacaacatc ttccattatc ggagctctga tcgagttcaa acgggtccat 1740
ccaaaccaca gaaaagaaga aatagaatac tcaatctcga atggaatacg ttatctcgag 1800
gaaacacaat ggcgatgatg ttcatggtat ggttattggg gagtatgctt catatatgga 1860
accttctttg ctctgagagc attaatgact gccggaaaag catataagaa taacgaagca 1920
gcctgtaaag gtgtcaaatt cttattgtcg atacaaaatg aagagggggg ttggggagag 1980
agcctactat cttgccctac tgaggatat acaccgttg atggaaatca gacaaatttg 2040
gtgcaaacat cgtgggctat gcttggccta ctttttggtg ggcaggtgga tagagatccc 2100
acgcctttac atagagcagc aaagctgttg attaatgctc agatggataa tggagatttt 2160
cctcaacagg aaattaccgg agtctacatg aagaattgct tgctgcttta cgcacaatac 2220
agaaacattt ttccactatg ggcacttggg gaataccgta aacgtgtttg gagcataaaa 2280
aagggtattc aaaattaa 2298

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<210> SEQ ID NO 8

<211> LENGTH: 765

<212> TYPE: PRT

<213> ORGANISM: Taraxacum koksaghyz

<400> SEQUENCE: 8

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Met Trp Glu Leu Lys Ile Ala Glu Gly Asp Gly Pro Tyr Leu Tyr Ser
1           5           10          15
Thr Asn Asn Phe Val Gly Arg Gln Phe Trp Glu Phe Asn Pro Asp Ala
20          25          30
Gly Thr Pro Glu Glu Lys Glu Glu Ile Glu Lys Val Arg Gln Lys Phe
35          40          45
Lys Asp Asn Arg Lys Lys Gly Gly Phe His Ala Cys Gly Asp Leu Leu
50          55          60
Met Arg Met Gln Leu Met Lys Glu Asn Ala Ile Asp Leu Thr Ser Ile
65          70          75          80
Leu Pro Val Arg Ile Ser Glu Gly Glu Gln Val Asn Tyr Glu Ala Thr
85          90          95
Thr Ile Ala Val Arg Lys Ala Val Arg Leu His Arg Gly Ile Gln Ala
100         105         110
Lys Asp Gly His Trp Pro Ala Glu Asn Ala Gly Pro Leu Phe Phe Thr
115         120         125
Pro Pro Leu Val Ile Ala Leu Tyr Ile Ser Gly Thr Ile Asn Thr Val
130         135         140
Leu Ser Glu Glu His Lys Lys Glu Met Ile Arg Tyr Phe Tyr Asn His
145         150         155         160
Gln Asn Glu Asp Gly Gly Trp Gly Phe Tyr Ile Glu Gly His Ser Thr
165         170         175

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Met	Ile	Gly	Ser 180	Val	Leu	Ser	Tyr	Val 185	Ala	Leu	Arg	Leu	Gly	Glu
Gly	Glu	Asp 195	Asp	Gly	Asp	Gly	Ala 200	Ile	Ala	Arg	Ala	Arg 205	Lys	Ile
Leu	Asp	His	Gly	Gly	Ala	Ala 215	Ser	Ile	Pro	Ser	Trp 220	Gly	Lys	Tyr
Leu 225	Ser	Val	Leu	Gly	Val 230	Tyr	Glu	Trp	Glu	Gly 235	Cys	Asn	Pro	Pro 240
Pro	Glu	Phe	Trp	Leu 245	Phe	Pro	Ser	Thr	Phe 250	Pro	Phe	His	Pro	Glu
Met	Trp	Cys	Tyr 260	Cys	Arg	Thr	Thr	Tyr 265	Met	Pro	Met	Ser	Tyr 270	Tyr
Gly	Lys	Arg 275	Ile	Gln	Gly	Pro	Leu 280	Thr	Pro	Leu	Val	Ser 285	Ser	Arg
Lys	Glu	Ile	His	Leu	Thr	Pro 295	Phe	Glu	Asp	Ile	Asn 300	Trp	Asn	Gln
Arg 305	Asn	Asn	Cys	Cys	Lys 310	Lys	Asp	Phe	Tyr	Tyr 315	Pro	His	Ser	Leu
Gln	Asp	Ala	Leu	Trp 325	His	Ser	Leu	His	Tyr 330	Leu	Thr	Glu	Pro	Leu
Lys	Tyr	Trp	Pro 340	Phe	Ser	Lys	Leu	Arg 345	Gly	Arg	Ser	Leu 350	Asp	Val
Val	Glu	Leu	Met 355	Arg	Tyr	Glu	Ser 360	Glu	Glu	Thr	Arg	Tyr 365	Met	Ile
Gly	Cys 370	Val	Glu	Lys	Ser	Leu 375	Gln	Met	Met	Cys	Trp 380	Trp	Ala	Asn
Pro 385	Asn	Gly	Asp	Glu	Phe 390	Lys	Tyr	His	Leu	Ala 395	Arg	Val	Pro	Tyr
Leu	Trp	Ile	Ala	Glu 405	Asp	Gly	Met	Thr	Met 410	His	Ser	Phe	Gly	Gln
Val	Trp	Asp 420	Cys	Ser	Leu	Ala	Thr	Gln 425	Ala	Ile	Ile	Ala	Ser 430	Met
Val	Glu	Glu 435	Tyr	Asp	Asp	Cys	Leu 440	Glu	Lys	Ala	His	Phe 445	Tyr	Arg
Glu	Ser	Gln	Val	Lys	Glu	Asn 455	Pro	Ser	Gly	Asp	Phe 460	Thr	Arg	Cys
Arg 465	Gln	Phe	Thr	Lys	Gly 470	Ser	Trp	Thr	Phe	Ser 475	Asp	Gln	Asp	Gly
Trp	Thr	Val	Ser	Asp 485	Cys	Thr	Ala	Glu	Ala 490	Leu	Lys	Cys	Leu 495	Leu
Leu	Ser	Asn	Met 500	Pro	Lys	Asn	Ile	Ala 505	Gly	Glu	Lys	Asp 510	Thr	Ala
Arg	Leu	Tyr 515	Glu	Ala	Val	Asn	Val 520	Leu	Leu	Tyr	Met	Gln 525	Ser	Val
Ser	Gly 530	Gly	Phe	Ala	Val	Trp 535	Glu	Pro	Pro	Ile	Pro 540	Lys	Pro	Leu
Gln 545	Leu	Leu	Asn	Pro	Ser 550	Glu	Ile	Phe	Ala	Asp 555	Ile	Val	Val	Lys
Glu	His	Val	Glu	Thr 565	Thr	Ser	Ser	Ile	Ile	Gly 570	Ala	Leu	Ile	Phe
Lys	Arg	Val	His 580	Pro	Asn	His	Arg	Lys 585	Glu	Glu	Ile	Glu	Tyr	Ile
Ser	Asn	Gly	Ile	Arg	Tyr	Leu	Glu	Glu	Thr	Gln	Trp	His	Asp	Ser

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595	600	605
Trp Tyr Gly Tyr Trp Gly Val Cys Phe Ile Tyr Gly Thr Phe Phe Ala 610 615 620		
Leu Arg Ala Leu Ser Thr Ala Gly Lys Thr Tyr Lys Asn Asn Glu Ala 625 630 635 640		
Ala Cys Lys Gly Val Lys Phe Leu Leu Ser Ile Gln Asn Glu Glu Gly 645 650 655		
Gly Trp Gly Glu Ser Leu Leu Ser Cys Pro Thr Glu Val Tyr Thr Pro 660 665 670		
Leu Asp Gly Asn Gln Thr Asn Leu Val Gln Thr Ser Trp Ala Met Leu 675 680 685		
Gly Leu Leu Phe Gly Gly Gln Val Asp Arg Asp Pro Thr Pro Leu His 690 695 700		
Arg Ala Ala Lys Leu Ile Asn Ala Gln Met Asp Asn Gly Asp Phe 705 710 715 720		
Pro Gln Gln Glu Ile Thr Gly Val Tyr Met Lys Asn Cys Leu Leu Leu 725 730 735		
Tyr Ala Gln Tyr Arg Asn Ile Phe Pro Leu Trp Ala Leu Gly Glu Tyr 740 745 750		
Arg Lys Arg Val Trp Ser Ile Lys Lys Gly Ile Gln Asn 755 760 765		

<210> SEQ ID NO 9

<211> LENGTH: 2286

<212> TYPE: DNA

<213> ORGANISM: Taraxacum koksaghyz

<400> SEQUENCE: 9

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atgtggagggt taagaatcgg gcaagggaac aaagacgata cttatttggt cagcacaat      60
aatTTtTgtg gaagacaaat atgggagtat gacgaaaact ataaagcaac tccagaagag      120
ttggaagaag ttaacaagc cccgatcttct ttctggaaca accgtcaca gcaaaggcca      180
tgtgatgata gtctttggcg ttttcagttt ctaagggaga agaatttcaa acaacaata      240
ccaagagaga ttatcgtgga tgaaagttag ataaagtatg aaactgtcga caatgcattg      300
aaaagggcag ttcgttattg gacggctcta caagcaagcg acggccactg gccttcagcc      360
aacaacggct gccactatTT tacaccaccc atcgtcatgt gtttTtatat aacaggacat      420
cttgatacat tcttctcgac ccaagataaa gaagaaatgc ttcgttatat atactgtcac      480
cagaatgaag atggaggatg gggatttcac cttgaaggcc atagcatcat gttttgcacc      540
gCGgtcaatt atatatgcat gcgaatgctt ggagaaggac ccgatggTgg cctcgatggT      600
gcgtgttcta gagcacgtaa gtggattctt gatcatggca gtgtcaccgc catccctca      660
tggggaaaga cttggttatc ggcacttggg ctatatgagt ggtctggaag caatccaatg      720
cctccagagt tttggcttct cccatcgttc cttcctatat cgccaggaaa gatatggTgt      780
tattgtcgga tgatttatat gccaatgtct tatttatatg ggaagagatt cattggTccg      840
atcactcctc taattctgca gttgagaaaa gagctttatt ctatacccta tcatgaagtc      900
aattggagga aagtaagaca tctttgcgcg aaggaggaca tatattatcc tcacccttg      960
ctacaagatc tctcttgTga tactcttcat ttggcagTgg agcctatcct aactcgtTgg      1020
ccattcaaga gcatgattcg agagaaagct cttaaaacca caatgagaca catacattat      1080
gaagacgaga atagtcgcta cattaccatc ggttgcgttg aaaaagcact atgtatgctt      1140
gcatgttggg tggaggatcc tcatggggat tatttcaaga aacacctaag cagaatccgc      1200

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gatatgattt gggtaacaaga agatggcatg acggttcaga gttttgtag ccagcaatgg 1260
gattcttctt tgagcgttct tgccttgata gattgcaata tgatagatga aactggatca 1320
acactaaaaa aaggacatga attcatcaaa aactctcagg ttcgggacaa cccttcaggc 1380
gacttcaaaa gcatgtatcg tcacatttct aaggagggtt ggacttattc cgaccaagat 1440
cacggatggc aagtttctga ttgcaactgct cacggggttaa tgtcttgccct ccttttgtea 1500
aagatgccac tggaaatagt tggagagaag atggaaactg agcgtttatt cgattgcata 1560
aatttactgc tttcactaca gtacaaaaat ggaggtttct cgggttgga accatgcagg 1620
gctccaaaat ggttgagat gcttaaccct tcggaatgt ttgcagacat tatgatcgaa 1680
attcaatacg ttgagtgtac atcatcgcg ttgcaggctt tgatattgtt taaaaagtta 1740
tatcctgaac atcgagtgca agaagttgca aattgcattt ctaatgctat tcggtttctt 1800
gaagacacac aatggccgga tggatcatgg tatggagagt ggggtgtatg cttcacatat 1860
gctacgtggt ttgtacaaa aggggttagct gcagctggca agacatatga gcaatctccg 1920
acaattcgta aggccactga gtttttactg aaacatcaac aagaagatgg tggttgggga 1980
gaaagctatc tttcttgccc taacttgaa tttacatctt tagaagaaag caggtcgaac 2040
gtggtgcaaa catcatggtg cttgatgagt ctaatccagt gtggacaggc agagagggat 2100
ttgacaccga ttcacgtgg agcaaagttt ctcataaatt ctcaaaggga agatggtgat 2160
ttccccaaa aggaaatcac ggggtctttt cgtaaaaatt gcatgttaca ctatgcttgc 2220
catagaaaca tatttccaat gtgggctctt gcagaataca agaatcgggt gtctaaaatc 2280
atctaa 2286

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<210> SEQ ID NO 10
<211> LENGTH: 761
<212> TYPE: PRT
<213> ORGANISM: Taraxacum koksaghyz

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<400> SEQUENCE: 10

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Met Trp Arg Leu Arg Ile Gly Gln Gly Asn Lys Asp Asp Pro Tyr Leu
1          5          10          15

Phe Ser Thr Asn Asn Phe Val Gly Arg Gln Ile Trp Glu Tyr Asp Glu
20        25        30

Asn Tyr Lys Ala Thr Pro Glu Glu Leu Glu Glu Val Lys Gln Ala Arg
35        40        45

Ser Ser Phe Trp Asn Asn Arg His Lys Gln Arg Pro Cys Asp Asp Arg
50        55        60

Leu Trp Arg Phe Gln Phe Leu Arg Glu Lys Asn Phe Lys Gln Thr Ile
65        70        75        80

Pro Arg Glu Ile Ile Val Asp Glu Ser Glu Ile Lys Tyr Glu Thr Val
85        90        95

Asp Asn Ala Leu Lys Arg Ala Val Arg Tyr Trp Thr Ala Leu Gln Ala
100       105       110

Ser Asp Gly His Trp Pro Ser Ala Asn Asn Gly Cys His Tyr Phe Thr
115       120       125

Pro Pro Ile Val Met Cys Leu Tyr Ile Thr Gly His Leu Asp Thr Phe
130       135       140

Phe Ser Thr Gln Asp Lys Glu Glu Met Leu Arg Tyr Ile Tyr Cys His
145       150       155       160

Gln Asn Glu Asp Gly Gly Trp Gly Phe His Leu Glu Gly His Ser Ile
165       170       175

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Met	Phe	Cys	Thr	Ala	Val	Asn	Tyr	Ile	Cys	Met	Arg	Met	Leu	Gly	Glu	
			180					185					190			
Gly	Pro	Asp	Gly	Gly	Leu	Asp	Gly	Ala	Cys	Ser	Arg	Ala	Arg	Lys	Trp	
		195					200					205				
Ile	Leu	Asp	His	Gly	Ser	Val	Thr	Ala	Ile	Pro	Ser	Trp	Gly	Lys	Thr	
	210					215					220					
Trp	Leu	Ser	Ala	Leu	Gly	Leu	Tyr	Glu	Trp	Ser	Gly	Ser	Asn	Pro	Met	
225					230					235					240	
Pro	Pro	Glu	Phe	Trp	Leu	Leu	Pro	Ser	Phe	Leu	Pro	Ile	Ser	Pro	Gly	
			245						250					255		
Lys	Ile	Trp	Cys	Tyr	Cys	Arg	Met	Ile	Tyr	Met	Pro	Met	Ser	Tyr	Leu	
		260						265					270			
Tyr	Gly	Lys	Arg	Phe	Ile	Gly	Pro	Ile	Thr	Pro	Leu	Ile	Leu	Gln	Leu	
		275					280					285				
Arg	Lys	Glu	Leu	Tyr	Ser	Ile	Pro	Tyr	His	Glu	Val	Asn	Trp	Arg	Lys	
	290					295					300					
Val	Arg	His	Leu	Cys	Ala	Lys	Glu	Asp	Ile	Tyr	Tyr	Pro	His	Pro	Trp	
305					310					315					320	
Leu	Gln	Asp	Leu	Ser	Trp	Asp	Thr	Leu	His	Leu	Ala	Val	Glu	Pro	Ile	
			325					330						335		
Leu	Thr	Arg	Trp	Pro	Phe	Lys	Ser	Met	Ile	Arg	Glu	Lys	Ala	Leu	Lys	
			340					345					350			
Thr	Thr	Met	Arg	His	Ile	His	Tyr	Glu	Asp	Glu	Asn	Ser	Arg	Tyr	Ile	
		355					360					365				
Thr	Ile	Gly	Cys	Val	Glu	Lys	Ala	Leu	Cys	Met	Leu	Ala	Cys	Trp	Val	
	370					375					380					
Glu	Asp	Pro	His	Gly	Asp	Tyr	Phe	Lys	Lys	His	Leu	Ser	Arg	Ile	Arg	
385					390					395					400	
Asp	Met	Ile	Trp	Val	Gln	Glu	Asp	Gly	Met	Thr	Val	Gln	Ser	Phe	Gly	
			405					410						415		
Ser	Gln	Gln	Trp	Asp	Ser	Ser	Leu	Ser	Val	Leu	Ala	Leu	Ile	Asp	Cys	
			420					425					430			
Asn	Met	Ile	Asp	Glu	Thr	Gly	Ser	Thr	Leu	Lys	Lys	Gly	His	Glu	Phe	
		435					440					445				
Ile	Lys	Asn	Ser	Gln	Val	Arg	Asp	Asn	Pro	Ser	Gly	Asp	Phe	Lys	Ser	
	450					455					460					
Met	Tyr	Arg	His	Ile	Ser	Lys	Gly	Gly	Trp	Thr	Tyr	Ser	Asp	Gln	Asp	
465					470					475					480	
His	Gly	Trp	Gln	Val	Ser	Asp	Cys	Thr	Ala	His	Gly	Leu	Met	Ser	Cys	
			485					490						495		
Leu	Leu	Leu	Ser	Lys	Met	Pro	Leu	Glu	Ile	Val	Gly	Glu	Lys	Met	Glu	
			500					505					510			
Thr	Glu	Arg	Leu	Phe	Asp	Cys	Ile	Asn	Leu	Leu	Leu	Ser	Leu	Gln	Tyr	
			515				520					525				
Lys	Asn	Gly	Gly	Phe	Ser	Gly	Trp	Glu	Pro	Ala	Gly	Ala	Pro	Lys	Trp	
	530					535					540					
Leu	Glu	Met	Leu	Asn	Pro	Ser	Glu	Met	Phe	Ala	Asp	Ile	Met	Ile	Glu	
545					550					555					560	
Ile	Gln	Tyr	Val	Glu	Cys	Thr	Ser	Ser	Ala	Leu	Gln	Ala	Leu	Ile	Leu	
			565						570					575		
Phe	Lys	Lys	Leu	Tyr	Pro	Glu	His	Arg	Ser	Gln	Glu	Val	Ala	Asn	Cys	
			580					585						590		

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Ile Ser Asn Ala Ile Arg Phe Leu Glu Asp Thr Gln Trp Pro Asp Gly
595 600 605

Ser Trp Tyr Gly Glu Trp Gly Val Cys Phe Thr Tyr Ala Thr Trp Phe
610 615 620

Ala Thr Lys Gly Leu Ala Ala Ala Gly Lys Thr Tyr Glu Gln Ser Pro
625 630 635 640

Thr Ile Arg Lys Ala Thr Glu Phe Leu Leu Lys His Gln Gln Glu Asp
645 650 655

Gly Gly Trp Gly Glu Ser Tyr Leu Ser Cys Pro Asn Leu Glu Phe Thr
660 665 670

Ser Leu Glu Glu Ser Arg Ser Asn Val Val Gln Thr Ser Trp Cys Leu
675 680 685

Met Ser Leu Ile Gln Cys Gly Gln Ala Glu Arg Asp Leu Thr Pro Ile
690 695 700

His Arg Gly Ala Lys Phe Leu Ile Asn Ser Gln Met Glu Asp Gly Asp
705 710 715 720

Phe Pro Gln Lys Glu Ile Thr Gly Ser Phe Arg Lys Asn Cys Met Leu
725 730 735

His Tyr Ala Cys His Arg Asn Ile Phe Pro Met Trp Ala Leu Ala Glu
740 745 750

Tyr Lys Asn Arg Val Ser Lys Ile Ile
755 760

<210> SEQ ID NO 11

<211> LENGTH: 2283

<212> TYPE: DNA

<213> ORGANISM: Taraxacum koksaghyz

<400> SEQUENCE: 11

atgtggaaat taaagatagc agaaggaggc aatgatccat atctocactc cacaaataac	60
ttcattggga ggcaaatatg ggagtttgac tcgaaccatg gcaccccaga agaccgagcc	120
gaggccgaac atgctcgaat caattttctgg aaccttagac atcgggtcaa gccagtagt	180
gatgtccttt ggcgcatgca gtttttaaaa gagaacaat tcaacaaac gatagcccaa	240
gtgaaaatag aggaatttga agagatatct tatgaaaaag ctacagtcac attgaagagg	300
tgtgtgaacc tttttgcagc tttgcaggcc gatgatggtc actggcctgc tgaaaatgcg	360
ggtccactat atttcatgca gccgttggtt atttgtttgt atatcactgg tcatcttgac	420
attgttttcc cagaagaaca ccgaaaagaa atcctacatt atatgtattg ccatcagaat	480
gaagacgggtg gttggggatt tcacatcgag ggtcatagca gcatgttctg caccacccta	540
agttacattt gtatgcggct tttaggagaa ggggtggatg gtggtttgaa tgggtgcatgc	600
accaaagccc gcaaatggat tctggatcat ggctctgtca ctgccattgc atcatggggc	660
aaaacatggc tttccatact tggagtgtgt gagtgggcag gaaccaaccc aatgccccca	720
gagttctgga tctcctcttc tttccttcct atgtatccag cgaaactatg gtgctattgt	780
cgactgggat acatgccaat gtcataccta tacggaaaaa gattcgttgg tcttattact	840
cctttaattt tacagcttag agacgaactt tacttacaac cttacaatga aatcaactgg	900
aagagcatac gacattttgtg tgctaaggaa gatctatact atcctcatcc attactacaa	960
gatttaattgt gggatagttt atatatttgc attgagcctc tattaaatcg ctggccatta	1020
aataaattgc gtcagaaagc acttgaagaa acaatgaaac atattcatta tgaagatgag	1080
aatagccgtt atatcaccat tggttcagtg gtaaagggtg tgtgtatgct tgcattgttg	1140

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gttgaagaac caaatggcgt atgtttcaag aagcatcttg ctggaatccc agattatata 1200
tggattgctg aagacggaat gaaatgcag agttttggca gtcaagggtg ggatgctagt 1260
ttagccatcc aagctctttt ggctactgat cttacagatg aaatcgggtc tactttgatg 1320
aaaggacaca aatttgtaa agcttcacag gttaaagata atccttctgg atactttaa 1380
aacatgcaca gacatatttc caaagggtca tggagctttt cagatcaaga ccatggatgg 1440
caaggttccg acaccactgc tgaagcattg aagtgttgtc tcctattctc aacaatgcca 1500
ccagaaattg ttggtgagaa aatgaaacct gagcaactga atgatgcggt caatgtgata 1560
ctttccctac agagcaaaaa cggcgggtta gcatcatggg aaccagcagg atcatcagaa 1620
tgggttgaga ttttcaatcc tacagagttc ttgctgaca tcgtcattga gcacgagtat 1680
gttgagtgc catcatcagc aattcaggca atcgttttat ttaacaagtt ataccgcaa 1740
catagaaaaa aagagataga aacttttctt acaggtgcaa gtggatacct agagaaatta 1800
cagacaaaag atggttcattg gtacggaaac tggggcgtgt gttttacata tggatcatgg 1860
tttggatttg gtgattgac agctgtcggg aaaacatttg aaaattgtca agcaattcaa 1920
aaagccggtt aatttctggt ggaacacaaa ctagaagatg gcggttgggg agaaagctac 1980
aaatcctgtc cagaaaaaat atacataccc ctagaaggag gccggtcaaa tttagtgcac 2040
acttcattgg ctatgattgg actaattcag tctcaatgga tggagagaga ctcaacaccc 2100
atacaciaag cagccagggt attgatcaat tcacagttag aaaacggtga ttttccacaa 2160
caggaaataa cgggagtatt catgaagaac tgcattgtac actacgcact ctacaggaac 2220
atatacccaa tgtgggccct agccgattat cgcaaaaagg tggtgtccac gtgtaaaaaa 2280
taa 2283

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<210> SEQ ID NO 12
<211> LENGTH: 760
<212> TYPE: PRT
<213> ORGANISM: Taraxacum koksaghyz

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<400> SEQUENCE: 12

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Met Trp Lys Leu Lys Ile Ala Glu Gly Gly Asn Asp Pro Tyr Leu His
1           5           10           15
Ser Thr Asn Asn Phe Ile Gly Arg Gln Ile Trp Glu Phe Asp Ser Asn
20          25          30
His Gly Thr Pro Glu Asp Arg Ala Glu Ala Glu His Ala Arg Ile Asn
35          40          45
Phe Trp Asn Leu Arg His Arg Val Lys Pro Ser Ser Asp Val Leu Trp
50          55          60
Arg Met Gln Phe Leu Lys Glu Lys Gln Phe Lys Gln Thr Ile Ala Gln
65          70          75          80
Val Lys Ile Glu Glu Phe Glu Glu Ile Ser Tyr Glu Lys Ala Thr Val
85          90          95
Thr Leu Lys Arg Cys Val Asn Leu Phe Ala Ala Leu Gln Ala Asp Asp
100         105         110
Gly His Trp Pro Ala Glu Asn Ala Gly Pro Leu Tyr Phe Met Gln Pro
115         120         125
Leu Val Ile Cys Leu Tyr Ile Thr Gly His Leu Asp Ile Val Phe Pro
130         135         140
Glu Glu His Arg Lys Glu Ile Leu His Tyr Met Tyr Cys His Gln Asn
145         150         155         160
Glu Asp Gly Gly Trp Gly Phe His Ile Glu Gly His Ser Ser Met Phe

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165							170					175				
Cys	Thr	Thr	Leu	Ser	Tyr	Ile	Cys	Met	Arg	Leu	Leu	Gly	Glu	Gly	Val	
180							185					190				
Asp	Gly	Gly	Leu	Asn	Gly	Ala	Cys	Thr	Lys	Ala	Arg	Lys	Trp	Ile	Leu	
195							200					205				
Asp	His	Gly	Ser	Val	Thr	Ala	Ile	Ala	Ser	Trp	Gly	Lys	Thr	Trp	Leu	
210							215					220				
Ser	Ile	Leu	Gly	Val	Cys	Glu	Trp	Ala	Gly	Thr	Asn	Pro	Met	Pro	Pro	
225							230					235				
Glu	Phe	Trp	Ile	Leu	Pro	Ser	Phe	Leu	Pro	Met	Tyr	Pro	Ala	Lys	Leu	
245							250					255				
Trp	Cys	Tyr	Cys	Arg	Leu	Val	Tyr	Met	Pro	Met	Ser	Tyr	Leu	Tyr	Gly	
260							265					270				
Lys	Arg	Phe	Val	Gly	Pro	Ile	Thr	Pro	Leu	Ile	Leu	Gln	Leu	Arg	Asp	
275							280					285				
Glu	Leu	Tyr	Leu	Gln	Pro	Tyr	Asn	Glu	Ile	Asn	Trp	Lys	Ser	Ile	Arg	
290							295					300				
His	Leu	Cys	Ala	Lys	Glu	Asp	Leu	Tyr	Tyr	Pro	His	Pro	Leu	Leu	Gln	
305							310					315				
Asp	Leu	Met	Trp	Asp	Ser	Leu	Tyr	Ile	Cys	Ile	Glu	Pro	Leu	Leu	Asn	
325							330					335				
Arg	Trp	Pro	Leu	Asn	Lys	Leu	Arg	Gln	Lys	Ala	Leu	Glu	Thr	Thr	Met	
340							345					350				
Lys	His	Ile	His	Tyr	Glu	Asp	Glu	Asn	Ser	Arg	Tyr	Ile	Thr	Ile	Gly	
355							360					365				
Ser	Val	Val	Lys	Val	Leu	Cys	Met	Leu	Ala	Cys	Trp	Val	Glu	Glu	Pro	
370							375					380				
Asn	Gly	Val	Cys	Phe	Lys	Lys	His	Leu	Ala	Arg	Ile	Pro	Asp	Tyr	Ile	
385							390					395				
Trp	Ile	Ala	Glu	Asp	Gly	Met	Lys	Met	Gln	Ser	Phe	Gly	Ser	Gln	Gly	
405							410					415				
Trp	Asp	Ala	Ser	Leu	Ala	Ile	Gln	Ala	Leu	Leu	Ala	Thr	Asp	Leu	Thr	
420							425					430				
Asp	Glu	Ile	Gly	Ser	Thr	Leu	Met	Lys	Gly	His	Lys	Phe	Val	Lys	Ala	
435							440					445				
Ser	Gln	Val	Lys	Asp	Asn	Pro	Ser	Gly	Tyr	Phe	Lys	Asn	Met	His	Arg	
450							455					460				
His	Ile	Ser	Lys	Gly	Ser	Trp	Ser	Phe	Ser	Asp	Gln	Asp	His	Gly	Trp	
465							470					475				
Gln	Gly	Ser	Asp	Thr	Thr	Ala	Glu	Ala	Leu	Lys	Cys	Cys	Leu	Leu	Phe	
485							490					495				
Ser	Thr	Met	Pro	Pro	Glu	Ile	Val	Gly	Glu	Lys	Met	Lys	Pro	Glu	Gln	
500							505					510				
Leu	Asn	Asp	Ala	Val	Asn	Val	Ile	Leu	Ser	Leu	Gln	Ser	Lys	Asn	Gly	
515							520					525				
Gly	Leu	Ala	Ser	Trp	Glu	Pro	Ala	Gly	Ser	Ser	Glu	Trp	Leu	Glu	Ile	
530							535					540				
Phe	Asn	Pro	Thr	Glu	Phe	Phe	Ala	Asp	Ile	Val	Ile	Glu	His	Glu	Tyr	
545							550					555				
Val	Glu	Cys	Thr	Ser	Ser	Ala	Ile	Gln	Ala	Ile	Val	Leu	Phe	Asn	Lys	
565							570					575				
Leu	Tyr	Pro	Gln	His	Arg	Lys	Lys	Glu	Ile	Glu	Thr	Phe	Leu	Thr	Gly	
580							585					590				

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Ala Ser Gly Tyr Leu Glu Lys Leu Gln Thr Lys Asp Gly Ser Trp Tyr
595 600 605

Gly Asn Trp Gly Val Cys Phe Thr Tyr Gly Thr Trp Phe Gly Ile Gly
610 615 620

Gly Leu Thr Ala Val Gly Lys Thr Phe Glu Asn Cys Gln Ala Ile Gln
625 630 635 640

Lys Ala Val Lys Phe Leu Leu Glu Thr Gln Leu Glu Asp Gly Gly Trp
645 650 655

Gly Glu Ser Tyr Lys Ser Cys Pro Glu Lys Ile Tyr Ile Pro Leu Glu
660 665 670

Gly Gly Arg Ser Asn Leu Val His Thr Ser Trp Ala Met Ile Gly Leu
675 680 685

Ile Gln Ser Gln Trp Met Glu Arg Asp Ser Thr Pro Ile His Lys Ala
690 695 700

Ala Arg Leu Leu Ile Asn Ser Gln Leu Glu Asn Gly Asp Phe Pro Gln
705 710 715 720

Gln Glu Ile Thr Gly Val Phe Met Lys Asn Cys Met Leu His Tyr Ala
725 730 735

Leu Tyr Arg Asn Ile Tyr Pro Met Trp Ala Leu Ala Asp Tyr Arg Lys
740 745 750

Lys Val Leu Ser Thr Cys Lys Lys
755 760

<210> SEQ ID NO 13

<211> LENGTH: 2277

<212> TYPE: DNA

<213> ORGANISM: Taraxacum koksaghyz

<400> SEQUENCE: 13

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tttatcgga gacaaatag ggagtttgac ccaaaccatg gaaccacaga agaccgagcc	120
gaggtcgaac aagctcgact cgatttctgg aaccatcgac atcagggtcaa gcctagcagt	180
gatgtccttt ggcgcattgca gtttttgaaa gagaaagaat tcaagcaaac gatagctcag	240
gtgaaaaatag aggattctga agagatatcg tatgaaaaaa ctacaaccac attgaggagg	300
tgtgtgagct tttttgcagc gttgcaagcc agtgatggtc actggcctgc tgaaaatgcg	360
ggccactgt atttcatgca gccattgggt atctgtttat atattactgg tcatcttaac	420
attgttttcc cagaagaata tcgaaaagaa attctacgtt acttatattg ccatcagaat	480
gaagacggtg gttggggatt tcacatcgag ggtcatagca ccatgtttgg gaccaccctc	540
agttacattt gtatgcggct ctaggagaa ggtccggatg gtggtctgaa tgggtgcatgc	600
accaaagccc gcaaatggat cctggatcat ggctcagtc cgcacattcc atcatggggc	660
aaaacatggc tttcaatact tgggtgatgt gagtgggcag gaaccaaccc aatgccccca	720
gagttctggc tccttccttc ttttcttct atgtgtccag caaagatgtg gtgctactgt	780
cgggtgggtg acatgccaat gtccctaccta tatggaaaaa gatttgtggg tcccatcact	840
cctttgggtc ttcaacttag agatgaatta tacgcacagc cctacgatga gatcaactgg	900
aaaagcatat gccatttggg tgcaaggaa gatttatact accctcatcc gttcctacaa	960
gatttaattg gggatagttt atatatttgc accgagcctc tattaatcg ctggccgtta	1020
aataaattac gtcaaaaagc acttgataca acaatgaac atattcatta tgaagatgag	1080
aatagtcgtt atatcaccat tggttcgggt gtaaagccac tgtgtatgct tgcattgttg	1140

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gttgaagatc caaatggcgt ttgtttcaaa aaacatattg cacgaatccc tgattacatc 1200
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tttgcgatcc aagctctatt ggctgctgat cttactgaag aaaatggatc tactttgatg 1320
aaaggacatg aatttattaa agcttcacag gttaaagata atccttctgg agactttaaa 1380
agcatgcaca ggcatatttc gaaagggcca tggactttct cagatcaaga tcacggatgg 1440
caaatttctg actgtactgc tgaaggacta aagtgttgct tactattctt aacaatgcc 1500
gcggaaatcg ttggtgaaca tatgaaacct gagcaattta atgacgcggt caatgtaata 1560
ctttccctac agagcaaaaa cggtgggcta gcggcatggg aaccagcagg atcgtctgaa 1620
tggttgagg ttcttaatcc tacggagtgc ttgctgaca ttgtaattga gcacgagtat 1680
gttgagtgc catcgtcagc aagtcaggca ctagtttgt ttaagaagtt ataccccgga 1740
caccggagaa aagagatcga aagttttctt acaggtgcaa gtggatacct ggagaaaata 1800
cagatggaag atggttcatg gtatggaaac tgggggggtg gttttacata tggtaacctg 1860
tttgccttg gaggactgac ggctgtcggg aagacgtttg aaaattgtct agcaatccga 1920
aaagccgtca agtttctgtt agaaactcag ctggaagatg gtgggtgggg agaaagtac 1980
aaatcttgcc ctgaaaagag atatgtacca ctggaagaag gccggtcaaa ttggtacac 2040
acggcatggg gtatgatggg actgattcat tctcaacaga tggagagaga cccgatgcc 2100
ctacacaaag cagcaaagtt attgatcaat tcccagttgg aaaatggtga tttccacaa 2160
cagaaaaatg ctggagtgtt caagatgaac tgcattgctac actatgcatt gtacaggaat 2220
atattcccaa tgtgggcctt tctgattat cgcaagcatg tctgaaagg gatctag 2277

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<210> SEQ ID NO 14

<211> LENGTH: 758

<212> TYPE: PRT

<213> ORGANISM: Taraxacum koksaghyz

<400> SEQUENCE: 14

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Met Trp Lys Leu Lys Ile Ala Glu Gly Gly Asn Asp Pro Tyr Leu His
1           5           10           15
Ser Thr Asn Asn Phe Ile Gly Arg Gln Ile Trp Glu Phe Asp Pro Asn
20          25          30
His Gly Thr Thr Glu Asp Arg Ala Glu Val Glu Gln Ala Arg Leu Asp
35          40          45
Phe Trp Asn His Arg His Gln Val Lys Pro Ser Ser Asp Val Leu Trp
50          55          60
Arg Met Gln Phe Leu Lys Glu Lys Glu Phe Lys Gln Thr Ile Ala Gln
65          70          75          80
Val Lys Ile Glu Asp Ser Glu Glu Ile Ser Tyr Glu Lys Thr Thr Thr
85          90          95
Thr Leu Arg Arg Cys Val Ser Phe Phe Ala Ala Leu Gln Ala Ser Asp
100         105         110
Gly His Trp Pro Ala Glu Asn Ala Gly Pro Leu Tyr Phe Met Gln Pro
115         120         125
Leu Val Ile Cys Leu Tyr Ile Thr Gly His Leu Asn Ile Val Phe Pro
130         135         140
Glu Glu Tyr Arg Lys Glu Ile Leu Arg Tyr Leu Tyr Cys His Gln Asn
145         150         155         160
Glu Asp Gly Gly Trp Gly Phe His Ile Glu Gly His Ser Thr Met Phe
165         170         175

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Gly Thr Thr	Leu Ser Tyr	Ile Cys Met	Arg Leu Leu	Gly Glu Gly	Pro
	180	185		190	
Asp Gly Gly	Leu Asn Gly	Ala Cys Thr	Lys Ala Arg	Lys Trp Ile	Leu
	195	200		205	
Asp His Gly	Ser Val Thr	Ala Ile Pro	Ser Trp Gly	Lys Thr Trp	Leu
	210	215		220	
Ser Ile Leu	Gly Val Cys	Glu Trp Ala	Gly Thr Asn	Pro Met Pro	Pro
	225	230		235	240
Glu Phe Trp	Leu Leu Pro	Ser Phe Leu	Pro Met Cys	Pro Ala Lys	Met
	245		250		255
Trp Cys Tyr	Cys Arg Leu	Val Tyr Met	Pro Met Ser	Tyr Leu Tyr	Gly
	260		265		270
Lys Arg Phe	Val Gly Pro	Ile Thr Pro	Leu Val Leu	Gln Leu Arg	Asp
	275		280		285
Glu Leu Tyr	Ala Gln Pro	Tyr Asp Glu	Ile Asn Trp	Lys Ser Ile	Arg
	290		295		300
His Leu Cys	Ala Lys Glu	Asp Leu Tyr	Tyr Pro His	Pro Phe Leu	Gln
	305		310		315
Asp Leu Met	Trp Asp Ser	Leu Tyr Ile	Cys Thr Glu	Pro Leu Leu	Asn
	325		330		335
Arg Trp Pro	Leu Asn Lys	Leu Arg Gln	Lys Ala Leu	Asp Thr Thr	Met
	340		345		350
Lys His Ile	His Tyr Glu	Asp Glu Asn	Ser Arg Tyr	Ile Thr Ile	Gly
	355		360		365
Ser Val Val	Lys Pro Leu	Cys Met Leu	Ala Cys Trp	Val Glu Asp	Pro
	370		375		380
Asn Gly Val	Cys Phe Lys	Lys His Ile	Ala Arg Ile	Pro Asp Tyr	Ile
	385		390		395
Trp Val Ala	Glu Asp Gly	Met Lys Met	Gln Ser Phe	Gly Ser Gln	Lys
	405		410		415
Trp Asp Ala	Gly Phe Ala	Ile Gln Ala	Leu Leu Ala	Ala Asp Leu	Thr
	420		425		430
Glu Glu Asn	Gly Ser Thr	Leu Met Lys	Gly His Glu	Phe Ile Lys	Ala
	435		440		445
Ser Gln Val	Lys Asp Asn	Pro Ser Gly	Asp Phe Lys	Ser Met His	Arg
	450		455		460
His Ile Ser	Lys Gly Ser	Trp Thr Phe	Ser Asp Gln	Asp His Gly	Trp
	465		470		475
Gln Ile Ser	Asp Cys Thr	Ala Glu Gly	Leu Lys Cys	Cys Leu Leu	Phe
	485		490		495
Leu Thr Met	Pro Ala Glu	Ile Val Gly	Glu His Met	Lys Pro Glu	Gln
	500		505		510
Phe Asn Asp	Ala Val Asn	Val Ile Leu	Ser Leu Gln	Ser Lys Asn	Gly
	515		520		525
Gly Leu Ala	Ala Trp Glu	Pro Ala Gly	Ser Ser Glu	Trp Leu Glu	Val
	530		535		540
Leu Asn Pro	Thr Glu Phe	Phe Ala Asp	Ile Val Ile	Glu His Glu	Tyr
	545		550		555
Val Glu Cys	Thr Ser Ser	Ala Ser Gln	Ala Leu Val	Leu Phe Lys	Lys
	565		570		575
Leu Tyr Pro	Gly His Arg	Arg Lys Glu	Ile Glu Ser	Phe Leu Thr	Gly
	580		585		590

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Ala Ser Gly Tyr Leu Glu Lys Ile Gln Met Glu Asp Gly Ser Trp Tyr
595 600 605

Gly Asn Trp Gly Val Cys Phe Thr Tyr Gly Thr Trp Phe Ala Leu Gly
610 615 620

Gly Leu Thr Ala Val Gly Lys Thr Phe Glu Asn Cys Leu Ala Ile Arg
625 630 635 640

Lys Ala Val Lys Phe Leu Leu Glu Thr Gln Leu Glu Asp Gly Gly Trp
645 650 655

Gly Glu Ser Tyr Lys Ser Cys Pro Glu Lys Arg Tyr Val Pro Leu Glu
660 665 670

Glu Gly Arg Ser Asn Leu Val His Thr Ala Trp Gly Met Met Gly Leu
675 680 685

Ile His Ser Gln Gln Met Glu Arg Asp Pro Met Pro Leu His Lys Ala
690 695 700

Ala Lys Leu Leu Ile Asn Ser Gln Leu Glu Asn Gly Asp Phe Pro Gln
705 710 715 720

Gln Lys Ile Ala Gly Val Phe Lys Met Asn Cys Met Leu His Tyr Ala
725 730 735

Leu Tyr Arg Asn Ile Phe Pro Met Trp Ala Leu Ala Asp Tyr Arg Lys
740 745 750

His Val Leu Lys Gly Ile
755

<210> SEQ ID NO 15

<211> LENGTH: 2277

<212> TYPE: DNA

<213> ORGANISM: Taraxacum koksaghyz

<400> SEQUENCE: 15

atgtggaagt tgaagatagc acaaggaaat gatccttatac tattttccac aaacaatttt	60
gttgccggc aaatttgga atttgatccc gatgctggaa caccgcgaga gcgacaagaa	120
gtagaagatg ctcgccagta ttttcgcaat aatcgaaggg aggggtgttca tccatgtggt	180
gatttgctta tgcgggcaca gttgatcaaa gagagtggaa ttgatgtatc gagcatacca	240
ccaacaagat tggaagaaaa cgaggaagtc aattatgatg cgctaacaac cgcgggttaa	300
aaagcgctcc gattaaaccg tgcaattcaa gcaaaagacg gtcattggcc tgcagaaaat	360
gctggcccta tgtttctcac tctccactc cttattgcta tgtacattag tggagccata	420
gatacgcatc taacgaaaga acacaagaca gaaatgatac gttacatcta caaccaccaa	480
aatgaagatg gaggggtggg attttatatac gaggggcaca gcacaatgat tggatctgca	540
ctaagctacg tagccctaag gttactggga gaaggaccga atgatgggaa tggatgcagt	600
acccaagcaa gagagtggat actgcaacat ggggggtgcaa cgctcgattcc ctcttggggc	660
aaaacttata tctctgtact tggggtatat gaatgggagg gatgtaatcc acttccacca	720
gaattttggc tattccctga atctttaccg tatcatccag caaaaatgtg gtgctattgt	780
cggacaactt atatgcccac gtcatacttg tacgggaaaa aataccatgg tccaatcacc	840
gatcttgttc ttcaacttcg aaaagaaatt catccaatcc catatcacia gattgattgg	900
aataaacagc gccacaactg ttgcaaggat gatttgtact accctcattc aacactacaa	960
gatctattgt gggatgtgtc taactacttt agcgagccac ttcttaagta ttggcctttt	1020
accaaattaa gagaaaaagg tctcaaaaaga gctgttgaa taatgcgata tagcgctgaa	1080
gagagcagat atatcaccat tggatgtgtt gaaaagagtt tgcaaatgat gtgttggtgg	1140

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gcgagagaacc caaatgggga tgagttaaag catcaccttg ctaggggttc cgattattta 1200
tggttagcag aagatggaat gaagatgcaa agttttggga gtcaaataat ggattgcaca 1260
ttagcaactc aagcaataat atctacggat atggtcgaag aatacgggga ttcccttaaa 1320
aaagcccatt tttatataca agaatctcaa attaaagaaa acccatccgg agattttagt 1380
aaaatgtgtc gccagtttac taaaggagcg tggactttct ctgaccaaga tcaaggttgg 1440
gttgtctcgg attgcacagc tgaagcactt aagtgtcttc tgttactatc acaaatgcca 1500
gaagaaatct cgggagaaaa ggcggataat gaaagattat atgaggctgt taatgtcctt 1560
ctctacttac aaagtctat tagtggaggt tttgctattt gggagccacc tgtcccgcaa 1620
ccatatttac agatgttgaa tccttcgaa attttcgccc acatagtgtt ggaaaaagag 1680
catgttgagt gtacgtcgtc aattattcaa gcccttttag ccttcaaacg attgcacccc 1740
ggtcacaggg agaagaaat cgaatatatc gtgtcaaaag cagttagttt ttggaggaa 1800
aaacaatggc acgatgggtc atggatgggt tattggggaa tatgcttcac atatgggacg 1860
ttttttactc taggaggggt aatttcggct ggaaaaacat ataacaatag tgaagcgggt 1920
cgtagagcgg ttaatttttt cctttcgaca caaaatgaag agggtaggat gggtagagac 1980
attaagtctt gccctagtga agtgtacaca ccgttgatg aaaataggac aaatttggtt 2040
caaacttcgt gggctatgct tgggcttatg ttaggtggac aggccgaaag agatccaaca 2100
ccattgcata aagcagcaaa gatattaatt aattcacaaa tggataacgg agattttcca 2160
caacaggaga ttactggagt atacatgaag aattgcatgc tacattatgc agagtacaga 2220
aacattttcc cactttgggc acttggggaa tatcgcaaaa gagtttgggt caattaa 2277

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<210> SEQ ID NO 16

<211> LENGTH: 758

<212> TYPE: PRT

<213> ORGANISM: Taraxacum koksaghyz

<400> SEQUENCE: 16

```

Met Trp Lys Leu Lys Ile Ala Gln Gly Asn Asp Pro Tyr Leu Phe Ser
1           5           10          15
Thr Asn Asn Phe Val Gly Arg Gln Ile Trp Glu Phe Asp Pro Asp Ala
20          25          30
Gly Thr Pro Ala Glu Arg Gln Glu Val Glu Asp Ala Arg Gln Tyr Phe
35          40          45
Arg Asn Asn Arg Arg Glu Gly Val His Pro Cys Gly Asp Leu Leu Met
50          55          60
Arg Ala Gln Leu Ile Lys Glu Ser Gly Ile Asp Val Ser Ser Ile Pro
65          70          75          80
Pro Thr Arg Leu Glu Glu Asn Glu Glu Val Asn Tyr Asp Ala Leu Thr
85          90          95
Thr Ala Val Lys Lys Ala Leu Arg Leu Asn Arg Ala Ile Gln Ala Lys
100         105         110
Asp Gly His Trp Pro Ala Glu Asn Ala Gly Pro Met Phe Phe Thr Pro
115         120         125
Pro Leu Leu Ile Ala Met Tyr Ile Ser Gly Ala Ile Asp Thr His Leu
130         135         140
Thr Lys Glu His Lys Thr Glu Met Ile Arg Tyr Ile Tyr Asn His Gln
145         150         155         160
Asn Glu Asp Gly Gly Trp Gly Phe Tyr Ile Glu Gly His Ser Thr Met
165         170         175

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Ile	Gly	Ser	Ala 180	Leu	Ser	Tyr	Val	Ala 185	Leu	Arg	Leu	Leu	Gly 190	Glu	Gly
Pro	Asn	Asp	Gly 195	Asn	Gly	Ala	Val 200	Thr	Gln	Ala	Arg	Glu 205	Trp	Ile	Leu
Gln	His	Gly	Gly 210	Ala	Thr	Ser 215	Ile	Pro	Ser	Trp	Gly 220	Lys	Thr	Tyr	Leu
Ser 225	Val	Leu	Gly	Val	Tyr 230	Glu	Trp	Glu	Gly	Cys 235	Asn	Pro	Leu	Pro	Pro 240
Glu	Phe	Trp	Leu	Phe 245	Pro	Glu	Ser	Leu	Pro 250	Tyr	His	Pro	Ala	Lys 255	Met
Trp	Cys	Tyr	Cys 260	Arg	Thr	Thr	Tyr	Met 265	Pro	Met	Ser	Tyr	Leu 270	Tyr	Gly
Lys	Lys	Tyr 275	His	Gly	Pro	Ile	Thr 280	Asp	Leu	Val	Leu	Gln 285	Leu	Arg	Lys
Glu 290	Ile	His	Pro	Ile	Pro	Tyr 295	His	Lys	Ile	Asp	Trp 300	Asn	Lys	Gln	Arg
His 305	Asn	Cys	Cys	Lys	Asp 310	Asp	Leu	Tyr	Tyr	Pro 315	His	Ser	Thr	Leu	Gln 320
Asp	Leu	Leu	Trp	Asp 325	Gly	Leu	Asn	Tyr	Phe 330	Ser	Glu	Pro	Leu	Leu 335	Lys
Tyr	Trp	Pro	Phe 340	Thr	Lys	Leu	Arg	Glu 345	Lys	Gly	Leu	Lys	Arg 350	Ala	Val
Glu	Leu	Met 355	Arg	Tyr	Ser	Ala	Glu 360	Glu	Ser	Arg	Tyr	Ile 365	Thr	Ile	Gly
Cys 370	Val	Glu	Lys	Ser	Leu	Gln 375	Met	Met	Cys	Trp 380	Trp	Ala	Glu	Asn	Pro
Asn 385	Gly	Asp	Glu	Phe	Lys 390	His	His	Leu	Ala	Arg 395	Val	Pro	Asp	Tyr	Leu 400
Trp	Leu	Ala	Glu	Asp 405	Gly	Met	Lys	Met	Gln 410	Ser	Phe	Gly	Ser	Gln 415	Ile
Trp	Asp	Cys	Thr 420	Leu	Ala	Thr	Gln	Ala 425	Ile	Ile	Ser	Thr	Asp 430	Met	Val
Glu	Glu	Tyr 435	Gly	Asp	Ser	Leu	Lys 440	Lys	Ala	His	Phe	Tyr 445	Ile	Gln	Glu
Ser	Gln	Ile	Lys	Glu	Asn 450	Pro 455	Ser	Gly	Asp	Phe 460	Ser	Lys	Met	Cys	Arg
Gln 465	Phe	Thr	Lys	Gly	Ala 470	Trp	Thr	Phe	Ser	Asp 475	Gln	Asp	Gln	Gly	Trp 480
Val	Val	Ser	Asp 485	Cys	Thr	Ala	Glu	Ala 490	Leu	Lys	Cys	Leu	Leu 495	Leu	Leu
Ser	Gln	Met	Pro 500	Glu	Glu	Ile	Ser	Gly 505	Glu	Lys	Ala	Asp 510	Asn	Glu	Arg
Leu	Tyr	Glu 515	Ala	Val	Asn	Val	Leu 520	Leu	Tyr	Leu	Gln	Ser 525	Pro	Ile	Ser
Gly 530	Gly	Phe	Ala	Ile	Trp	Glu 535	Pro	Pro	Val	Pro	Gln	Pro 540	Tyr	Leu	Gln
Met 545	Leu	Asn	Pro	Ser	Glu 550	Ile	Phe	Ala	Asp	Ile 555	Val	Val	Glu	Lys	Glu 560
His	Val	Glu	Cys 565	Thr	Ser	Ser	Ile	Ile	Gln	Ala 570	Leu	Leu	Ala	Phe 575	Lys
Arg	Leu	His	Pro 580	Gly	His	Arg	Glu	Lys 585	Glu	Ile	Glu	Ile 590	Ser	Val	Ser
Lys	Ala	Val	Ser	Phe	Leu	Glu	Glu	Lys	Gln	Trp	His	Asp	Gly	Ser	Trp

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595	600	605
Tyr Gly Tyr Trp Gly Ile Cys Phe Ile Tyr Gly Thr Phe Phe Thr Leu 610 615 620		
Gly Gly Leu Ile Ser Ala Gly Lys Thr Tyr Asn Asn Ser Glu Ala Val 625 630 635 640		
Arg Arg Ala Val Asn Phe Phe Leu Ser Thr Gln Asn Glu Glu Gly Gly 645 650 655		
Trp Gly Glu Ser Ile Lys Ser Cys Pro Ser Glu Val Tyr Thr Pro Leu 660 665 670		
Asp Glu Asn Arg Thr Asn Leu Val Gln Thr Ser Trp Ala Met Leu Gly 675 680 685		
Leu Met Leu Gly Gly Gln Ala Glu Arg Asp Pro Thr Pro Leu His Lys 690 695 700		
Ala Ala Lys Ile Leu Ile Asn Ser Gln Met Asp Asn Gly Asp Phe Pro 705 710 715 720		
Gln Gln Glu Ile Thr Gly Val Tyr Met Lys Asn Cys Met Leu His Tyr 725 730 735		
Ala Glu Tyr Arg Asn Ile Phe Pro Leu Trp Ala Leu Gly Glu Tyr Arg 740 745 750		
Lys Arg Val Trp Val Asn 755		

<210> SEQ ID NO 17

<211> LENGTH: 2277

<212> TYPE: DNA

<213> ORGANISM: Taraxacum koksaghyz

<400> SEQUENCE: 17

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atgtggaagc taaagatagc agaagggaat gatccttact tgtttagcac caacaacttt      60
gttgccgcc aaatttggga gttcgatccc gatgcaggaa cacctgaaga acggcaagaa      120
gtagagaatg ctcgcaaaa ttttcttaat cgacaaaagg aagggtttca gacatcgggc      180
gatttactca tgcggataca gttaatcagg gagaatggaa ttgatttact gagcatacca      240
cctgcgaggt tgggagaaaa agaggatgtg gattatgaag cagtgcgcac atcaataaga      300
aaagctgttc gatacagttg tgccattcaa gcaaaagatg gtcattgggc tactgaacat      360
ggtggccctt tgtttctcac tcctccctt attattatct tatacattag tggagccatt      420
gatacacatt taacaaaaga acacaagaag gagatgaaac gttttatcca caaccatcaa      480
aatgaagatg gaggatgggg attccatc gaaggacata gcaccatggt tgtaactgtc      540
ttaagttagt tgtccctacg attgctaggg gaagaaaaac acgacacaaa tgttgcatg      600
attcgtgcc aaaaatggat acttgaccat ggtggtgcga ccaatgtacc atcttggggc      660
aaactttatc tctcggttct tgggttatat gaatgggaag gatgcaaccc aataccacct      720
gaattttgga tttccctga gttcttacca ttcatccag caaaaatgtg gtgctattgt      780
cgaacggcct atatgccaat gtcatacttg tacgggagaa aattccaagg cccaatcact      840
gatcttgtgt tacaactgag aaaagagatt tatcttacac catacaacga aataaattgg      900
aatgaacatc gccataaatg ttgcaaggaa gacctctatt accctcatic aatagtccaa      960
gatttggtat gggatgttct ttattacttg agtgagccac tcttcaagtt ttggcctttt     1020
accaaactaa gagaaaaagc cctcaaaaga acacttgagt tgactcgcta taatgctgaa     1080
gaaagcaggt acattacat ggcaagtatt gaaaagggtt ttcaaatgat gtgttggtgg     1140
gcggcgaacc caaacggaga tgaattcaag catcaccttg ctagactacc agattactta     1200

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tggttagcag aagatggaat gaagtcgcaa acttttggtg gtcaattatg gtgtagcgca 1260
tttgcaacac aagcaataat cgcaagtaat atgcccgaag aatatgggga ttctctcaaa 1320
aaggctcatt tctttctcaa agaatctcag gttaaacaaa acccaaaggg agattttact 1380
aaaatgtgtc gacagtttag caaaggatcg tggacgttca ccgatcaaga tcatggttgg 1440
cctgtctcag attgtacagg agaagcattg aaatgcctgc ttttattatc ccaaagccc 1500
gaagaaattt caggagaaaa agtagataat cagcggctat atgatgctgt taacttctc 1560
cttcacgtac aaagtcctac aacaggaggt ttgtctgttt gggaaaaacc gatccctcat 1620
ccatatttac agacgttaaa tccttcagaa atgttcgccg acattgttgt tgaaagagag 1680
catgttgagt gcacaacttc agtgatgcaa gctctcatcg cattccaaca ctccactcc 1740
gggcacaggg aaaaagaaat agaaaacgct attgcaaatg cagtacgtta tctagaggag 1800
atacaacggg aagatggttc atggtatggc tattggggta tatgtttcat atatggcaca 1860
ttcttttctg taggaggtt agaatcagct ggaaaaacat ataatgattg tgaagcaatt 1920
cgaaaaggag ccaagttttt actttcgaca caaaatgaag aaggtggatg gggagagagc 1980
tataaatctt gccctagtga agtgtagaca ccgctagatg ggaataaac taatatagtt 2040
caaacttcat gggctatgct tgggtctcatg tcatcggggc aggctgaacg agatccaaca 2100
ccgttacata aagcagcaag gatcttgatt aatgcacaaa tggataatgg agattttccg 2160
caacaggaga tgactggagc tgcgatgagg aattgcattg tacattatcc gttatatagt 2220
aatatcttcc ccttattggc tctttccaaa tatcgccaac aattttggga taaatga 2277

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<210> SEQ ID NO 18

<211> LENGTH: 758

<212> TYPE: PRT

<213> ORGANISM: Taraxacum koksaghyz

<400> SEQUENCE: 18

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Met Trp Lys Leu Lys Ile Ala Glu Gly Asn Asp Pro Tyr Leu Phe Ser
1             5             10             15

Thr Asn Asn Phe Val Gly Arg Gln Ile Trp Glu Phe Asp Pro Asp Ala
20            25            30

Gly Thr Pro Glu Glu Arg Gln Glu Val Glu Asn Ala Arg Gln Asn Phe
35            40            45

Leu Asn Arg Gln Lys Glu Gly Phe Gln Thr Cys Gly Asp Leu Leu Met
50            55            60

Arg Ile Gln Leu Ile Arg Glu Asn Gly Ile Asp Leu Leu Ser Ile Pro
65            70            75            80

Pro Ala Arg Leu Gly Glu Lys Glu Asp Val Asp Tyr Glu Ala Val Thr
85            90            95

Thr Ser Ile Arg Lys Ala Val Arg Tyr Ser Cys Ala Ile Gln Ala Lys
100           105           110

Asp Gly His Trp Ala Thr Glu His Gly Gly Pro Leu Phe Phe Thr Pro
115           120           125

Pro Leu Ile Ile Ile Leu Tyr Ile Ser Gly Ala Ile Asp Thr His Leu
130           135           140

Thr Lys Glu His Lys Lys Glu Met Lys Arg Phe Ile His Asn His Gln
145           150           155           160

Asn Glu Asp Gly Gly Trp Gly Phe His Ile Glu Gly His Ser Thr Met
165           170           175

Phe Val Thr Val Leu Ser Tyr Val Ser Leu Arg Leu Leu Gly Glu Glu

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180							185					190				
Lys	His	Asp	Thr	Asn	Val	Ala	Leu	Ile	Arg	Ala	Arg	Lys	Trp	Ile	Leu	
		195					200					205				
Asp	His	Gly	Gly	Ala	Thr	Asn	Val	Pro	Ser	Trp	Gly	Lys	Leu	Tyr	Leu	
		210				215					220					
Ser	Val	Leu	Gly	Val	Tyr	Glu	Trp	Glu	Gly	Cys	Asn	Pro	Ile	Pro	Pro	
225					230					235				240		
Glu	Phe	Trp	Ile	Phe	Pro	Glu	Phe	Leu	Pro	Phe	His	Pro	Ala	Lys	Met	
				245					250					255		
Trp	Cys	Tyr	Cys	Arg	Thr	Ala	Tyr	Met	Pro	Met	Ser	Tyr	Leu	Tyr	Gly	
			260					265					270			
Arg	Lys	Phe	Gln	Gly	Pro	Ile	Thr	Asp	Leu	Val	Leu	Gln	Leu	Arg	Lys	
		275					280					285				
Glu	Ile	Tyr	Leu	Thr	Pro	Tyr	Asn	Glu	Ile	Asn	Trp	Asn	Glu	His	Arg	
		290				295					300					
His	Lys	Cys	Cys	Lys	Glu	Asp	Leu	Tyr	Tyr	Pro	His	Thr	Ile	Val	Gln	
305					310					315				320		
Asp	Leu	Leu	Trp	Asp	Gly	Leu	Tyr	Tyr	Leu	Ser	Glu	Pro	Leu	Phe	Lys	
				325					330					335		
Phe	Trp	Pro	Phe	Thr	Lys	Leu	Arg	Glu	Lys	Ala	Leu	Lys	Arg	Thr	Leu	
			340					345					350			
Glu	Leu	Thr	Arg	Tyr	Asn	Ala	Glu	Glu	Ser	Arg	Tyr	Ile	Thr	Met	Ala	
		355				360						365				
Ser	Ile	Glu	Lys	Gly	Phe	Gln	Met	Met	Cys	Trp	Trp	Ala	Ala	Asn	Pro	
		370				375					380					
Asn	Gly	Asp	Glu	Phe	Lys	His	His	Leu	Ala	Arg	Leu	Pro	Asp	Tyr	Leu	
385					390					395				400		
Trp	Leu	Ala	Glu	Asp	Gly	Met	Lys	Ser	Gln	Thr	Phe	Gly	Ser	Gln	Leu	
				405					410					415		
Trp	Cys	Ser	Ala	Phe	Ala	Thr	Gln	Ala	Ile	Ile	Ala	Ser	Asn	Met	Pro	
			420					425					430			
Glu	Glu	Tyr	Gly	Asp	Ser	Leu	Lys	Lys	Ala	His	Phe	Phe	Leu	Lys	Glu	
		435				440					445					
Ser	Gln	Val	Lys	Gln	Asn	Pro	Lys	Gly	Asp	Phe	Thr	Lys	Met	Cys	Arg	
		450				455					460					
Gln	Phe	Ser	Lys	Gly	Ser	Trp	Thr	Phe	Thr	Asp	Gln	Asp	His	Gly	Trp	
				470					475					480		
Pro	Val	Ser	Asp	Cys	Thr	Gly	Glu	Ala	Leu	Lys	Cys	Leu	Leu	Leu	Leu	
				485				490					495			
Ser	Gln	Met	Pro	Glu	Glu	Ile	Ser	Gly	Glu	Lys	Val	Asp	Asn	Gln	Arg	
			500					505					510			
Leu	Tyr	Asp	Ala	Val	Asn	Phe	Leu	Leu	His	Val	Gln	Ser	Pro	Thr	Thr	
		515				520						525				
Gly	Gly	Phe	Ala	Val	Trp	Glu	Lys	Pro	Ile	Pro	His	Pro	Tyr	Leu	Gln	
		530				535					540					
Thr	Leu	Asn	Pro	Ser	Glu	Met	Phe	Ala	Asp	Ile	Val	Val	Glu	Arg	Glu	
545					550					555				560		
His	Val	Glu	Cys	Thr	Thr	Ser	Val	Met	Gln	Ala	Leu	Ile	Ala	Phe	Gln	
				565				570						575		
His	Phe	His	Ser	Gly	His	Arg	Glu	Lys	Glu	Ile	Glu	Asn	Ala	Ile	Ala	
			580					585					590			
Asn	Ala	Val	Arg	Tyr	Leu	Glu	Glu	Ile	Gln	Arg	Glu	Asp	Gly	Ser	Trp	
		595				600						605				

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Tyr Gly Tyr Trp Gly Ile Cys Phe Ile Tyr Gly Thr Phe Phe Ser Leu
 610 615 620
 Gly Gly Leu Glu Ser Ala Gly Lys Thr Tyr Asn Asp Cys Glu Ala Ile
 625 630 635 640
 Arg Lys Gly Ala Lys Phe Leu Leu Ser Thr Gln Asn Glu Glu Gly Gly
 645 650 655
 Trp Gly Glu Ser Tyr Lys Ser Cys Pro Ser Glu Val Tyr Thr Pro Leu
 660 665 670
 Asp Gly Asn Lys Thr Asn Ile Val Gln Thr Ser Trp Ala Met Leu Gly
 675 680 685
 Leu Met Ser Ser Gly Gln Ala Glu Arg Asp Pro Thr Pro Leu His Lys
 690 695 700
 Ala Ala Arg Ile Leu Ile Asn Ala Gln Met Asp Asn Gly Asp Phe Pro
 705 710 715 720
 Gln Gln Glu Met Thr Gly Ala Ala Met Arg Asn Cys Met Leu His Tyr
 725 730 735
 Pro Leu Tyr Ser Asn Ile Phe Pro Leu Leu Ala Leu Ser Lys Tyr Arg
 740 745 750
 Gln Gln Phe Trp Asp Lys
 755

<210> SEQ ID NO 19

<211> LENGTH: 2277

<212> TYPE: DNA

<213> ORGANISM: Taraxacum koksaghyz

<400> SEQUENCE: 19

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cacgtcggcc gtcagcactg gcagtttgat ccggatgctg gaaccgaaga ggaacgtgct    120
cagattgaaa agattcgtct caacttcaaa cttaatcgtt ttcaattcaa acaaagtgcc    180
gacttgtaa tgcgtactca actaagaaaa gagaacccaa tcaataaaat accggatgca    240
ataaaattga atgaacaga agaagtgaca aatgacgcag tgtcaactac actcaaaaga    300
gccattagct ttactccac cattcaagcc catgatgggc actggccagc tgagtctgct    360
ggccctttgt tcttcttcc tccattggtg atagcactat atgtgactgg agcaatgaat    420
gatattctaa caccgcaca tcagctagaa ataaacggtt acatatacaa tcatcagaat    480
gaagatggag gttggggatt acatatagag ggtcatagca caatatttgg atcagtactt    540
agttacataa ctttaagatt acttggggaa gaagctgata gtgttcgaga ggacatggct    600
ttgggttaag ggcgtaaatg gatccttgac catggtggtg cagttgggat tccttcgtgg    660
ggaaagtttt ggcttacgat acttgagta tacgaatggg gaggctgtaa tcctatgcca    720
cccgaatttt ggctcatgcc taagttttcc ccaattcatc caggcaaaat gttgtgttat    780
tgtcgcttag ttacatgcc catgctgtac ttatacggca aaagatttgt gggaaaaata    840
accgagttgg ttcgcgacct aaggcaagag ctttatacag acccttatga tgagattaat    900
tggaataaag cacgaaacac gtgtgcaaag gaagatctct actatccaca cccttttgtt    960
caagatatgg tatggggtgt acttcataat gtttttgaac ctgtattaac aagtcgtccg    1020
ctttccacac taagagaaaa ggctttgaaa gtcgcaatgg atcatgttca ctatgaagat    1080
aagagtagta gatatctttg cattggatgt gtggaaaagg tggtatgctt gattgcaacg    1140
tgggtggaag atccaaatgg tgatgcatat aaacgtcatc ttgctagaat tcctgactac    1200

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ttttgggttg ctgaagatgg gatgaaaatg cagagttttg gatgtcaaat gtgggatgcg 1260
gcctttgcta ttcaagctat tttatctagt aatctagccg aagaatacgg cccgactctt 1320
aaaaaagcac acgagtttgt aaaagcatca caggttcgtg ataatccgcc gggagatttc 1380
agtaaaatgt acagacacac ttctaagggt gcatggacat tttccatata agatcatggt 1440
tggcaagtct ctgattgtac ggctgaaggc ttgaagggtg cacttttgta ctcccaaagt 1500
agcccagaac ttgtggcgga aaaacttgaa actgagcatc tctacgatgc tgtcaatgtc 1560
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acacaagaaa gtgatggttc atggtatggt tgttggggaa tttgctacac ctatggtaca 1860
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catacatcat gggctttaat atcacttatt aaagcgggac aggctgaaat tgatcctaca 2100
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caacaggaaa tcacaggagt gttcatgaag aactgtaacc tcaattactc atcatatcga 2220
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<210> SEQ ID NO 20

<211> LENGTH: 758

<212> TYPE: PRT

<213> ORGANISM: Taraxacum koksaghyz

<400> SEQUENCE: 20

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Met Trp Lys Leu Lys Ile Ala Glu Gly Ser Asp Asp Glu Trp Leu Thr
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Thr Thr Asn Asn His Val Gly Arg Gln His Trp Gln Phe Asp Pro Asp
20        25        30

Ala Gly Thr Glu Glu Glu Arg Ala Gln Ile Glu Lys Ile Arg Leu Asn
35        40        45

Phe Lys Leu Asn Arg Phe Gln Phe Lys Gln Ser Ala Asp Leu Leu Met
50        55        60

Arg Thr Gln Leu Arg Lys Glu Asn Pro Ile Asn Lys Ile Pro Asp Ala
65        70        75        80

Ile Lys Leu Asn Glu Thr Glu Glu Val Thr Asn Asp Ala Val Ser Thr
85        90        95

Thr Leu Lys Arg Ala Ile Ser Phe Tyr Ser Thr Ile Gln Ala His Asp
100       105       110

Gly His Trp Pro Ala Glu Ser Ala Gly Pro Leu Phe Phe Leu Pro Pro
115       120       125

Leu Val Ile Ala Leu Tyr Val Thr Gly Ala Met Asn Asp Ile Leu Thr
130       135       140

Pro Ala His Gln Leu Glu Ile Lys Arg Tyr Ile Tyr Asn His Gln Asn
145       150       155       160

Glu Asp Gly Gly Trp Gly Leu His Ile Glu Gly His Ser Thr Ile Phe
165       170       175

Gly Ser Val Leu Ser Tyr Ile Thr Leu Arg Leu Leu Gly Glu Glu Ala
180       185       190

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Asp	Ser	Val	Ala	Glu	Asp	Met	Ala	Leu	Val	Lys	Gly	Arg	Lys	Trp	Ile
		195					200					205			
Leu	Asp	His	Gly	Gly	Ala	Val	Gly	Ile	Pro	Ser	Trp	Gly	Lys	Phe	Trp
	210					215					220				
Leu	Thr	Ile	Leu	Gly	Val	Tyr	Glu	Trp	Gly	Gly	Cys	Asn	Pro	Met	Pro
	225				230					235					240
Pro	Glu	Phe	Trp	Leu	Met	Pro	Lys	Phe	Phe	Pro	Ile	His	Pro	Gly	Lys
				245					250					255	
Met	Leu	Cys	Tyr	Cys	Arg	Leu	Val	Tyr	Met	Pro	Met	Ser	Tyr	Leu	Tyr
			260					265					270		
Gly	Lys	Arg	Phe	Val	Gly	Lys	Ile	Thr	Glu	Leu	Val	Arg	Asp	Leu	Arg
		275					280					285			
Gln	Glu	Leu	Tyr	Thr	Asp	Pro	Tyr	Asp	Glu	Ile	Asn	Trp	Asn	Lys	Ala
	290					295					300				
Arg	Asn	Thr	Cys	Ala	Lys	Glu	Asp	Leu	Tyr	Tyr	Pro	His	Pro	Phe	Val
	305				310					315					320
Gln	Asp	Met	Val	Trp	Gly	Val	Leu	His	Asn	Val	Phe	Glu	Pro	Val	Leu
				325					330					335	
Thr	Ser	Arg	Pro	Leu	Ser	Thr	Leu	Arg	Glu	Lys	Ala	Leu	Lys	Val	Ala
			340					345					350		
Met	Asp	His	Val	His	Tyr	Glu	Asp	Lys	Ser	Ser	Arg	Tyr	Leu	Cys	Ile
		355					360					365			
Gly	Cys	Val	Glu	Lys	Val	Leu	Cys	Leu	Ile	Ala	Thr	Trp	Val	Glu	Asp
	370					375					380				
Pro	Asn	Gly	Asp	Ala	Tyr	Lys	Arg	His	Leu	Ala	Arg	Ile	Pro	Asp	Tyr
	385				390					395					400
Phe	Trp	Val	Ala	Glu	Asp	Gly	Met	Lys	Met	Gln	Ser	Phe	Gly	Cys	Gln
			405						410					415	
Met	Trp	Asp	Ala	Ala	Phe	Ala	Ile	Gln	Ala	Ile	Leu	Ser	Ser	Asn	Leu
			420					425					430		
Ala	Glu	Glu	Tyr	Gly	Pro	Thr	Leu	Lys	Lys	Ala	His	Glu	Phe	Val	Lys
		435					440					445			
Ala	Ser	Gln	Val	Arg	Asp	Asn	Pro	Pro	Gly	Asp	Phe	Ser	Lys	Met	Tyr
	450					455					460				
Arg	His	Thr	Ser	Lys	Gly	Ala	Trp	Thr	Phe	Ser	Ile	Gln	Asp	His	Gly
	465				470				475						480
Trp	Gln	Val	Ser	Asp	Cys	Thr	Ala	Glu	Gly	Leu	Lys	Val	Ala	Leu	Leu
				485					490					495	
Tyr	Ser	Gln	Met	Ser	Pro	Glu	Leu	Val	Gly	Glu	Lys	Leu	Glu	Thr	Glu
			500					505					510		
His	Leu	Tyr	Asp	Ala	Val	Asn	Val	Ile	Leu	Ser	Leu	Gln	Ser	Glu	Asn
		515					520					525			
Gly	Gly	Phe	Pro	Ala	Trp	Glu	Pro	Gln	Arg	Ala	Tyr	Ala	Trp	Leu	Glu
	530					535					540				
Lys	Phe														

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Tyr Gly Cys Trp Gly Ile Cys Tyr Thr Tyr Gly Thr Trp Phe Ala Val
 610 615 620
 Asp Ala Leu Val Ala Cys Gly Lys Asn Tyr His Asn Ser Pro Ala Leu
 625 630 635 640
 Gln Lys Ala Cys Lys Phe Leu Leu Ser Lys Gln Leu Pro Asp Gly Gly
 645 650 655
 Trp Gly Glu Ser Tyr Leu Ser Ser Ser Asn Lys Val Tyr Thr Asn Leu
 660 665 670
 Glu Gly Asn Arg Ser Asn Leu Val His Thr Ser Trp Ala Leu Ile Ser
 675 680 685
 Leu Ile Lys Ala Gly Gln Ala Glu Ile Asp Pro Thr Pro Ile Ser Asn
 690 695 700
 Gly Val Arg Leu Leu Ile Asn Ser Gln Met Glu Glu Gly Asp Phe Pro
 705 710 715 720
 Gln Gln Glu Ile Thr Gly Val Phe Met Lys Asn Cys Asn Leu Asn Tyr
 725 730 735
 Ser Ser Tyr Arg Asn Ile Phe Pro Ile Trp Ala Leu Gly Glu Tyr Arg
 740 745 750
 Arg Ile Val Gln Asn Ile
 755

<210> SEQ ID NO 21
 <211> LENGTH: 30
 <212> TYPE: DNA
 <213> ORGANISM: Artificial
 <220> FEATURE:
 <223> OTHER INFORMATION: TkOSC1-RNAi-fwd-NcoI

<400> SEQUENCE: 21

aaaccatggg cggaattgat cttataagcg

30

<210> SEQ ID NO 22
 <211> LENGTH: 29
 <212> TYPE: DNA
 <213> ORGANISM: Artificial
 <220> FEATURE:
 <223> OTHER INFORMATION: TkOSC1-RNAi-rev-XhoI

<400> SEQUENCE: 22

aaactcgaga tccaagctt gaatcgcac

29

<210> SEQ ID NO 23
 <211> LENGTH: 29
 <212> TYPE: DNA
 <213> ORGANISM: Artificial
 <220> FEATURE:
 <223> OTHER INFORMATION: TkOSC-RNAi-fwd-NcoI

<400> SEQUENCE: 23

aaaccatgga acaagaaaat ggttcttg

29

<210> SEQ ID NO 24
 <211> LENGTH: 29
 <212> TYPE: DNA
 <213> ORGANISM: Artificial
 <220> FEATURE:
 <223> OTHER INFORMATION: TkOSC-RNAi-rev-XhoI

<400> SEQUENCE: 24

aaactcgagt ctccaagcgc ccatagcgg

29

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<210> SEQ ID NO 25
<211> LENGTH: 21
<212> TYPE: DNA
<213> ORGANISM: Artificial
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<223> OTHER INFORMATION: TkeF1alpha-fw-realtime

<400> SEQUENCE: 25

cgagagattc gagaaggaag c 21

<210> SEQ ID NO 26
<211> LENGTH: 21
<212> TYPE: DNA
<213> ORGANISM: Artificial
<220> FEATURE:
<223> OTHER INFORMATION: TkeF1alpha-rv-realtime

<400> SEQUENCE: 26

ctgtgcagta gtacttggtg g 21

<210> SEQ ID NO 27
<211> LENGTH: 22
<212> TYPE: DNA
<213> ORGANISM: Artificial
<220> FEATURE:
<223> OTHER INFORMATION: TkLUP-fw-realtime

<400> SEQUENCE: 27

gctgaccacc accaacaacc ac 22

<210> SEQ ID NO 28
<211> LENGTH: 23
<212> TYPE: DNA
<213> ORGANISM: Artificial
<220> FEATURE:
<223> OTHER INFORMATION: TkLUP-rv-realtime

<400> SEQUENCE: 28

agcacgttcc tcttcggttc cag 23

<210> SEQ ID NO 29
<211> LENGTH: 22
<212> TYPE: DNA
<213> ORGANISM: Artificial
<220> FEATURE:
<223> OTHER INFORMATION: TkOSC1-fw-realtime

<400> SEQUENCE: 29

actcctccct tgataattgc cc 22

<210> SEQ ID NO 30
<211> LENGTH: 26
<212> TYPE: DNA
<213> ORGANISM: Artificial
<220> FEATURE:
<223> OTHER INFORMATION: TkOSC1-rv-realtime

<400> SEQUENCE: 30

ttgtgcttct gcctgatata tagaac 26

<210> SEQ ID NO 31
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial
<220> FEATURE:
<223> OTHER INFORMATION: TkOSC2-fw-realtime

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<400> SEQUENCE: 31
ccggtgagaa ggtggaagtt 20

<210> SEQ ID NO 32
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial
<220> FEATURE:
<223> OTHER INFORMATION: TkOSC2-rv-realtime

<400> SEQUENCE: 32
ggaaccggta cctcccaaac 20

<210> SEQ ID NO 33
<211> LENGTH: 21
<212> TYPE: DNA
<213> ORGANISM: Artificial
<220> FEATURE:
<223> OTHER INFORMATION: TkOSC3-fw-realtime

<400> SEQUENCE: 33
tccatccaaa ccacagaaaa g 21

<210> SEQ ID NO 34
<211> LENGTH: 22
<212> TYPE: DNA
<213> ORGANISM: Artificial
<220> FEATURE:
<223> OTHER INFORMATION: TkOSC3-rv-realtime

<400> SEQUENCE: 34
atgaagcata ctccccaata ac 22

<210> SEQ ID NO 35
<211> LENGTH: 22
<212> TYPE: DNA
<213> ORGANISM: Artificial
<220> FEATURE:
<223> OTHER INFORMATION: TkOSC4-fw-realtime

<400> SEQUENCE: 35
ccgacaattc gtaaggccac tg 22

<210> SEQ ID NO 36
<211> LENGTH: 20
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<213> ORGANISM: Artificial
<220> FEATURE:
<223> OTHER INFORMATION: TkOSC4-rv-realtime

<400> SEQUENCE: 36
tgtttgcacc acgttcgacc 20

<210> SEQ ID NO 37
<211> LENGTH: 24
<212> TYPE: DNA
<213> ORGANISM: Artificial
<220> FEATURE:
<223> OTHER INFORMATION: TkOSC5-fw-realtime

<400> SEQUENCE: 37
gaaacacaac tagaagatgg cggt 24

<210> SEQ ID NO 38
<211> LENGTH: 22

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<212> TYPE: DNA
<213> ORGANISM: Artificial
<220> FEATURE:
<223> OTHER INFORMATION: TkOSC5-rv-realtime

<400> SEQUENCE: 38

catagcccat gaagtgtgca ct 22

<210> SEQ ID NO 39
<211> LENGTH: 21
<212> TYPE: DNA
<213> ORGANISM: Artificial
<220> FEATURE:
<223> OTHER INFORMATION: TkOSC6-fw-realtime

<400> SEQUENCE: 39

ggtcatagca ccatgtttgg g 21

<210> SEQ ID NO 40
<211> LENGTH: 22
<212> TYPE: DNA
<213> ORGANISM: Artificial
<220> FEATURE:
<223> OTHER INFORMATION: TkOSC6-rv-realtime

<400> SEQUENCE: 40

ggtgactgag ccatgatcca gg 22

<210> SEQ ID NO 41
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial
<220> FEATURE:
<223> OTHER INFORMATION: TkOSC7-fw-realtime

<400> SEQUENCE: 41

cgatgctgga acacccgc 18

<210> SEQ ID NO 42
<211> LENGTH: 23
<212> TYPE: DNA
<213> ORGANISM: Artificial
<220> FEATURE:
<223> OTHER INFORMATION: TkOSC7-rv-realtime

<400> SEQUENCE: 42

gttggtggta tgctcgatac atc 23

<210> SEQ ID NO 43
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial
<220> FEATURE:
<223> OTHER INFORMATION: TkOSC8-fw-realtime

<400> SEQUENCE: 43

ctgagcatat cacctgcgag 20

<210> SEQ ID NO 44
<211> LENGTH: 27
<212> TYPE: DNA
<213> ORGANISM: Artificial
<220> FEATURE:
<223> OTHER INFORMATION: TkOSC8-rv-realtime

<400> SEQUENCE: 44

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cagcttttct tattgatgtc gtcactg 27

<210> SEQ ID NO 45
 <211> LENGTH: 22
 <212> TYPE: DNA
 <213> ORGANISM: Artificial
 <220> FEATURE:
 <223> OTHER INFORMATION: TkRP-fw-realtime

<400> SEQUENCE: 45

cgtcgatctc aaggatgttg tc 22

<210> SEQ ID NO 46
 <211> LENGTH: 22
 <212> TYPE: DNA
 <213> ORGANISM: Artificial
 <220> FEATURE:
 <223> OTHER INFORMATION: TkRP-rv-realtime

<400> SEQUENCE: 46

ggagctttga gaagaaccaa cg 22

<210> SEQ ID NO 47
 <211> LENGTH: 445
 <212> TYPE: DNA
 <213> ORGANISM: Artificial
 <220> FEATURE:
 <223> OTHER INFORMATION: Sense sequence in TkOSC hpRNAi construct

<400> SEQUENCE: 47

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 tcgtgttaca aggattggta gcttgtggac aaacatatga aaatagtga gcagttcgga 120
 aagctgtcaa gtttttcctc tcgacacaaa actcagaagg tggttggga gagaactttg 180
 agtcatgccc acaagagaaa ttatacctt tggaggga cagaacaaat tttgtgcaaa 240
 cttcatgggc catgcttgggt cttctatgtg gtggacaggc tgaagagat ccgacaccat 300
 tacacaaggc agcaaaattg ctcataaatg gacaaatgga taatggagac tttcctcaac 360
 aggaaataac aggagtgtac atgaagaatt gcatgttaca ttatgcagaa tataggaaca 420
 cttttccgct atgggcgctt ggaga 445

<210> SEQ ID NO 48
 <211> LENGTH: 125
 <212> TYPE: DNA
 <213> ORGANISM: Artificial
 <220> FEATURE:
 <223> OTHER INFORMATION: Sense sequence in TkOSC1 hpRNAi construct

<400> SEQUENCE: 48

gcggaattga tcttataagc gtacctccag tgagacttgg agacgacgaa gatgtgaact 60
 atgaagcggg gacaacagcc gtgagaaagg cagtttagact gaaccgtgcg attcaagctt 120
 gggat 125

<210> SEQ ID NO 49
 <211> LENGTH: 5889
 <212> TYPE: DNA
 <213> ORGANISM: Artificial
 <220> FEATURE:
 <223> OTHER INFORMATION: TkOSC-RNAi expression vector
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: (3361)..(3361)
 <223> OTHER INFORMATION: n is a, c, g, or t

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<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (3574)..(3574)
<223> OTHER INFORMATION: n is a, c, g, or t

<400> SEQUENCE: 49

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gacgttttta atgtactgaa ttaacgccga attaattcga gctcggatct gataatttat      120
ttgaaaatto ataagaaaag caaacgttac atgaattgat gaaacaatac aaagacagat      180
aaagccacgc acatttagga tattggccga gattactgaa tattgagtaa gatcacggaa      240
tttctgacag gagcatgtct tcaattcagc ccaaatggca gttgaaatac tcaaaccgcc      300
ccatatgcag gagcggatca ttcatgtgtt gtttggttgc ctttgccaac atgggagtcc      360
aagattctgc agtcaaactc cggtgacggg caggaccgga cggggcggtg ccggcaggct      420
gaagtccagc tggcagaaac ccacgtcatg ccagttcccg tgcttgaagc cggccgcccg      480
cagcatgccg cggggggcat atccgagcgc ctctgcatg cgcacgctcg ggtcgttggg      540
cagcccgatg acagcgacca cgctcttgaa gccctgtgcc tccagggact tcagcaggtg      600
ggtgtagagc gtggagccca gtcccgtccg ctggtggcgg ggggagacgt acacggtcga      660
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ggccgggcgt cgttctgggc tcatcgattc gatttgggtg atcgagattg gttatgaaat      960
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aattcaggcc cgggtgccat gtccacacg ccgaaataaa cgaccaaatt agtagaaaaa     1260
taaaaactga ctcgatact tacgtcacgt cttgcgcact gatttgaata atctcagaat     1320
taagcttccc gggattgttt catgcgcacg gaaacgttaa gagcttcgga ttatcatttt     1380
gataaagaaa ttaataaaag ataacgaaat aaatttaact ttaaaatttt gaaaatttat     1440
caggattctt atgaaagaaa ctgtatcagg attcttagga caaaaactag ctagtctctt     1500
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atcatatgtg atttctgcat ggaacatctt atacacatgc agcatctcat tcacatgtaa     1620
ttgaactcgt cagaaaagtc attaaagtca ttaactaac aatcaagata aactaaaaag     1680
ttatttaaat ttctttaaaa aaaatttga acgacaacta ttgtaactta atcttaattt     1740
tacacatctc ggttcaaaca tgaataaat attctaccca aacaataaat ggtaacttaa     1800
tttagtttct ttaatttgt tgattgctta taatctctta gataatataa atacgtcaaa     1860
tcagaagtgt tggtatccac aaaaacgtgt tttaatgaat tcaactagggt aaaatcccg     1920
gtaatacacg ggttaaaaat ttaataaca tataaataat ataattttat gtttgataat     1980
tcttaatatt tataacataa catgaacaat tttagaatat attaagattt aaaataagat     2040
aatagtaatt aatagttacc ataacaataa ttgaaaaata ttgtcaaaaa ttaggaaatt     2100
tcaattttga ttaaatgggc gggaaaataa aattgcgaaa gattatggtg ataacacgtg     2160

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tcctatacgt attcttttat cagtagtaga ttatatgcaa gaaatgctaa atgacaaaaa	2220
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The invention claimed is:

1. A method of reducing the content of triterpenes, triterpenoids, or both triterpenes and triterpenoids, of latex or rubber from a laticiferous plant or part thereof, comprising genetically modifying the laticiferous plant or part thereof, such that the activity of at least one oxidosqualene cyclase is reduced, compared to a like laticiferous plant or part thereof that has not been genetically modified by at least one heterologous nucleotide sequence that is capable of reducing the activity of the at least one oxidosqualene cyclase,

wherein the laticiferous plant or part thereof is genetically modified by at least one heterologous nucleotide sequence that is capable of reducing the activity of the at least one oxidosqualene cyclase,

wherein the at least one heterologous nucleotide sequence encodes at least one RNAi that interferes with the mRNA encoding the at least one oxidosqualene cyclase, with the proviso that the at least one RNAi is capable of reducing the activity of the at least one oxidosqualene cyclase,

wherein the at least one RNAi is specifically directed against the mRNA transcript of the at least one oxidosqualene cyclase and thereby affects the presence of the protein of the at least one oxidosqualene cyclase, wherein the at least one oxidosqualene cyclase is selected from the group consisting of an amino acid sequence being at least 99% identical to any one of SEQ ID NOs: 4, 6, 8, 10, 12, 14, 16, 18, or 20, and

wherein the at least one RNAi comprises or consists of a nucleotide sequence being at least 99% identical to SEQ ID NO: 1 or SEQ ID NO: 2,

wherein the at least one RNAi attaches individually or binds to the mRNA transcript of the at least one oxidosqualene cyclase,

wherein the at least one RNAi comprising or consisting of a nucleotide sequence being at least 99% identical to SEQ ID NO: 1 is specifically directed against the mRNA transcript of the at least one oxidosqualene cyclase selected from the group consisting of an amino acid sequence being at least 99% identical to any one of SEQ ID NOs: 4, 6, 8, 10, 12, 14, 16, 18, or 20, and wherein the at least one RNAi comprising or consisting of a nucleotide sequence being at least 99% identical to SEQ ID NO: 2 is specifically directed against the mRNA transcript of the oxidosqualene cyclase of an amino acid sequence being at least 99% identical to SEQ ID NO: 4.

2. The method according to claim 1, wherein the laticiferous plant or part thereof is selected from the group of plant families consisting of Apocynaceae, Asteraceae, Euphorbiaceae and Moraceae.

3. The method according to claim 1, wherein the laticiferous plant or part thereof is selected from the group consisting of *Agoseris glauca*, *Apocynum venetum*, *Chrysanthamnus graveolens*, *Chrysanthamnus nauseosus*, *Chrysanthamnus viscidiflorus*, *Funtumia elastica*, *Parthenium argentatum*, *Scorzonera acanthoclada*, *Scorzonera albicaulis*, *Scorzonera divaricate*, *Scorzonera hissaricata*, *Scorzonera racemosa*, *Scorzonera tau-saghyz*, *Scorzonera tragapogonoides*, *Scorzonera turkestanica*, *Scorzonera virgate*,

Solidago canadensis scabra, *Solidago fistulosa*, *Solidago leavenworthii*, *Solidago rigida*, *Sonchus oleraceus*, *Taraxacum brevicorniculatum*, *Taraxacum hybernum*, *Taraxacum koksaghyz*, *Taraxacum megalorrhizon*, *Taraxacum officinale* and crossings thereof.

4. The method according to claim 1, wherein the at least one oxidosqualene cyclase is selected from the group consisting of oxidosqualene cyclase 1 according to SEQ ID NO: 4, the oxidosqualene cyclase 2 according to SEQ ID NO: 6, the oxidosqualene cyclase 3 according to SEQ ID NO: 8, the oxidosqualene cyclase 4 according to SEQ ID NO: 10, the oxidosqualene cyclase 5 according to SEQ ID NO: 12, the oxidosqualene cyclase 6 according to SEQ ID NO: 14, the oxidosqualene cyclase 7 according to SEQ ID NO: 16, the oxidosqualene cyclase 8 according to SEQ ID NO: 18, and the lupeol synthase according to SEQ ID NO: 20.

5. The method according to claim 1, wherein the part of the laticiferous plant is selected from the group consisting of cells, tissues, organs, roots, stems, branches, leaves, exudates, latex and mixtures thereof.

6. The method according to claim 1, wherein the at least one RNAi comprises or consists of the nucleotide sequence according to SEQ ID NO: 1, or comprises or consists of the nucleotide sequence according to SEQ ID NO: 2.

7. The method according to claim 3, wherein the laticiferous plant or part thereof is selected from the group consisting of *Parthenium argentatum*, *Chrysanthamnus nauseosus*, *Taraxacum koksaghyz*, *Taraxacum officinale* and crossings thereof.

8. The method according to claim 4, wherein the at least one oxidosqualene cyclase is selected from the group consisting of the oxidosqualene cyclase 1 according to SEQ ID NO: 4, the oxidosqualene cyclase 2 according to SEQ ID NO: 6, the oxidosqualene cyclase 3 according to SEQ ID NO: 8, and the oxidosqualene cyclase 5 according to SEQ ID NO: 12, and an amino acid sequence being at least 99% identical to any one of SEQ ID NOs: 4, 6, 8 or 12.

9. The method according to claim 1, wherein the at least one RNAi comprising or consisting of a nucleotide sequence being identical to SEQ ID NO: 1 is specifically directed against the mRNA transcript of at least one oxidosqualene cyclase selected from the group consisting of an amino acid sequence being identical to any one of SEQ ID NOs: 4, 6, 8, 10, 12, 14, 16, 18, or 20, and wherein the at least one RNAi comprising or consisting of a nucleotide sequence being identical to SEQ ID NO: 2 is specifically directed against the mRNA transcript of the oxidosqualene cyclase of an amino acid sequence being identical to SEQ ID NO: 4.

10. The method according to claim 1, wherein the at least one RNAi comprising or consisting of a nucleotide sequence being identical to SEQ ID NO: 1 is specifically directed against the mRNA transcript of the at least one oxidosqualene cyclase selected from the group consisting of an amino acid sequence being identical to any one of SEQ ID NOs: 4, 6, 8, 10, 12, 14, or 20.

11. The method according to claim 1, wherein the at least one RNAi comprising or consisting of a nucleotide sequence being identical to SEQ ID NO: 2 is specifically directed against the mRNA transcript of the oxidosqualene cyclase of an amino acid sequence being identical to SEQ ID NO: 4.

119

12. The method according to claim 1, wherein taraxasterol and/or β -amyirin, is/are reduced in the lactiferous plant or part thereof.

13. The method according to claim 1, wherein the plant is (a) *Taraxacum koksaghyz*, (b) wherein the part of the plant is the plant root, or both (a) and (b).

14. A method of increasing the content of a triterpene precursor, a triterpenoid precursor, or both triterpene and triterpenoid precursors, of latex or rubber from a laticiferous plant or part thereof, comprising genetically modifying the laticiferous plant or part thereof, such that the activity of at least one oxidosqualene cyclase is reduced, compared to a like laticiferous plant or part thereof that has not been genetically modified by at least one heterologous nucleotide sequence that is capable of reducing the activity of the at least one oxidosqualene cyclase,

wherein the laticiferous plant or part thereof is genetically modified by at least one heterologous nucleotide sequence that is capable of reducing the activity of the at least one oxidosqualene cyclase,

wherein the at least one heterologous nucleotide sequence encodes at least one RNAi that interferes with the mRNA encoding the at least one oxidosqualene cyclase, with the proviso that the at least one RNAi is capable of reducing the activity of the at least one oxidosqualene cyclase,

wherein the at least one RNAi is specifically directed against the mRNA transcript of the at least one oxidosqualene cyclase and thereby affects the presence of the protein of the at least one oxidosqualene cyclase,

120

wherein the at least one oxidosqualene cyclase is selected from the group consisting of an amino acid sequence being at least 99% identical to any one of SEQ ID NOs: 4, 6, 8, 10, 12, 14, 16, 18, or 20, and

wherein the at least one RNAi comprises or consists of the nucleotide sequence being at least 99% identical to SEQ ID NO: 1 or SEQ ID NO: 2,

wherein the at least one RNAi attaches individually or binds to the mRNA transcript of the at least one oxidosqualene cyclase,

wherein the at least one RNAi comprising or consisting of a nucleotide sequence being at least 99% identical to SEQ ID NO: 1 is specifically directed against the mRNA transcript of the at least one oxidosqualene cyclase selected from the group consisting of an amino acid sequence being at least 99% identical to any one of SEQ ID NOs: 4, 6, 8, 10, 12, 14, 16, 18, or 20, and

wherein the at least one RNAi comprising or consisting of a nucleotide sequence being at least 99% identical to SEQ ID NO: 2 is specifically directed against the mRNA transcript of the oxidosqualene cyclase of an amino acid sequence being at least 99% identical to SEQ ID NO: 4.

15. The method according to claim 14, wherein one or more of 2,3-oxidosqualene, cycloartenol, 24-methylene cycloartenol is increased in the lactiferous plant or part thereof.

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